StratoCNC

Machine Description Manual

Comprehensive Hardware and I/O Documentation

SFERRI Ltd, BULGARIA

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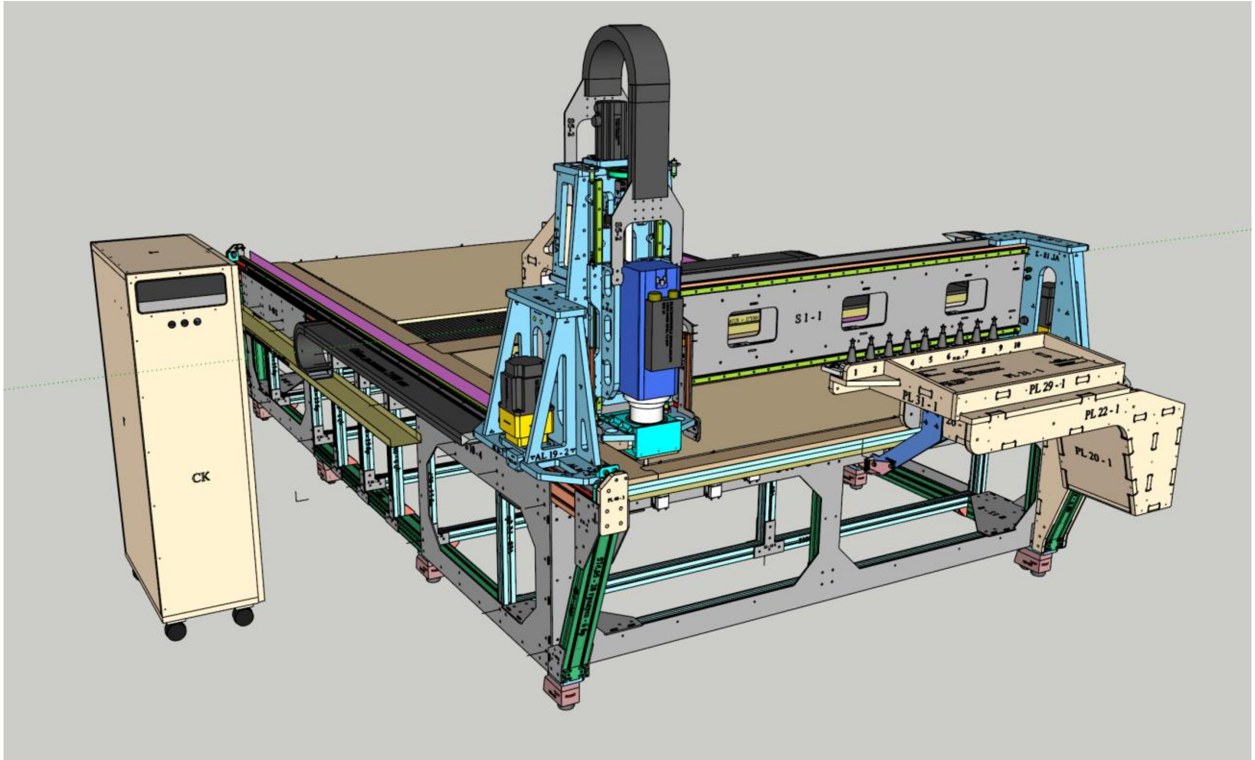

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1. Introduction

The StratoCNC is a professional-grade CNC Router developed and manufactured by SFERRI Ltd, Bulgaria. Engineered for demanding production environments, the StratoCNC combines industrial-quality servo motion control, an advanced automatic tool change system, and a comprehensive software macro architecture to deliver precision, reliability, and operator efficiency in a single integrated platform.

The main purpose of the system is to manipulate two sheets of 1525x1525 plywood simultaneously and independently with the laser and the spindle. A sheet enters one side of the system for engraving, then exits via the other after being cut:

I. Load with ease a sheet of plywood (or any other stock) on the one side of the machine by utilizing the integrated ball bearing rollers.

II. Position it quickly and precisely with the positioning pins.

III. Run the integrated laser engraver that will engrave the individual elements with bar codes, markings, assembly tips, and instructions.

IV. Slide the engraved sheet to the cutting area using the rollers and position it again with the pins on its new working position.

V. Engage the sheet with the vacuum.

VI. Run the cutting cycle – the system will automatically start/stop the spindle, automatically adjust the RPMs, automatically load/change/measure tools. Cutting a file with a whole plywood sheet of different elements, multiple toolpaths/operations, and many tool changes is a ONE CLICK-task – just load the file and press CYCLE START!

VII. Offload with ease the sheet with the help of the rollers.

At its core, the machine is built around the proven UCCNC control software paired with the high-performance UC300ETH Ethernet motion controller, delivering real-time pulse generation at up to 400 kHz with negligible jitter. The four-axis servo drive system — powered by DELTA high-inertia AC servo motors with 17-bit and 24-bit encoders — provides closed-loop accuracy and dynamic performance that far exceeds conventional stepper-based routers. An HSD 7.5 kW air-cooled electrospindle with ISO30 automatic tool change capability, a retractable 10-position tool rack with integrated tool length probe, an 8-zone vacuum table with independent vacuum motors, and a full suite of pneumatically actuated subsystems round out the hardware platform.

Every aspect of the StratoCNC — from its epoxy-granite-filled aluminium extrusion frame to its multi-point adjustment capabilities for gantry levelling, squaring, and spindle tramming — has been designed to meet the rigorous demands of precision woodworking, composite machining, and light metalwork. A proprietary macro system running over 20 integrated C# macros provides automated homing and squaring, tool measurement, spindle safety monitoring, and comprehensive health diagnostics, ensuring safe and efficient operation from startup to shutdown.

Regardless of the stellar characteristics and capabilities, the StratoCNC is designed with affordability in mind. It is not a monolithic heavy monster with a 'black box' logic and components that can only be maintained by a professional specialist at high cost. It is an open-code style modular system that can be delivered as a kit, and it can be easily assembled, modified, operated, and maintained with fun and confidence by everyone. At the same time, it is not a hobby machine but a serious workhorse built to deliver and to last.

This manual provides a comprehensive description of the machine's hardware, electrical I/O configuration, key components, mechanical systems, and adjustment capabilities. It is intended to serve as the primary technical reference for operators, integrators, and maintenance personnel working with the StratoCNC platform.

2. Machine Overview

The StratoCNC is distinguished by the following key features and capabilities:

- Modern and reliable UCCNC software paired with a compatible UC300ETH motion controller for smooth and reliable toolpath generation
- Custom UCCNC screenset with custom buttons and features
- Intelligent and comprehensive 'operating system' of custom macros for thoroughly managing and controlling the daily operations and the health of the machine
- Powerful DELTA 0,9 KW Servo Motors paired with solid Planetary Gearbox Reducers for extreme precision, low to zero backlash, high efficiency, and a monstrous **torque of 60 NM**
- Industrial grade HSD 7,5 KW ATC Spindle paired with DELTA VFD
- High-end drive system for smooth, precise, and reliable operations that last - Helical rack GAMBINI MOD2 and pinions DOGUS CALIP for X/Y axes, TBI ballscrew with a nut for the Z axis.
- High-end guide rails and blocks TBI TR25 (25 mm) for efficient and smooth movements along the axes
- An independent CNC Laser engraver is mounted on the machine for labeling and engraving of the workpiece
- Enormous working area of **1559 /3264 mm (X/Y)**
- Maximum thickness of the workpiece – 45 mm
- Smart Decentralized Vacuum table with 8 individual zones. Each zone has its own vacuum motor instead of one big expensive vacuum pump for the whole table. The upper surface of the table is covered with 1 mm perforated rubber for improved holding characteristics of the vacuum table. Low maintenance, hassle-free, respectable workpiece holding!
- Spring-loaded pusher – integrated with a retractable dust collector camera. The engagement of the pusher is dynamically controlled by a dedicated macro, so when the machine is not cutting, the pusher is retracted
- Pneumatically retractable Tool Holder Rack with 10 tool slots
- Plunger Type Tool Length Probe (mounted on the tool rack)
- Touchplate for quick and precise measurement of G54 Z0 (the top of the workpiece, regardless of its thickness and the tool length)
- Pneumatic positioning pins – for quick placement of the workpiece on the **G54 X0/Y0** position
- Pneumatic transportation rollers (ball bearings) for easy handling of the sheets (loading and unloading)
- Vibration dampening - the main structure of the machine is designed with aluminum extruded profiles filled with epoxy and sand for vibration dampening and added stiffness. The structure is reinforced with 5 mm steel plates
- Built-in mechanical adjustment capability – the structure of the machine is designed to be adjustable in all the possible meaningful ways – there are provisions for leveling the whole machine, leveling the gantry assembly itself, squaring the gantry mechanically (in addition to a software-based macro squaring on a daily routine), and tramming the spindle in all directions

- Modular design – the system is not a monolithic one and can be easily transported, assembled, and deployed in confined and constrained or hard-to-reach spaces. No cranes or wide industrial doors needed
- Rigidity and robustness – Strato CNC is not yet another aluminum extrusion CNC. Indeed Its main structure is made with epoxy-granite-filled extrusions, but in all the 8 corners of the parallelepiped, there are sets of self-locking 5-mm-thick steel plates that form a box-like geometry. All the main assemblies are made from 20 mm aluminium with grooves for improved geometric accuracy, stiffness, and overall strength
- Ease of assembly – all the structural components are designed to be easily assembled, with no special tools required, no welding, and no specialized equipment
- Last but not least – a comprehensive documentation with lots of manuals, diagrams, data, catalogues, and software code

3. Dimensions and Coordinate System

Dimensions

	X (Width)	Y (Length)	Z (Height)
Overall Machine Dimensions	2615 mm	4350 mm	2220 mm

Weights

	Machine	Y-moving assembly	X-moving assembly	Z-moving assembly
Weight	1500 kg	340 kg	130 kg	48 kg

Travel Limitations

The machine's travel on each axis is constrained by three progressively tighter boundaries, providing layered protection against over-travel conditions.

Travel Boundary	X Axis (mm)	Y Axis (mm)	Z Axis (mm)
Physical (mechanical bumpers)	1575	3280	290
Usable (proximity switches – G53 home)	1569	3274	284
Working (UCCNC soft limits)	1559	3264	236

Coordinate System Conventions

Axis	Positive Direction	Negative Direction
X Axis	RIGHT	LEFT
Y Axis	AFT (away from operator)	FRONT (towards operator)
Z Axis	UP	DOWN

Homing Switch Positions

The proximity limit switches that define the Machine Origin (G53 X0 Y0 Z0) are positioned as follows:

Axis	Switch Location	Homing Behavior
X	Left side of the gantry	Spindle moves LEFT (negative) until the switch triggers. G53 X0 established; only positive X values used in normal operation.
Y	Front-left and front-right corners of the table	Gantry moves FRONT (negative) until both switches trigger. G53 Y0 established; only positive Y values used in normal operation. Both Y-axis switches enable gantry squaring.
Z	Upper side of the spindle assembly	Spindle moves UP (positive) until the switch triggers. G53 Z0 established; only negative Z values used in normal operation.

Work Offsets and Key Coordinates

The G54 work offset represents the upper front-left corner of the workpiece sheet. For a typical 18 mm plywood workpiece, the theoretical G54 offsets from the G53 machine origin are:

Work Offset	Value	Description
G54 X0	G53 X0 + 22 mm	22 mm to the RIGHT of Machine Home
G54 Y0	G53 Y0 + 17 mm	17 mm AFT of Machine Home
G54 Z0	G53 Z0 − 240 mm (est.)	Approx. 240 mm BELOW Machine Home. Precise value determined by the M20303 touch plate and tool probe calibration macro.

Tool Rack Coordinates (G53)

Parameter	Value	Notes
X of the first tool	740 mm	740 mm from Machine Home along X
Tool spacing (X)	80 mm	Tool 2 at 820 mm, Tool 3 at 900 mm, etc.
Y (rack extended)	17 mm	All tools at 17 mm AFT from Machine Home
Z for tool change	−92 mm	ISO30 holder aligned with rack clamps
Z for lateral positioning	−7 mm	Safe clearance above tools for X-axis traversal

Tool Length Probe Coordinates (G53)

Parameter	Value	Notes
Probe X	1550 mm	Center of probe sensitive surface
Probe Y	17 mm	Center of probe sensitive surface
Probe Z	≈ −30 mm	Approximate – precise value determined by M20303 calibration macro
Z for probing start	−4 mm	Starting height for the probing routine

Relationship between the different travel limitations and coordinate systems

- The machine travels on each of three axis are restricted physically by mechanical bumpers mounted on the main structure.
- In normal circumstances, the machine should never hit these bumpers, and it should be stopped by the proximity switches at a distance of 3 mm from the bumpers. The sensitivity of the proximity switches is 4 mm, which means that when the machine triggers the limit switch and is stopped, it should be at exactly 4 mm from the limit switch. If, for some reason, the switch doesn’t trigger and the machine continues its movement after 3 mm, it will hit the bumper and stop. This gives a 1 mm safety clearance for the limit switch.

- The machine is allowed to travel to the limit switches only during homing and squaring routines. During normal operations, its travel is limited by soft limits set in the UCCNC software. For X and Y, these soft limits are set to 5 mm from the proximity switches limits in both directions. For Z, the soft limit is 3 mm from the proximity switches.
- In theory, if we are using an 18 mm thick plywood as our work-piece, the G54 working offsets, which give us the coordinates of the UPPER, FRONT, LEFT corner of the work piece sheet, are calculated as follows:

  G54 X0 = G53 X0+22 (22 mm to the RIGHT from the machine Home)

  G54 Y0 = G53 Y+17 (17 mm AFT rom the machine Home)

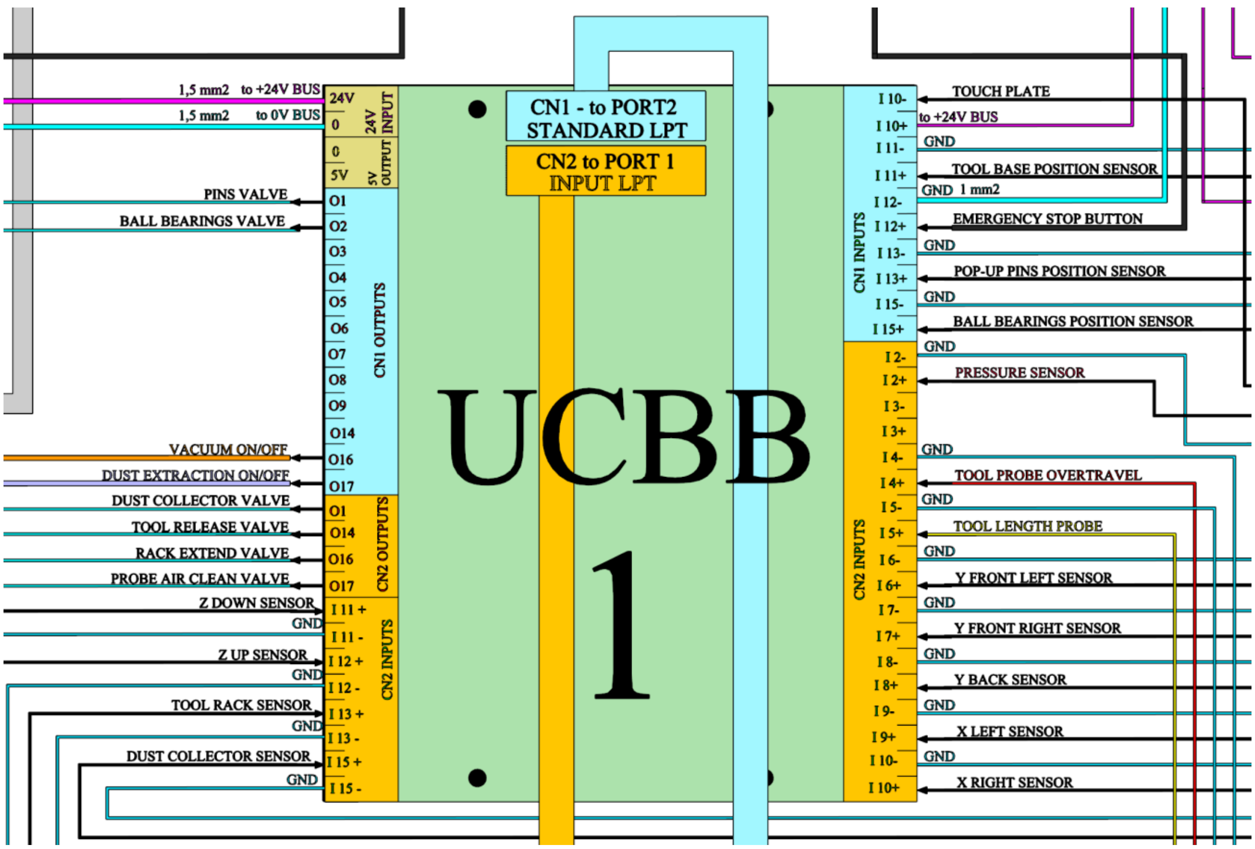
  G54 Z0 = G53 Z0 - 240 (240 mm BENEATH the machine Home) – this is an estimated value. With the use of a Touch Plate and a macro called M20303, the real G54 Z0 is found for the current tool length and the current workpiece thickness.

4. Pin Assignment Tables

The StratoCNC uses two UCBB Breakout Boards connected to the UC300ETH motion controller. Each board provides two I/O ports, for a total of 36 digital inputs and 32 digital outputs across 4 ports. The following tables document the complete pin assignment for the system.

UCBB-1: Sensors, Solenoids, and I/O

UCBB-1 handles all solenoid valve outputs, proximity sensors, touch plate, tool probe, emergency stop, and pump relays. Port CN2 connects to UC300ETH Port1 via an IDC26 ribbon cable; Port CN1 connects to UC300ETH Port 2 via an IDC26 ribbon cable. The board is powerd with 24VDC.



Power Connections

Function	UCBB-1 Terminal	Notes
+24 V DC power supply	+24V	Min 2A capacity, powers sensors and valves
GND	0V	Common ground for all sensors and valves

Solenoid Valve Outputs

Component	Function	UCBB-1 Terminal	UC300ETH Port/Pin	UCCNC Pin	Active Low	Specification
Tool Release Valve	Controls tool release	OUT14 (CN2-14)	Port 1, Pin 14	LED14	No	24V DC, NC
Rack Extend Valve	Tool rack extension	OUT16 (CN2-16)	Port 1, Pin 16	LED16	No	24V DC, NC
Probe Air Clean Valve	Cleans the probe surface	OUT17 (CN2-17)	Port 1, Pin 17	LED17	No	24V DC, NC
Dust Collector Down	Lowers the dust collector	OUT1 (CN2-1)	Port 1, Pin 1	LED1	No	24V DC, NC
Pop-up Pins Valve	Extends pop-up pins	OUT1 (CN1-1)	Port 2, Pin 1	LED69	No	24V DC, NC
Ball Bearings Valve	Extends ball bearings	OUT2 (CN1-2)	Port 2, Pin 2	LED70	No	24V DC, NC

Tool Probe, Touch Plate, and Overtravel

Component	Function	UCBB-1 Terminal	UC300ETH Port/Pin	UCCNC Pin	Active Low	Specification
Tool Length Probe	Tool length measurement	IN5+ (CN2-5)	Port 1, Pin 5	LED5	Yes	24V PNP NC
Probe Overtravel	Overtravel alarm	IN4+ (CN2-4)	Port 1, Pin 4	LED4	Yes	24V PNP NC
Touch Plate	G54 Z=0 detection	IN10+ (CN1-10)	Port 2, Pin 10	LED78	No	NO switch

Axis Limit Switches (PNP NC Serial Pairs)

Function	UCBB-1 Terminal	UC300ETH Port/Pin	UCCNC Pin	Active Low	Notes
Y Front-Left Limit	IN6+ (CN2-6)	Port 1, Pin 6	LED6	Yes	Y negative limit/home (LEFT)
Y Front-Right Limit	IN7+ (CN2-7)	Port 1, Pin 7	LED7	Yes	Y negative limit/home (RIGHT)
Y Back Limit	IN8+ (CN2-8)	Port 1, Pin 8	LED8	Yes	Y positive limit
X Left Limit	IN9+ (CN2-9)	Port 1, Pin 9	LED9	Yes	X negative limit/home
X Right Limit	IN10+ (CN2-10)	Port 1, Pin 10	LED10	Yes	X positive limit
Z Up Limit	IN12+ (CN2-12)	Port 1, Pin 12	LED12	Yes	Z upper limit
Z Down Limit	IN11+ (CN2-11)	Port 1, Pin 11	LED11	Yes	Z lower limit

Position Proximity Sensors (PNP NC)

Function	UCBB-1 Terminal	UC300ETH Port/Pin	UCCNC Pin	Active Low	Notes
Tool Rack Position	IN13+ (CN2-13)	Port 1, Pin 13	LED13	Yes	Rack Extended/Retracted
Dust Collector Position	IN15+ (CN2-15)	Port 1, Pin 15	LED15	Yes	Collector Up/Down
Tool Base Position	IN11+ (CN1-11)	Port 2, Pin 11	LED79	Yes	Rotating base In/Out
Pop-up Pins Position	IN13+ (CN1-13)	Port 2, Pin 13	LED81	Yes	Pins Extended/Retracted
Ball Bearing Position	IN15+ (CN1-15)	Port 2, Pin 15	LED83	Yes	Bearings Extended/Retracted

Emergency Stop

Function	UCBB-1 Terminal	UC300ETH Port/Pin	UCCNC Pin	Notes
Emergency Stop Button	IN12+ (CN1-12)	Port 2, Pin 12	LED80 (Active Low: Yes)	Normally Closed

Pneumatic Pressure Sensor

Function	UCBB-1 Terminal	UC300ETH Port/Pin	UCCNC Pin	Notes
Pneumatic Pressure Sensor	IN2+ (CN2-2)	Port 1, Pin 2	LED2 (Active Low: Yes)	PNP NC, 24V DC

Vacuum and Dust Pump Relays

Function	UCBB-1 Terminal	UC300ETH Port/Pin	UCCNC Pin	Notes
Vacuum Pump Relay	OUT16 (CN1-16)	Port 2, Pin 16	LED84 (Active Low: No)	Turns ON vacuum pump
Dust Pump Relay	OUT17 (CN1-17)	Port 2, Pin 17	LED85 (Active Low: No)	Turns ON dust extraction pump

Available (unused) UCBB-1 Pins for Future Expansion

The following pins are currently unassigned and available for future peripheral expansion:

(8 OUTPUT pins and 1 INPUT pin in total)

CN1 – OUT 3
CN1 – OUT 4
CN1 – OUT 5

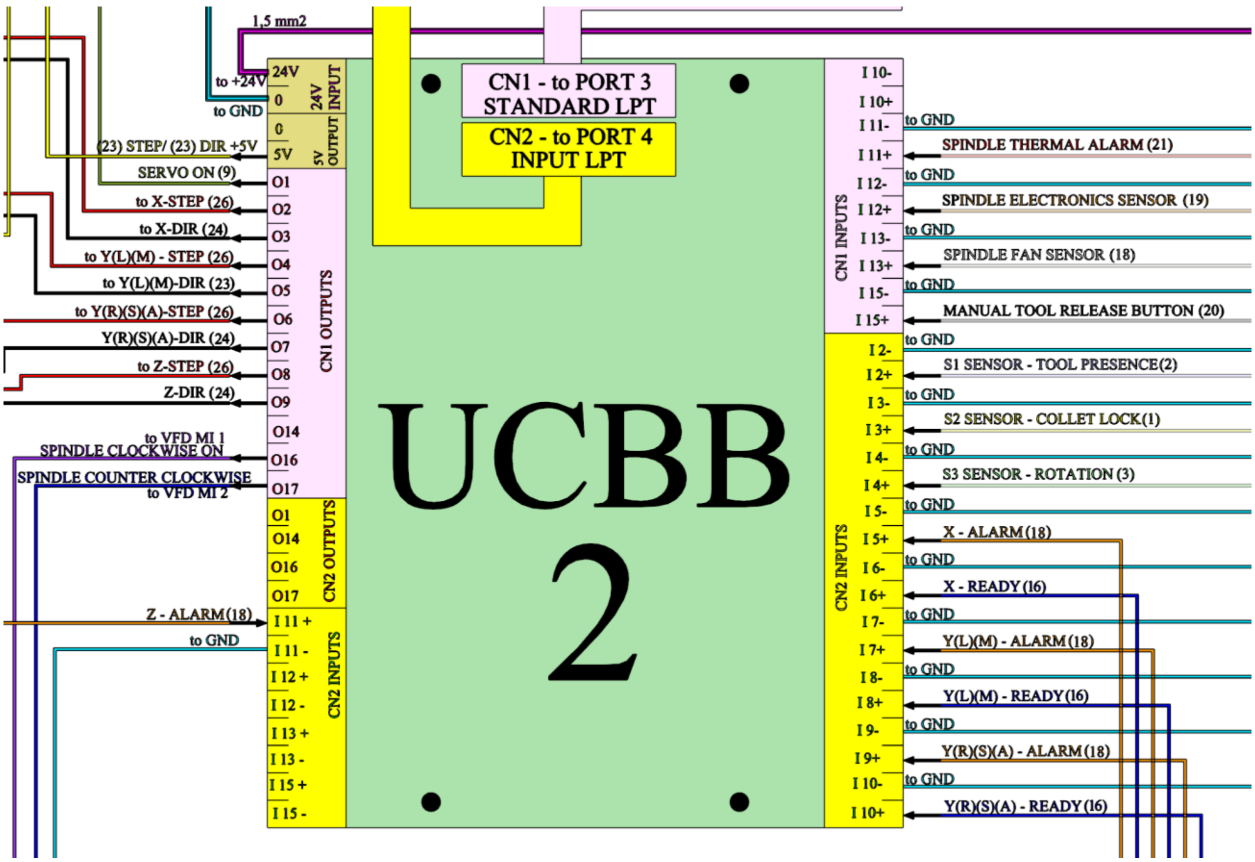
CN1 – OUT 6
CN1 – OUT 7
CN1 – OUT 8

CN1 – OUT 9
CN1 – OUT 14

CN2 – IN 3+

UCBB-2: Servo Drivers and Spindle Control

UCBB-2 handles all servo motor control signals (STEP/DIR/ENABLE), servo monitoring (ALARM/READY), spindle control, and spindle sensor inputs. Port CN1 connects to UC300ETH Port3 via a ribbon IDC26 cable; Port CN2 connects to UC300ETH Port4 v via a ribbon IDC26 cable. The board is powered with 24VDC.



Servo Motion Control (STEP/DIR/ENABLE)

Function	UCBB-2 Terminal	UC300ETH Port/Pin	UCCNC Pin	Active Low	Notes
Enable (All Servos)	OUT1 (CN1-1)	Port 3, Pin 1	LED86	No	Connected to all 4 servo drives
X-Axis STEP	OUT2 (CN1-2)	Port 3, Pin 2	LED87	No	Pulse signal to X servo
X-Axis DIR	OUT3 (CN1-3)	Port 3, Pin 3	LED88	No	Direction signal to X servo
Y-Axis STEP	OUT4 (CN1-4)	Port 3, Pin 4	LED89	No	Pulse to Y servo (Master)
Y-Axis DIR	OUT5 (CN1-5)	Port 3, Pin 5	LED90	No	Direction to Y servo (Master)
A-Axis STEP	OUT6 (CN1-6)	Port 3, Pin 6	LED91	No	Pulse to A servo (Y Slave)
A-Axis DIR	OUT7 (CN1-7)	Port 3, Pin 7	LED92	No	Direction to A servo (Y Slave)
Z-Axis STEP	OUT8 (CN1-8)	Port 3, Pin 8	LED93	No	Pulse signal to Z servo
Z-Axis DIR	OUT9 (CN1-9)	Port 3, Pin 9	LED94	No	Direction signal to Z servo

Servo Monitoring (ALARM and READY)

All servo monitoring signals use Sourcing logic. They are Active HIGH from the internal transistor perspective (the transistor conducts when the condition is present), and are considered Active HIGH from the UCCNC perspective because the servo output sends +24V to the input when active.

Function	UCBB-2 Terminal	UC300ETH Port/Pin	UCCNC Pin	Active Low	Notes
X-Axis ALARM	IN5+ (CN2-5)	Port 4, Pin 5	LED107	NO	Alarm signal of X servo
X-Axis READY	IN6+ (CN2-6)	Port 4, Pin 6	LED108	NO	Ready signal of X servo
Y-Axis ALARM	IN7+ (CN2-7)	Port 4, Pin 7	LED109	NO	Alarm – Y servo (Master)
Y-Axis READY	IN8+ (CN2-8)	Port 4, Pin 8	LED110	NO	Ready – Y servo (Master)
A-Axis ALARM	IN9+ (CN2-9)	Port 4, Pin 9	LED111	NO	Alarm – A servo (Slave)
A-Axis READY	IN10+ (CN2-10)	Port 4, Pin 10	LED112	NO	Ready – A servo (Slave)
Z-Axis ALARM	IN11+ (CN2-11)	Port 4, Pin 11	LED113	NO	Alarm signal of the Z servo

Note: The Z-axis READY signal has been removed from the system since there are not enough pins on the 26-pin CN1 I/O connector on the servo drive. The pin on the CN1 servo signal port has been reassigned to control the BRAKE of the Z-axis Servo.

Spindle Control via VFD

Function	UCBB-2 Terminal	UC300ETH Port/Pin	UCCNC Pin	Active Low	Notes
Spindle ON CW	OUT16 (CN1-16)	Port 3, Pin 16	LED101	No	Clockwise spindle start
Spindle ON CCW	OUT17 (CN1-17)	Port 3, Pin 17	LED102	No	Counter-clockwise spindle start

Spindle Sensor Inputs and Button

Component	Function	UCBB-2 Terminal	UC300ETH Port/Pin	UCCNC Pin	Active Low	Specification
S1 Sensor	Tool presence detection	IN2+ (CN2-2)	Port 4, Pin 2	LED104	No	24V PNP NO
S2 Sensor	Collet open status	IN3+ (CN2-3)	Port 4, Pin 3	LED105	No	24V PNP NO
S3 Sensor	The shaft stopped status	IN4+ (CN2-4)	Port 4, Pin 4	LED106	No	24V PNP NO
Spindle Button	Manual tool release	IN15+ (CN1-15)	Port 3, Pin 15	LED100	No	24V NO

Spindle Monitoring Sensors

Component	Function	UCBB-2 Terminal	UC300ETH Port/Pin	UCCNC Pin	Active Low	Specification
Thermal Alarm	High temp detection	IN11+ (CN1-11)	Port 3, Pin 11	LED96	No	24V NC switch
Fan Status	Fan monitoring	IN13+ (CN1-13)	Port 3, Pin 13	LED98	No	24V PNP NO
Electronics Status	Electronics monitoring	IN12+ (CN1-12)	Port 3, Pin 12	LED97	No	24V PNP NO

Available (unused) UCBB-2 Pins for Future Expansion

The following pins are currently unassigned and available for future peripheral expansion:

(5 OUTPUT pins and 3 INPUT pins in total)

CN1 – OUT 14
CN2 – OUT 1
CN2 – OUT 14

CN2 – OUT 16
CN2 – OUT 17

CN1 – IN 10+
CN2 – IN 13+
CN2 – IN 15+

I/O Addressing in UCCNC

UCCNC uses different approaches for addressing input and output pins, which is essential to understand when writing or interpreting macros.

Output Pin Addressing

Output pins in UCCNC are specified by both their Port Number on the UC300ETH and their PIN within that port. For example, the X-axis STEP signal on UCBB-2 OUT2 (CN1) corresponds to UC300ETH Port 3, Pin 2. In macro code, two separate definitions are required: one for the port number and one for the PIN.

Input Pin Addressing (LED Mapping)

For input pins, UCCNC does not read the pin address directly. Instead, it reads specifically assigned internal software LEDs. The mapping convention is:

UC300ETH Port	Pin Range	UCCNC LED Range
Port 1	Pins 1–17	LEDs 1–17
Port 2	Pins 1–17	LEDs 69–85
Port 3	Pins 1–17	LEDs 86–102
Port 4	Pins 1–17	LEDs 103–119
Port 5	Pins 1–17	LEDs 120–136

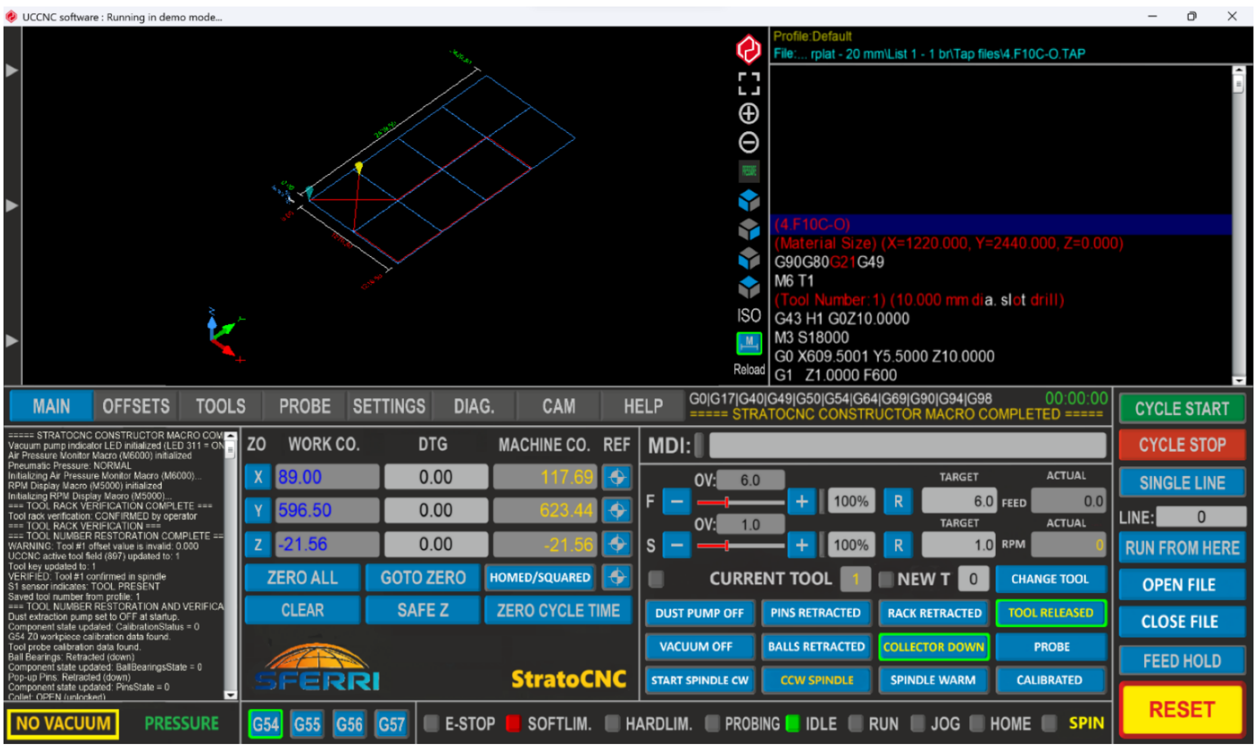
In macro code, input pins are always referenced by their corresponding LED number using the exec.GetLED() function.

5. Main Components

Control Software – UCCNC

UCCNC is a professional machine control software that uses external motion controllers (in the StratoCNC, the UC300ETH) to generate coordinated, real-time motion signals for up to 6 machine axes. It interprets RS274 G-code and M-code, manages tool tables and work offsets, and provides extensive customization through its macro scripting engine and visual screen editor.

StratoCNC uses its own modified screenset with custom buttons and features. An extensive system of custom macros is taking care of component manipulation (tool rack, dust collector, positioning pins, transportation ball bearings, dust pump, vacuum pump), tool management (tool loading, offloading, changing, measurement), system initialization (homing, squaring, spindle warm-up), system health monitoring (servo health, spindle health, sensors health, pneumatic pressure, vacuum pressure), spindle management (start, stop, RPM display), and many more.



Key capabilities relevant to the StratoCNC include:

- Controls up to 6 axes
- Executes RS274 G and M codes
- An unlimited number of user text macros with a flexible script engine
- Up to 400 kHz operation. Note - maximum step frequency depends on the motion controller used.

- Exact stop and constant velocity interpolations with a highly configurable, advanced look-ahead function
- G54..G59 work offsets and G52/G92 temporary offset
- Realtime 3D toolpath viewer
- OpenGL screen optimized for fast screen update rates with low CPU/GPU usage
- Built-in visual screen editor, which allows the screen to be fully customized by the end-user
- 48 configurable hotkeys
- 48 configurable inputs trigger to assign physical input pins to UCCNC function calls
- 48 configurable outputs trigger to assign virtual LEDs to physical output pins
- Parametric programming using internal variables and programming mathematical expressions
- Controls THC control for plasma cutters. (M205 and M206 macros)
- Screen sets for mills/routers and plasma cutters
- Modbus TCP, RTU, and ASCII communication support via the Modbus plugin
- M10/M11 macros for fast synchronous outputs for laser cutting applications
- G33 and G76 synchronous thread cutting with free-running spindle and encoder feedback to the motion controller
- G33.1 and G33.2 rigid tapping with the Z-axis synchronized to the spindle motor
- Supports wired MPGs and the UCR200, UCR201, XHC HB04, and other 3rd party wireless pendants via plugin
- Built-in basic CAM module with DXF file import. (Limited functionality!)
- Stop-less smooth backlash compensation with adjustable backlash acceleration
- Plugin interface with Visual C# example plugin template. The plugin works with C#, VB, and C++ programming languages
- Fast communication with optimized buffering technique producing low button press to event execution time
- Fast laser scan type engraving plugin, web camera plugin, Xbox360 controller plugin, 3D printer plugin, etc.
- Extremely low jitter on the step and direction signals for smooth, high-speed, and precise motor operation
- Runs in demo (simulation mode) without a software license key
- Supported on UCCNC forum and via e-mail by a dedicated support team
- Compatible OP systems: Windows XP, 7, 8, 8.1,10 on both 32 and 64bit versions

Custom MACRO/SCREENSET system – StratoCNC

The StratoCNC router features an integrated proprietary system of a customized screenset, numerous Macros, UCCNC settings, and global keys that work together to provide automated operations, machine health monitoring, safety monitoring, and enhanced functionality. The system is designed with redundant safety features to protect both the machine and the operator.

Configuration parameters and shared helper logic are embedded directly within the individual macros or managed through UCCNC's standard configuration files (.pro) and global keys, promoting self-contained macro functionality. All macros use the M20xxx numbering convention for GUI button macros, standard M-code numbers (M3, M4, M5, M6) for G-code compatible overrides, and Mxxxx numbering for background monitoring loops.

System Architecture

The StratoCNC system consists of several integrated layers:

1. UCCNC Configuration

Standard UCCNC profile files (.pro) store core settings like pin configurations, motor tuning, steps per unit, tool tables, and work offsets. Global keys stored within the profile (accessed via exec.Readkey/exec.Writekey) are used extensively for state management (e.g., SystemHomed, RackExtended, CurrentToolInSpindle, SpindleRunning) and inter-macro communication (e.g., M20301Running, CalledFromMacro, SoftwareColletOperation).

2. Operational Macros

A collection of specialized .txt files located in the profile's Macro_Default folder, each designed for a specific task. These macros contain C# code executed by UCCNC. The macros are organized into the following functional groups:

Startup and Initialization

- **M99998** (Constructor Macro) – Automatically runs on UCCNC startup, initializes all system states and starts background loops

Homing and Setup Macros

- **M20301** (Homing and Squaring) – GUI button macro for complete machine homing and gantry squaring
- **M20303** (G54 Z0 and Tool Probe Calibration) – GUI button macro for workpiece Z0 and tool probe calibration
- **M20305** (Spindle Warm-up) – GUI button macro for controlled spindle warm-up routine

Component Toggle Macros

- **M20710** (Tool Rack Operation) – GUI toggle for tool rack extend/retract
- **M20720** (Dust Collector Operation) – GUI toggle for dust collector lower/lift
- **M20730** (Pop-up Pins Operation) – GUI toggle for pop-up pins extend/retract
- **M20740** (Ball Bearings Operation) – GUI toggle for ball bearings extend/retract
- **M20750** (Dust Pump Operation) – GUI toggle for dust extraction pump on/off
- **M20760** (Vacuum Pump Operation) – GUI toggle for vacuum pump on/off

Spindle Control Macros

- **M3** (Start Spindle Clockwise) – G-code/MDI override macro with tool presence safety check and GUI synchronization
- **M4** (Start Spindle Counter-Clockwise) – G-code/MDI override macro with tool presence safety check and GUI synchronization
- **M5** (Stop Spindle) – G-code/MDI override macro with stop timestamp recording and GUI synchronization
- **M20601** (Spindle CCW Toggle) – GUI button for manual counter-clockwise spindle start/stop with two-step confirmation
- **M20602** (Spindle CW Toggle) – GUI button for manual clockwise spindle start/stop
- **M5000** (Spindle RPM Display) – Running in a Loop - Real-time spindle speed display from VFD analog feedback

Tool Management Macros

- **M20006** (Tool Change) – GUI button for complete automated tool change, including integrated offload and load
- **M6** (Tool Change) – G-code/MDI compatible tool change command
- **M20031** (Tool Length Measurement) – GUI button for automated tool length probing and offset storage
- **M20770** (Manual Tool Manipulation) – GUI button for timed collet release with automatic close and tool change detection
- **M8001** (Manual Tool Release Button) –Running in a Loop - Monitors physical spindle button for manual collet operation
- **M4000** (Collet Override Monitor) – Running in a Loop - Detects solenoid override button use, prompts for tool registration

Monitoring Macros (Background Loops)

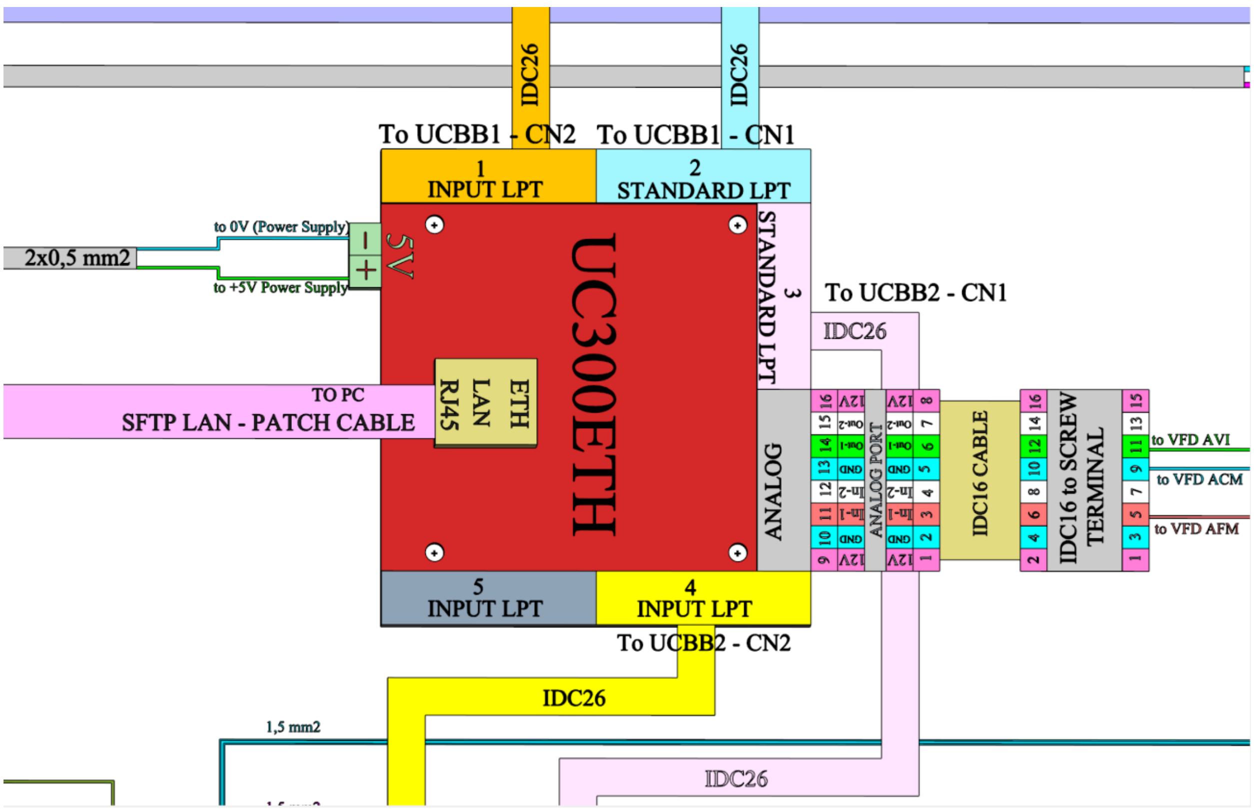
- **M6000** (Air Pressure Monitor) – Continuous pneumatic system pressure monitoring
- **M8000** (General Safety Monitor) – Pre-cycle safety checks and cycle start coordination
- **M10001** (Servo Health Monitoring) – Continuous servo driver ALARM/READY signal monitoring
- **M10002** (Spindle Health Monitoring) – Continuous spindle thermal, fan, and electronics monitoring
- **M10003** (Sensor Health Monitoring) – Continuous limit switch anomaly detection
- **M9000** (Dynamic Dust Collector Control) – Automatic dust collector positioning based on Z-axis height during cutting

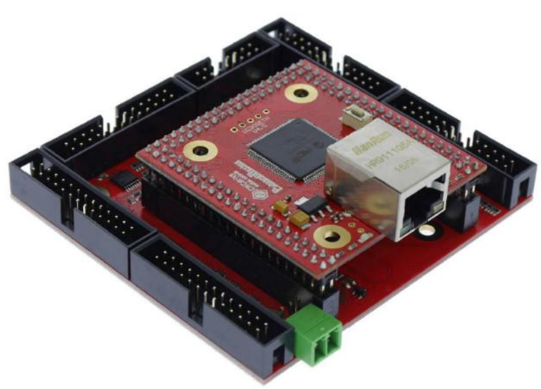
3. StratoCNC custom screenset

The screenset is designed with custom buttons, textfields, and entities which fully encompass the unique features, components, functions, and capabilities of the machine.

Motion Controller – UC300ETH 5LPT

The UC300ETH is the dedicated real-time motion controller that bridges the gap between the PC-based UCCNC software and the precision timing demands of the machine hardware. While UCCNC handles G-code interpretation, trajectory planning, and user interface management, the UC300ETH takes ownership of the time-critical tasks that a standard PC operating system cannot reliably perform.





The controller uses Ethernet interface (offering advantages over USB or parallel port connections in terms of noise resistance and distance from the control computer) and 5 IDC26 ports. It works with UCCNC, Mach3, and Mach4, providing flexibility for different CNC control software. Supports Windows XP, 7, 8, 8.1, and 10 (32-bit and 64-bit). It can control up to 6 axes simultaneously, with a maximum step frequency of 400 kHz, using step and direction interfaces for stepper or servo motor controls. It is suitable for industrial applications with complex sensor and actuator setups, such as multi-axis machines with many limit switches, safety circuits, and tool changers. Requires a 5V external power supply. The controller also has an Analog port with 2 Analog Inputs and 2 Analog Output pins. In StratoCNC this port is used for controlling the Spindle’s RPM and for reading the real RPMs achieved.

I/O Type	Count	Typical Usage
Digital Inputs	49	Limit switches, probe, E-stop, sensor monitoring
Digital Outputs	36	Step/Dir, relays, solenoids, spindle enable
Analog Inputs	2	Spindle RPM feedback from VFD
Analog Outputs	2	Spindle speed command (0–10V) to VFD

Here's a breakdown of its specific roles:

Real-Time Pulse Generation: This is its most critical function.

- UCCNC calculates how many steps and in which direction each axis needs to move. It sends this information in larger chunks (buffered commands) over Ethernet to the UC300ETH.
- The UC300ETH has its own dedicated processor and specialized hardware (often an FPGA - Field-Programmable Gate Array) designed specifically to generate the extremely fast and precisely timed electrical Step and Direction pulses required by the servo drivers.
- A standard PC operating system (like Windows) cannot guarantee the microsecond-level timing accuracy needed for smooth, reliable motor control, especially at high speeds. The UC300ETH takes over this demanding real-time task. It can generate pulses up to 400 kHz reliably, coordinating multiple axes simultaneously.

Communication Interface (PC <-> Machine):

- It receives motion commands, configuration data, and I/O instructions from UCCNC running on your PC via the Ethernet connection. Ethernet is robust, electrically isolated, and allows for longer distances compared to older parallel or USB ports.
- It translates these commands into the electrical signals needed by the machine hardware (Step, Direction, On/Off signals for relays, etc.) and sends them out through its physical ports (the IDC26 connectors going to your UCBB boards).

- It also reads the status of inputs from the machine (limit switches, probe, E-stop, servo alarms via the UCBB boards) and sends this information back to UCCNC over Ethernet.

Hardware Input/Output (I/O) Management:

- The UC300ETH directly manages the physical input and output pins defined in its ports.
- When a limit switch is triggered, the signal goes to the UCBB, then to the UC300ETH. The UC300ETH detects this change immediately (in real-time) and can react (e.g., stop motion based on its firmware or UCCNC's configuration) much faster than if the PC had to process it. It then reports the input status change back to UCCNC.
- Similarly, when UCCNC commands an output (like turning on the spindle relay or extending the tool rack), it sends the command over Ethernet to the UC300ETH, which then physically changes the voltage on the corresponding output pin going to the UCBB.

Buffering:

- The UC300ETH typically has a buffer where it stores a sequence of upcoming motion commands sent by UCCNC. This ensures that motion continues smoothly even if there are minor delays or fluctuations in the Ethernet communication or PC processing speed.

Dedicated Processing:

- By handling the real-time tasks, the UC300ETH frees up your PC's CPU. UCCNC can focus on G-code interpretation, trajectory planning, user interface updates, and look-ahead calculations without being bogged down by the need to generate every single pulse precisely on time.

In short, the UC300ETH is essential because it guarantees the real-time performance, timing accuracy, and reliable I/O handling that a standard PC running UCCNC cannot provide on its own, ensuring smooth, precise, and safe machine operation.

There are 4 LEDs in total on the UC300ETH-5LPT board. Two of these LEDs are inside the RJ45 jack. The green LED is the 'Link' Led which lights when there is an active connection to the computer or switch/router. The orange LED is the 'Activity' LED which lights when packets are being received or transmitted by the device.

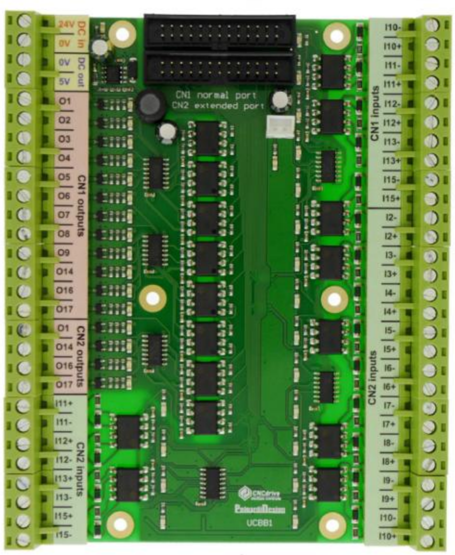
The other 2 LEDs are on 2 sides RJ45 connector. The green LED is the power. The blue LED on the board is the communication one and this LED lights when there is a connection to the software on the host PC and there is an active communication ongoing.

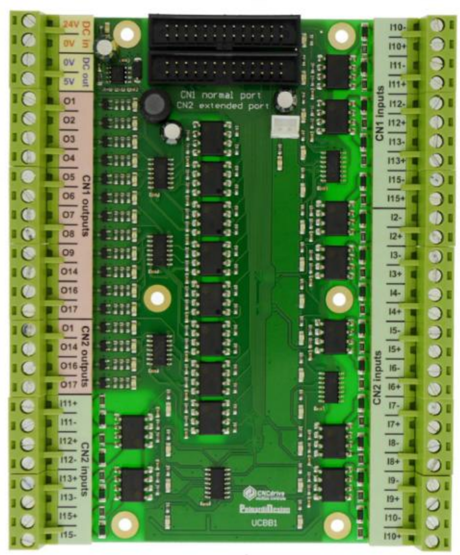
Note – When powering up the machine wait at least 3 min for the UC300ETH to initialize properly and establish its communication network with the PC before starting the UCCNC.

Break-out Boards – UCBB

The StratoCNC uses two UCBB break-out boards to interface between the UC300ETH controller and the machine's electrical components. All I/O connections are optically isolated with fast 10 Mbit/sec optocouplers, providing robust noise immunity in the electrically demanding CNC environment. The boards are powered by 24 VDC and are connected to the UC300ETH controller by standard IDC26 cables. The combination of the two boards gives 36 digital inputs and 32 digital outputs.

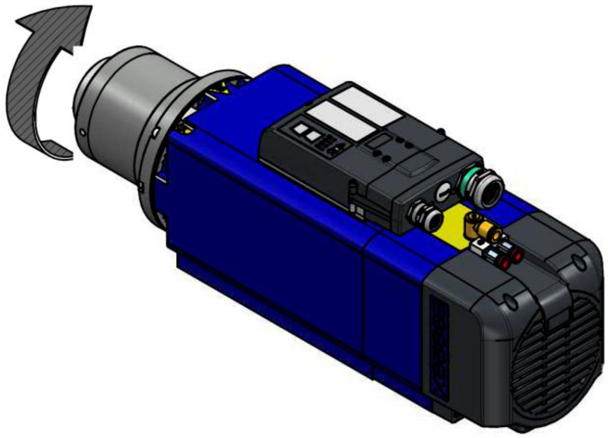
Board	Function	Port Connections	I/O Capacity
UCBB-1	Solenoids, sensors, probes, E-stop, pumps	CN2 → UC300ETH Port 1; CN1 → UC300ETH Port 2	18 inputs, 16 outputs
UCBB-2	Servo control, spindle control, servo monitoring	CN1 → UC300ETH Port 3; CN2 → UC300ETH Port 4	18 inputs, 16 outputs



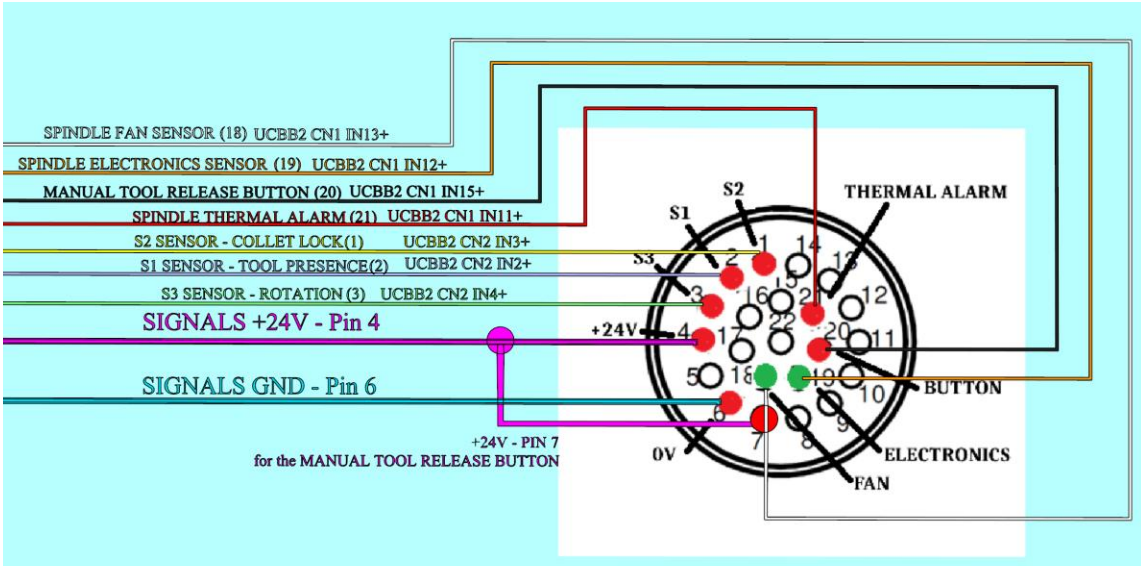


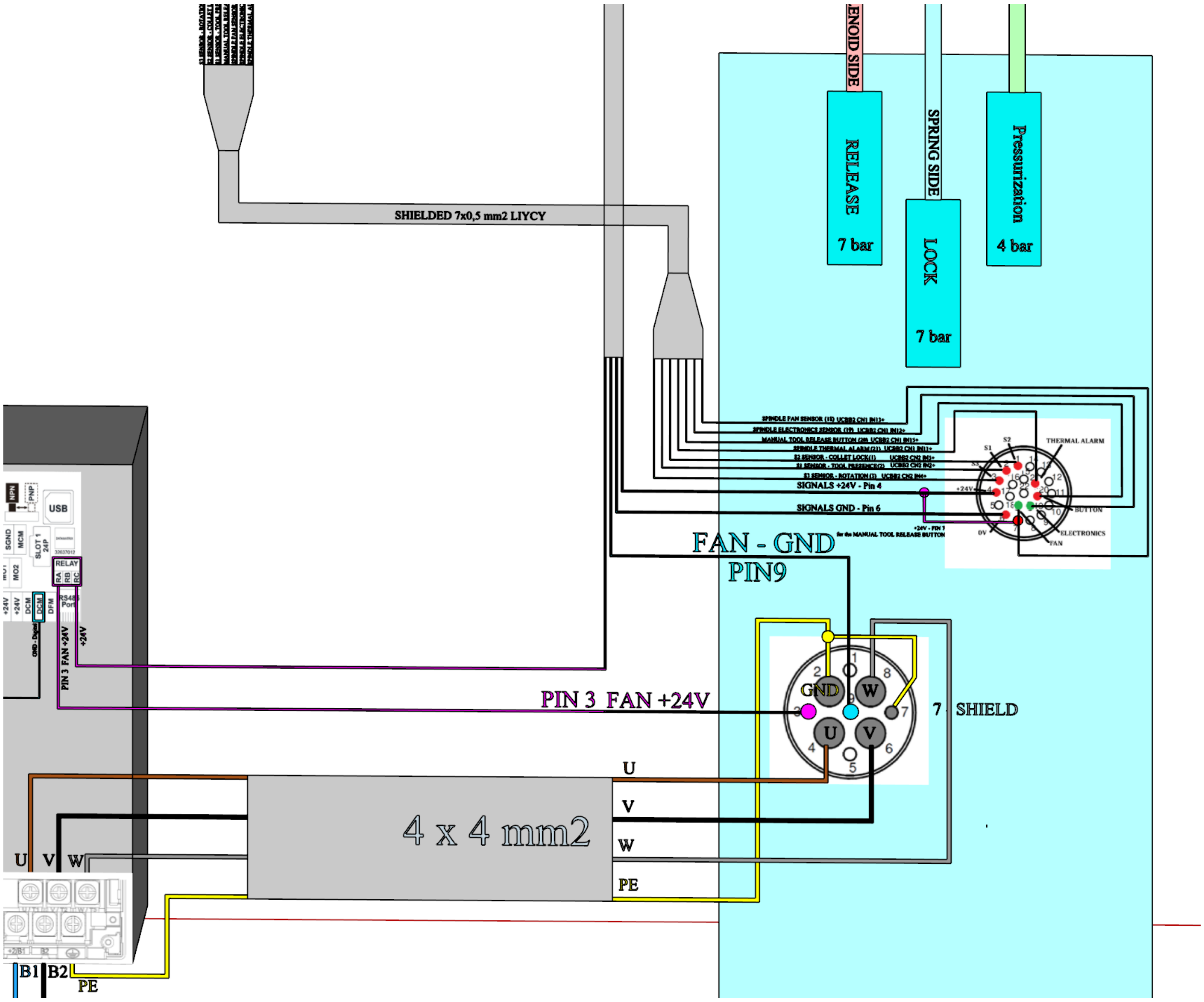
Spindle – HSD Electrospindle ES929X

An industrial-grade high quality powerful spindle with ATC capability and an enhanced electronic suite for control and diagnostics



Parameter	Value
Model	HSD ES929X A 0712 S (H6161H2145)
Power	7.5 kW
Maximum Speed	24,000 RPM
Tool Interface	ISO30
Nose Configuration	Long
Voltage	380V, 4-pole
Cooling	Air-cooled
Protocol	Digital
Encoder	None





- In StratoCNC the spindle is controlled via the UCCNC and the UC300ETH. There is a subsystem of custom macros that fully manages and controls all the aspects of the spindle operations – CW start, CCW start, STOP, warm-up, tool-locking collet manipulation, run time, idle time, RPM settings, RPM display, and tool load/offload/change/measure. The spindle can be started/stopped/controlled by several dedicated buttons on the UCCNC main custom screen, by the G-code commands in the files, or by MDI commands.
  When the operator starts a cutting file, the system automatically starts/stops the spindle, automatically sets the RPMs, automatically changes and measures tools. Cutting a file with a whole plywood sheet of different elements, multiple toolpaths/operations, and many tool changes is a ONE CLICK-task – just load the file and press CYCLE START! The system will take care of all the necessary operations.
- The spindle has a double-acting cylinder for locking/unlocking the tool. The cylinder is controlled with a 24V SINGLE 24V 5/2 solenoid valve with its spring-loaded position corresponding to the Tool Locked state (when the solenoid is activated with 24V power, the tool will be unlocked).
- The spindle is equipped with sensors for Tool presence detection (**S1**), Collet open status (**S2**), and Shaft stopped status (**S3**), which use PNP NO logic. All three sensors are wired

in UC300ETH via the UCBB-2 and are extensively and repeatedly used throughout the whole StratoCNC macro system.

- There is a **Button for manual release** of the tool. The button will only release the tool if the spindle is not rotating. This is achieved with the use of UCCNC. The button is wired in such a way as to be connected to an input pin on UC300ETH via the UCBB, and it is being controlled from UCCNC. A special macro M8001 in a loop is looking for the button signal. When there is a signal, the macro checks if the spindle is rotating (by using the S3 sensor), and if the spindle is idle, the solenoid for tool unlock is activated. The macro also recognizes if a tool is being offloaded/loaded/changed, and takes care of updating the system logs accordingly.
- There are three additional monitoring functions available through their dedicated pins on the SIGNALS port of the spindle – **Thermal Alarm** (NC bimetallic switch), **Fan Status** (PNP NO logic), and **Electronics Status** (PNP NO logic). All the signals are monitored by UCCNC through a dedicated spindle health monitoring M10002 macro.
- The technical documentation of the spindle prescribes a warming-up routine before the daily operations or when the spindle is cold. Special macro M20305 with a button assigned to it is responsible for warming up the spindle accordingly before operation.

WARM-UP

When starting-up the electrospindle for the first time each day, allow it to run a short warm-up cycle:

- *50% maximum rated speed for 2 minutes*
- *75% maximum rated speed for 2 minutes*
- *100% maximum rated speed for 1 minutes*

- Another dedicated macro M5000 is constantly reading the analog voltage provided by the VFD and is calculating and displaying the real RPMs. It is also lighting up the screen SPIN LED when the spindle is rotating.
- If the operator opens the spindle collet by pressing the red OVERRIDE button on the pneumatic solenoid that controls it, the system is managing the situation by a specialized macro and is updating the system information for the tool inserted/offloaded accordingly.

- Required pneumatic pressure

DESCRIPTION	**NOTE**
SPINDLE PRESSURIZATION - CONE CLEANING	Ø 8mm - 4÷5 bar
TOOLHOLDER LOCKING AIR INLET	Ø 8mm - 6÷7 bar
TOOLHOLDER RELEASE AIR INLET	Ø 8mm - 6÷7 bar

Variable Frequency Drive – DELTA MS300

The **Variable Frequency Drive (VFD)** is an electronic controller primarily used to regulate the **speed, torque, and direction** of the spindle motor. It serves as the bridge between the CNC controller and the AC motor, allowing the machine to perform tasks ranging from high-speed engraving to high-torque heavy milling.

Primary Roles of a VFD

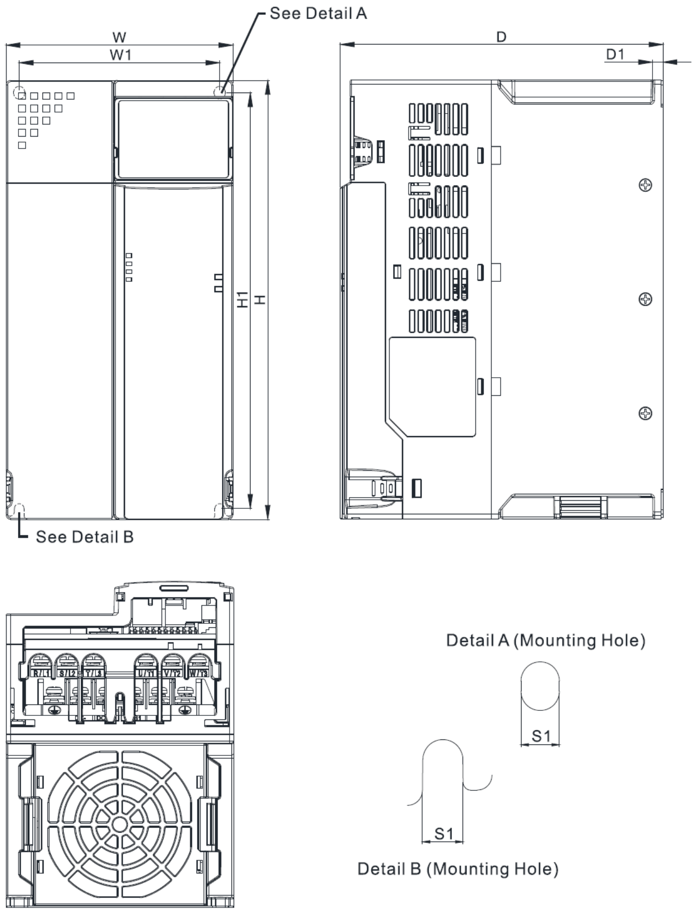
- **Precise Speed Control:** The VFD converts fixed-frequency AC power into a variable-frequency output. By adjusting this frequency (in our case, from 0 to 800 Hz), it allows the spindle to operate across a wide range of RPMs, such as 0 to 24,000 RPM.

- **Torque Management:** It ensures the motor maintains a consistent cutting force. Using advanced "vector control," a VFD can independently regulate torque even at low speeds, which is essential for machining harder materials like aluminum or steel without stalling.

- **Soft Start and Stop:** Unlike traditional starters that "blast" a motor with full voltage, VFDs use a gradual acceleration ramp. This "soft start" reduces mechanical stress on bearings and gears and prevents sudden current surges that could damage the machine or tool.

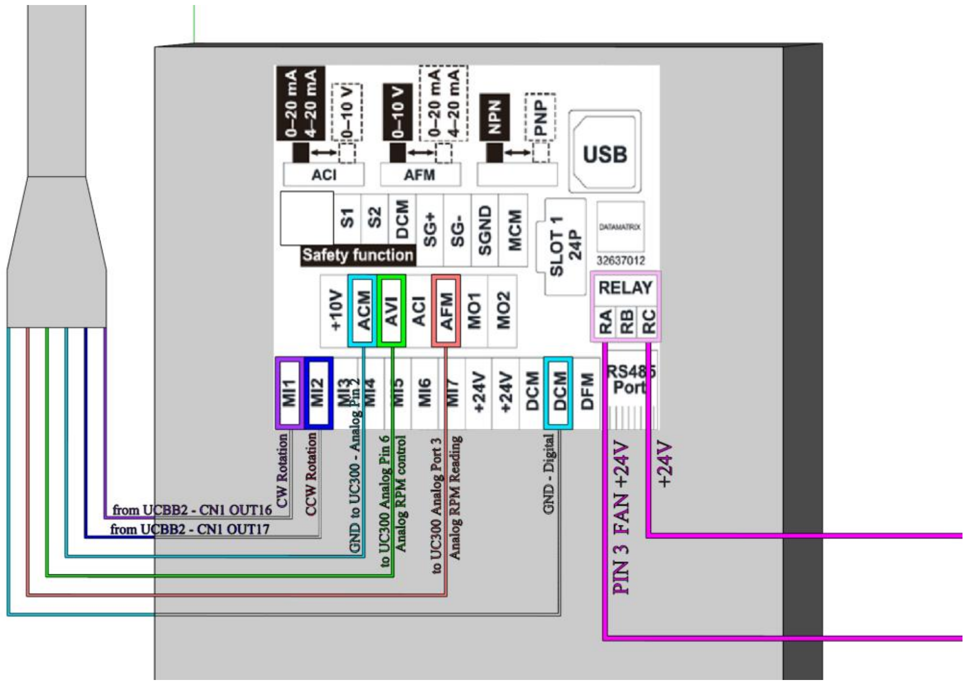
The VFD is controlled by UCCNC through the UC300ETH analog output for speed control (0–10V) and digital outputs for direction/enable. A **Regenerative resistor** is installed to safely dissipate kinetic energy during spindle deceleration, preventing overvoltage faults on the VFD DC bus.

Parameter	Value
Model	DELTA MS300 (VFD25AMS43ANSHA)
Power Rating	11 kW
Voltage	460V, 3-Phase
Current	25A
Protection	IP20, No EMC filter
Frame Size	E1

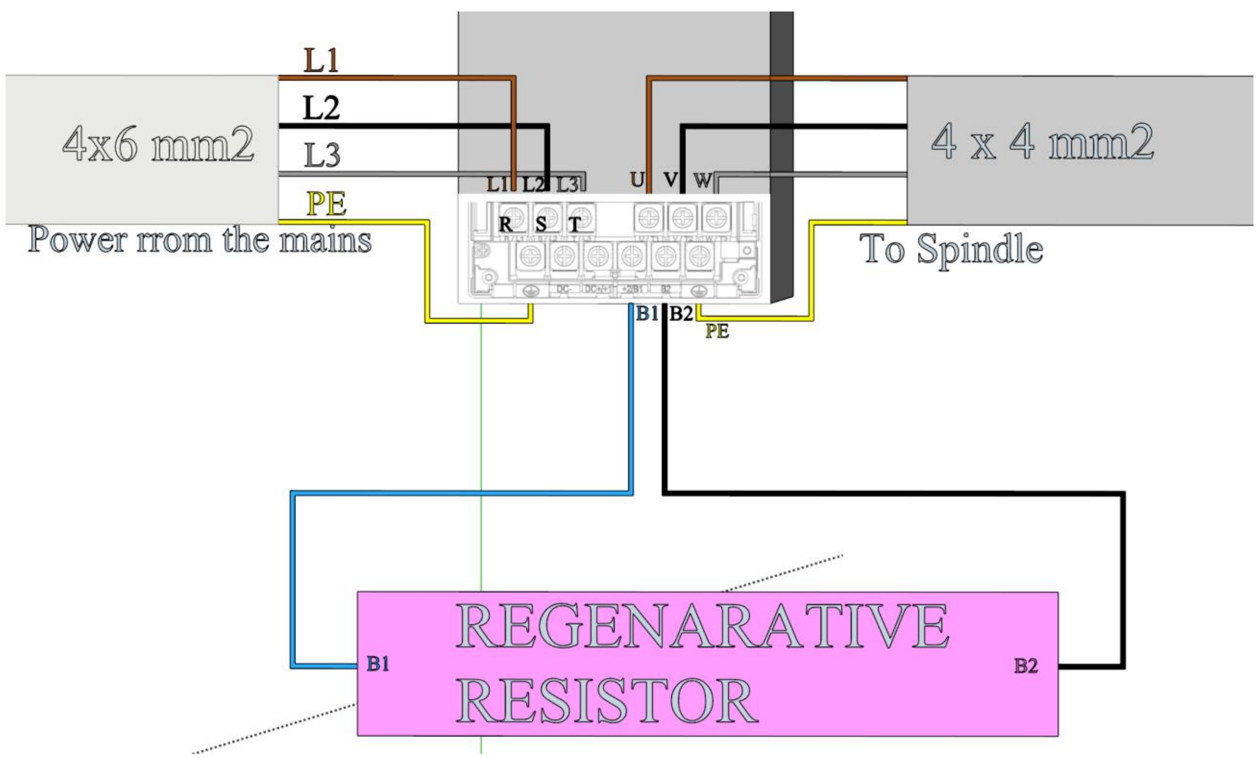


Unit: mm [inch]

Frame	W	H	D	W1	H1	D1	S1
E1	130.0 [5.12]	250.0 [9.84]	185.0 [7.83]	115.0 [4.53]	236.8 [9.32]	6.0 [0.24]	5.5 [0.22]
E2	130.0 [5.12]	250.0 [9.84]	219.0 [8.62]	115.0 [4.53]	236.8 [9.32]	6.0 [0.24]	5.5 [0.22]



0–20 mA
4–20 mA
0–10 V
ACI
0–10 V
0–20 mA
4–20 mA
AFM
NPN
PNP
USB
S1
S2
DCM
SG+
SG-
SGND
MCM
SLOT 1
24P
Safety function
32637012
RELAY
RA
RB
RC
+10V
ACM
AVI
ACI
AFM
MO1
MO2
MI1
MI2
MI3
MI4
MI5
MI6
MI7
+24V
+24V
DCM
DCM
DFM
RS485
Port
CW Rotation
CCW Rotation
from UCBB2 - CN1 OUT16
from UCBB2 - CN1 OUT17
GND to UC300 - Analog Pin 2
to UC300 Analog Pin 6
Analog RPM control
to UC300 Analog Port 3
Analog RPM Reading
GND - Digital
PIN 3 FAN +24V
+24V

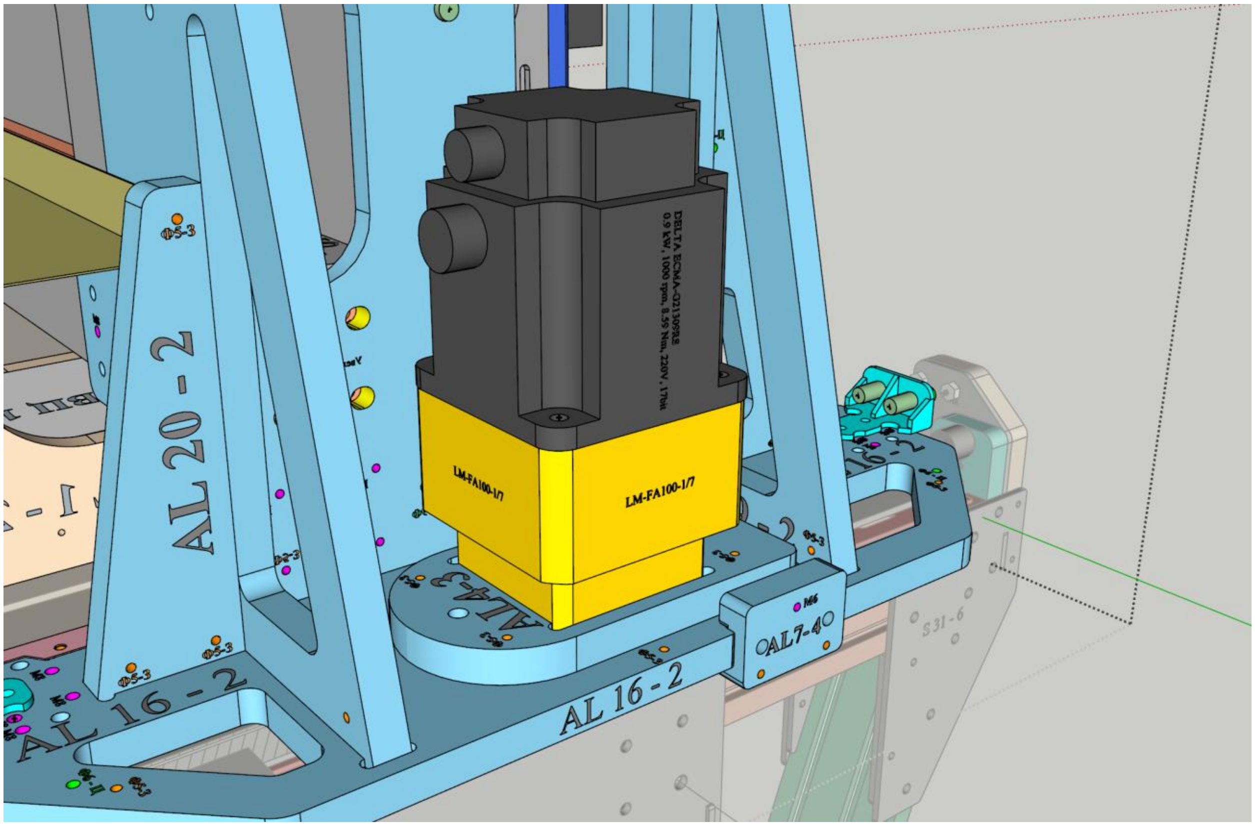


L1
L2
L3
PE
4x6 mm2
Power rrom the mains
L1
L2
L3
R
S
T
U
V
W
4 x 4 mm2
To Spindle
B1
B2
PE
REGENARATIVE
RESISTOR
B1
B2

Servo Motors and Drivers

X and Y Axis Servo Motors – DELTA ECMA-G21309RS

Three identical high-inertia AC servo motors are used for the X and Y axes. Two motors drive the gantry along the Y axis in a Master/Slave configuration (Y-Master on the left, A-Slave on the right), and one drives the spindle assembly along the X axis.



Parameter	Value
Power	0.9 kW
Torque	8.59 Nm
Speed	1000 RPM (nominal) / 2000 RPM (max)
Current	7.5A (nominal) / 22.5A (max)
Encoder	17-bit incremental
Voltage	220V
Features	Oil seal, keyway axis, 130 mm flange, no brake

Model: ECMA Series	E△13				E△18			G△13		
	05	10	15	20	20	30	35	03	06	09
Rated power (kW)	0.5	1.0	1.5	2.0	2.0	3.0	3.5	0.3	0.6	0.9
Rated torque (N-m)[*1]	2.39	4.77	7.16	9.55	9.55	14.32	16.71	2.86	5.73	8.59
Maximum torque (N-m)	7.16	14.3	21.48	28.65	28.65	42.97	50.13	8.59	17.19	21.48
Rated speed (r/min)	2000							1000		
Maximum speed (r/min)	3000							2000		
Rated current (A)	2.9	5.6	8.3	11.01	11.22	16.1	19.2	2.5	4.8	7.5
Maximum current (A)	8.7	16.8	24.9	33.03	33.66	48.3	57.6	7.5	14.4	22.5
Power rating (kW/s)	7.0	27.1	45.9	62.5	26.3	37.3	50.8	10.0	39.0	66.0
Rotor inertia (x10-4kg-m^2)(without brake)	8.17	8.41	11.18	14.59	34.68	54.95	54.95	8.17	8.41	11.18
Mechanical constant (ms)	1.91	1.51	1.10	0.96	1.62	1.06	1.08	1.84	1.40	1.06
Torque constant-KT (N-m/A)	0.83	0.85	0.87	0.87	0.85	0.89	0.87	1.15	1.19	1.15
Voltage constant-KE(mV/(r/min)	30.9	31.9	31.8	31.8	31.4	32.0	32	42.5	43.8	41.6
Armature resistance (Ohm)	0.57	0.47	0.26	0.174	0.119	0.052	0.052	1.06	0.82	0.43
Armature inductance (mH)	7.39	5.99	4.01	2.76	2.84	1.38	1.38	14.29	11.12	6.97
Electric constant (ms)	12.96	12.88	15.31	15.86	23.87	26.39	26.39	13.55	13.50	16.06
Insulation class	Class A (UL), Class B (CE)									
Insulation resistance	>100 MΩ, 500 V_{DC}									
Insulation strength	1.8k V_{AC}, 1 sec									
Weight (kg) (without brake)	6.8	7.0	7.5	7.8	13.5	18.5	18.5	6.8	7.0	7.5
Weight (kg) (with brake)	8.2	8.4	8.9	9.2	17.5	22.5	22.5	8.2	8.4	8.9
Max. radial shaft load (N)	490	490	490	490	1176	1470	490	490	490	490
Max. thrust shaft load (N)	98	98	98	98	490	490	98	98	98	98
Power rating (kW/s) (with brake)	6.4	24.9	43.1	57.4	24.1	35.9	48.9	9.2	35.9	62.1
Rotor inertia (x10-4kg-m^2) (with brake)	8.94	9.14	11.90	15.88	37.86	57.06	57.06	8.94	9.14	11.9
Mechanical constant (ms) (with brake)	2.07	1.64	1.19	1.05	1.77	1.10	1.12	2.0	1.51	1.13
Brake holding torque [Nt-m (min)][*2]	10.0	10.0	10.0	10.0	25.0	25.0	25.0	10.0	10.0	10.0
Brake power consumption (at 20 °C) [W]	19.0	19.0	19.0	19.0	20.4	20.4	20.4	19.0	19.0	19.0
Brake release time [ms (Max)]	10	10	10	10	10	10	10	10	10	10
Brake pull-in time [ms (Max)]	70	70	70	70	70	70	70	70	70	70
Vibration grade (μm)	15									
Operating temperature (°C)	0 °C to 40 °C (32 °F to 104 °F)									
Storage temperature (°C)	-10 °C to 80 °C (-14 °F to 176 °F)									
Operating humidity	20 to 90% RH (non-condensing)									
Storage humidity	20 to 90% RH (non-condensing)									
Vibration capacity	2.5 G									
IP Rating	IP65 (when waterproof connectors are used, or when an oil seal is used to be fitted to the rotating shaft (an oil seal model is used))									
Certifications	CE cULus									

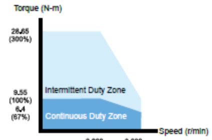
Footnote:

* 1 Rate torque values are continuous permissible values at 0~40 °C ambient temperature when attaching with the sizes of heatsinks listed below:

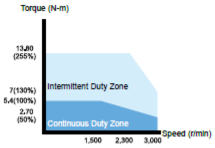
ECMA-__04 / 06 / 08 : 250 mm × 250 mm × 6 mm
ECMA-__10 : 300 mm × 300 mm × 12 mm
ECMA-__13 : 400 mm × 400 mm × 20 mm
ECMA-__18 : 550 mm × 550 mm × 30 mm
ECMA-__22 : 650 mm × 650 mm × 30 mm
Material type : Aluminum F40, F60, F80, F100, F130, F180, F220

*2 The holding brake is used to hold the motor shaft, not for braking the rotation. Never use it for decelerating or stopping the machine.

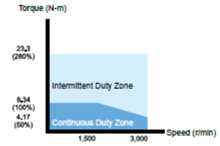
Speed-Torque Curves (T-N Curves)



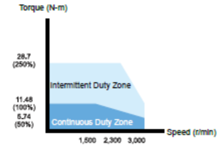
ECMA-E △ 1320 □ S



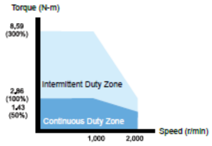
ECMA-F △ 1308 □ S



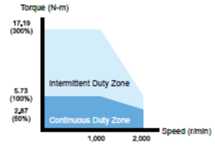
ECMA-F △ 1313 □ S



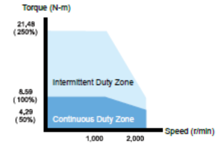
ECMA-F △ 1318 □ S



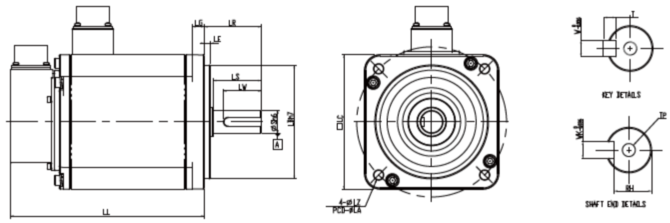
ECMA-G △ 1303 □ S



ECMA-G △ 1306 □ S



ECMA-G △ 1309 □ S



Units: mm

Model	E△1320□S	F△1308□S	F△1313□S	F△1318□S	G△1303□S	G△1306□S	G△1309□S
LC	130	130	130	130	130	130	130
LZ	9	9	9	9	9	9	9
LA	145	145	145	145	145	145	145
S	$22\,(^{+0}_{-0.013})$	$22\,(^{+0}_{-0.013})$	$22\,(^{+0}_{-0.013})$	$22\,(^{+0}_{-0.013})$	$22\,(^{+0}_{-0.013})$	$22\,(^{+0}_{-0.013})$	$22\,(^{+0}_{-0.013})$
LB	$110\,(^{+0}_{-0.035})$	$110\,(^{+0}_{-0.035})$	$110\,(^{+0}_{-0.035})$	$110\,(^{+0}_{-0.035})$	$110\,(^{+0}_{-0.035})$	$110\,(^{+0}_{-0.035})$	$110\,(^{+0}_{-0.035})$
LL (without brake)	187.5	152.5	187.5	202	147.5	147.5	163.5
LL (with brake)	216	181	216	230.7	183.5	183.5	198
LS	47	47	47	47	47	47	47
LR	55	55	55	55	55	55	55
LE	6	6	6	6	6	6	6
LG	11.5	11.5	11.5	11.5	11.5	11.5	11.5
LW	36	36	36	36	36	36	36
RH	18	18	18	18	18	18	18
WK	8	8	8	8	8	8	8
W	8	8	8	8	8	8	8
T	7	7	7	7	7	7	7
TP	M6 Depth 20	M6 Depth 20	M6 Depth 20	M6 Depth 20	M6 Depth 20	M6 Depth 20	M6 Depth 20

NOTE

1) Dimensions are in millimeters. Actual measured values are in metric units.
2) Dimensions of the servo motor may be updated without prior notice.
3) The boxes (□) in the model names represent shaft end/brake or the number of oil seal.
4) The boxes (△) in the model names represent encoder types (△=1: Incremental encoder, 20-bit; △=2: Incremental encoder, 17-bit).

X and Y Axis Servo Drivers – DELTA ASD-B3-1021-E

AC Servo drive, 1.0 kW, 220V, B3 series. Features 6 pulse inputs, analog voltage control, PR (position register) mode, RS-485 communication, and STO (Safe Torque Off) safety function.

Servo Drive Model Information

ASD-B3 Series Servo Drive

ASD - B3 - 04 21 - L

Product Name	AC Servo Drive
Product Series	B3 series B3A series B3B series
Rated Power Output	01: 100W 02: 200W 04: 400W 07: 750W 10: 1kW 15: 1.5kW 20: 2kW 30: 3kW
Input Voltage and Phase	21: 220V single-/three-phase 23: 220V three-phase
Model Code	

ASD-B3

Code	PT Mode Pulse Input	PR Mode	RS-485	Analog Voltage Control	CANopen	DMCNET	EtherCAT	STO
L	√	√	√	√	-	-	-	-
M	-	√	-	√	√	-	-	-
F	-	√	-	√	-	√	-	-
E	-	√	-	√	-	-	√	-

ASD-B3A[*1]

Code	PT Mode Pulse Input	PR Mode	RS-485	Analog Voltage Control	CANopen	DMCNET	EtherCAT	STO[*2]
L	√	√	√	√	-	-	-	√
M	√	√	√	√	√	-	-	√
F	√	√	-	√	-	√	-	√
E	√	√	-	√	-	-	√	√

Note: The model information is for reference only. Not all kinds of permutations are available. Please contact the distributor near your region or Delta for the details.

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Send Quote Requests to info@automatedpt.com
Call +1(800)985-6929 To Order or Order Online At Deltaacdrives.com

Servo Drive Specifications

ASD-B3			100 W 01	200 W 02	400 W 04	750 W 07	1 kW 10	1.5 kW 15	2 kW 20	3 kW 30
Power Supply	Phase / Voltage		Single-phase / Three-phase 220V_{AC}						Three-phase 220V_{AC}	
Power Supply	Permissible Voltage		Single-phase / Three-phase 200 - 230V_{AC}, -15% to 10%						Three-phase 200 - 230V_{AC}, -15% to 10%	
Power Supply	Input Current (3PH) (Unit: Arms)		0.88	1.29	2.04	3.52	5.72	6.33	7.6	10.3
Power Supply	Input Current (1PH) (Unit: Arms)		1.47	2.35	3.74	6.47	10.4	11.7		
Power Supply	Continuous Output Current (Unit: Arms)		0.9	1.55	2.65	5.1	7.3	8.3	13.4	19.4
Power Supply	Max. Instantaneous Output Current (Unit: Arms)		3.88	7.07	10.6	16.4	21.21	27	38.3	58.9
Cooling Method			Natural cooling				Fan cooling			
Drive Resolution			24-bit (16,777,216 pls/rev)							
Main Circuit Control			SVPWM control							
Tuning Mode			Auto / Manual							
Regenerative Resistor			N/A		Built-in					
Position Control Mode	Pulse Type (only for pulse control mode)		Pulse + Direction; A phase + B phase; CCW pulse +CW pulse							
Position Control Mode	Max. Output Pulse Frequency (only for pulse control mode)		Pulse + direction: 4 Mpps; CCW pulse + CW pulse: 4 Mpps; A phase + B phase: single-phase 2 Mpps; Open collector: 200 Kpps							
Position Control Mode	Command Source		External pulse (only for pulse control mode) / Internal register (PR mode)							
Position Control Mode	Smoothing Method		Low-pass, S-curve, and moving filters							
Position Control Mode	E-Gear Ratio		E-Gear ratio: N / M times, limited to (1 / 4 < N / M < 262144) N: 1 - 536870911 / M: 1 - 2147483647							
Position Control Mode	Torque Limit		Parameter settings							
Position Control Mode	Feed Forward Compensation		Parameter settings							
Speed Control Mode	Analog Command Input	Voltage Range	0 to ±10 V_{DC}							
Speed Control Mode	Analog Command Input	Resolution	12-bit							
Speed Control Mode	Analog Command Input	Input Impedance	1 MΩ							
Speed Control Mode	Analog Command Input	Time Constant	25μs							
Speed Control Mode	Speed Control Range [*1]		1 : 6000							
Speed Control Mode	Command Source		External analog command / Internal register							
Speed Control Mode	Smoothing Method		Low-pass and S-curve filters							
Speed Control Mode	Torque Limit		Parameter settings or analog input							
Speed Control Mode	Bandwidth		Maximum 3.1 kHz							
Speed Control Mode	Speed Calibration Ratio [*2]		±0.01% at 0% to 100% load fluctuation							
Speed Control Mode	Speed Calibration Ratio [*2]		±0.01% at ±10% power fluctuation							
Speed Control Mode	Speed Calibration Ratio [*2]		±0.01% at 0°C to 50°C ambient temperature fluctuation							
Torque Control Mode	Analog Command Input	Voltage Range	0 to ±10 V_{DC}							
Torque Control Mode	Analog Command Input	Input Impedance	1 MΩ							
Torque Control Mode	Analog Command Input	Time Constant	25μs							
Torque Control Mode	Command Source		External analog command / Internal register							
Torque Control Mode	Smoothing Method		Low-pass filter							
Torque Control Mode	Speed Limit		Parameter settings or analog input							
Analog Monitor Output			Monitoring signal can be set with parameters (voltage output range: ±8V); resolution: 10-bit							
Digital Input / Output	Input		Servo on, Fault reset, Gain switch, Pulse clear, Zero speed clamping, Command input reverse control, Internal position command trigger, Torque limit, Speed limit, Internal position command selection, Motor stop, Speed command selection, Speed / Position mode switching, Speed / Torque command switching, Torque / Position mode switching, PT / PR command switching, Emergency stop, Forward / reverse limit, Original point, Forward / reverse operation torque limit, Homing activated, Forward / reverse JOG input, Event trigger, E-Gear N selection, Pulse input prohibition *The DI mentioned above are only used in pulse control mode. When controlling through communication, it is suggested that you use communication for DI input. DI only supports emergency stop, forward / reverse limit, and homing.							
Digital Input / Output	Output		A, B, Z line driver output							
Digital Input / Output	Output		Servo ready, Servo on, Zero speed detection, Target speed reached, Target position reached, Torque limiting, Servo alarm, Magnetic brake control, Homing complete, Early warning for overload, Servo warning, Position command overflows, Software limit (reverse direction), Software limit (forward direction), Internal position command complete, Servo procedure complete, Capture procedure complete							
Protection Function			Overcurrent, Overvoltage, Undervoltage, Overheat, Regeneration error, Overload, Excessive speed deviation, Excessive position deviation, Encoder error, Adjustment error, Emergency stop, Forward / reverse limit error, Serial communication error, RST leak phase, Serial communication timeout, Short-circuit protection for terminals U, V, W							
Communication Interface			USB / RS-485 / CANopen / DMCNET / EtherCAT							
Environment	Installation Site		Indoors (avoid direct sunlight), no corrosive vapor (avoid fumes, flammable gases, and dust)							
Environment	Altitude		Altitude 2000 m or lower above sea level							
Environment	Atmospheric Pressure		86 kPa - 106 kPa							
Environment	Operating Temperature		0°C to 55°C (If operating temperature is above 45°C, forced cooling is required)							
Environment	Storage Temperature		-20°C to 65°C							
Environment	Humidity		0 to 90% RH (non-condensing)							
Environment	Vibration		9.80665 m/s^2 (1 G) less than 20 Hz, 5.88 m/s^2 (0.6 G) 20 to 50 Hz							
Environment	IP Rating		IP20							
Environment	Power System		TN system [*3*4]							
Environment	Certifications		IEC / EN / UL 61800-5-1							

Notes:
*1. Within the rated load, the speed ratio is: the minimum speed (smooth operation) / rated speed.
*2. Within the rated speed, the speed calibration ratio is: (rotational speed with no load - rotational speed with full load) / rated speed.
*3. TN system: the neutral point of the power system connects directly to the ground. The exposed metal components connect to the ground through the protective ground conductor.
*4. Use a single-phase three-wire power system for the single-phase power model.

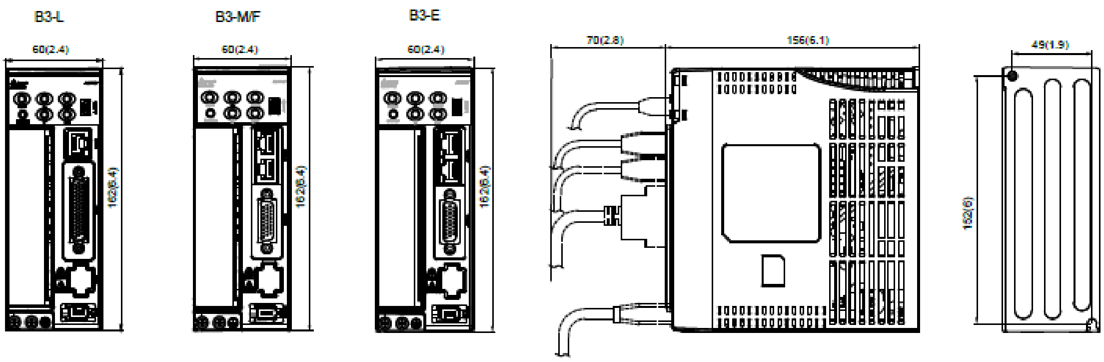
Servo Drive Specifications

Dimensions

100 W / 200 W / 400 W

Weight
0.9 kg

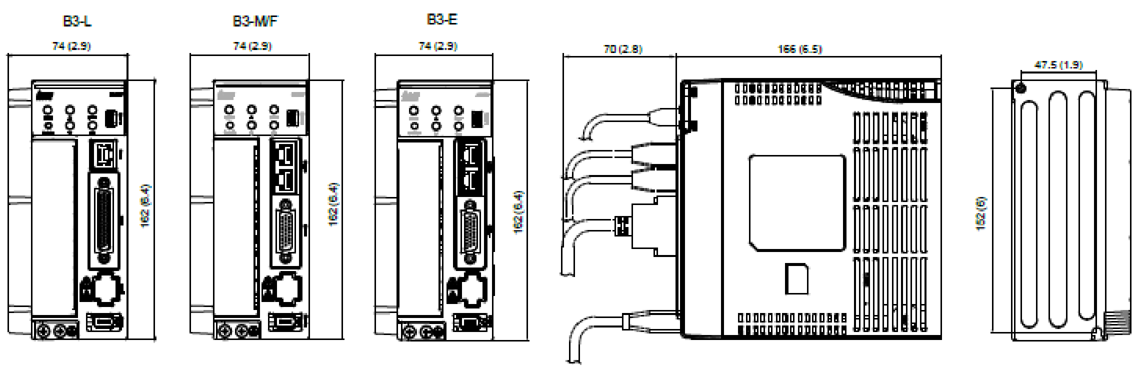
Unit: mm (inch)



750 W

Weight
1.2 kg

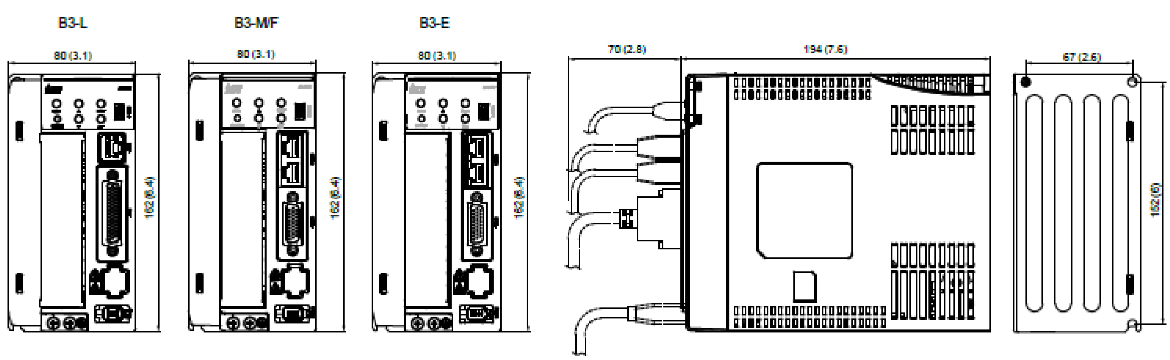
Unit: mm (inch)



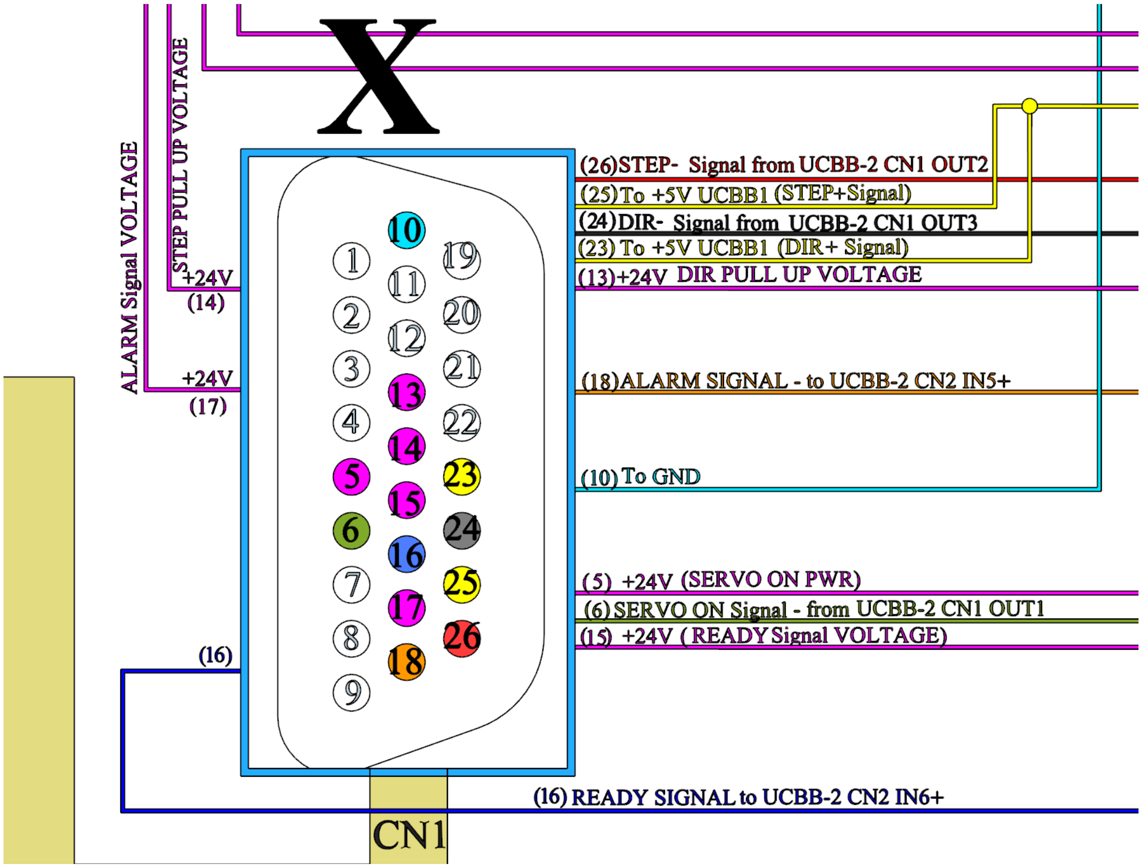
1 kW / 1.5 kW

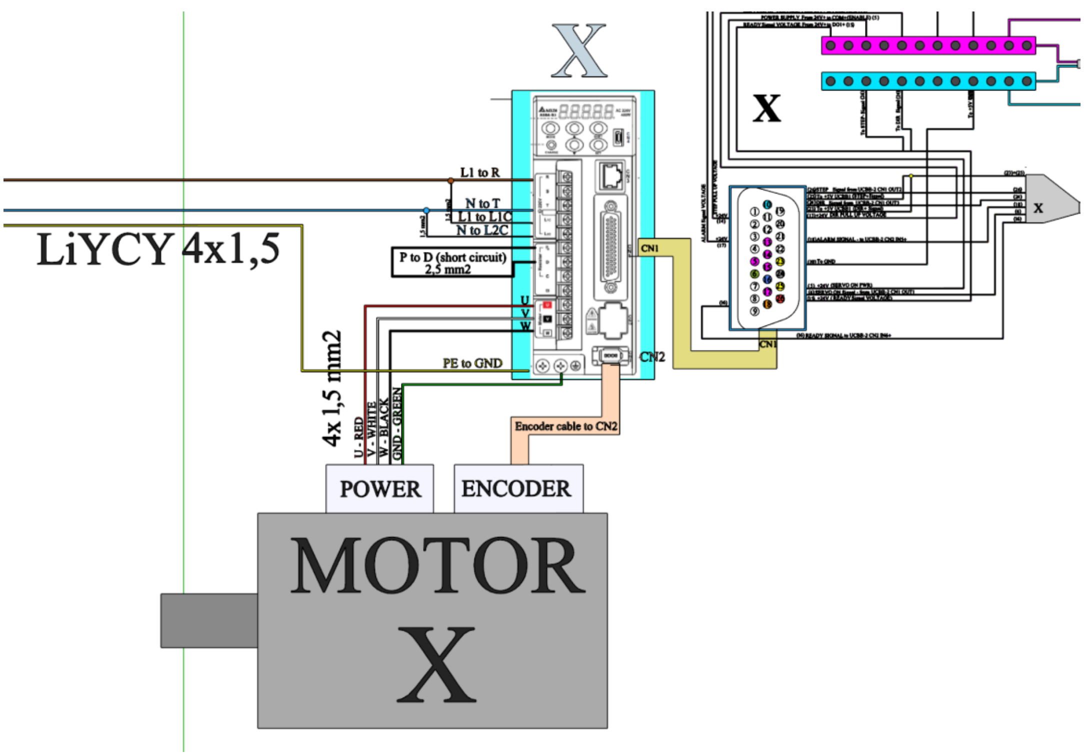
Weight
1.8 kg

Unit: mm (inch)

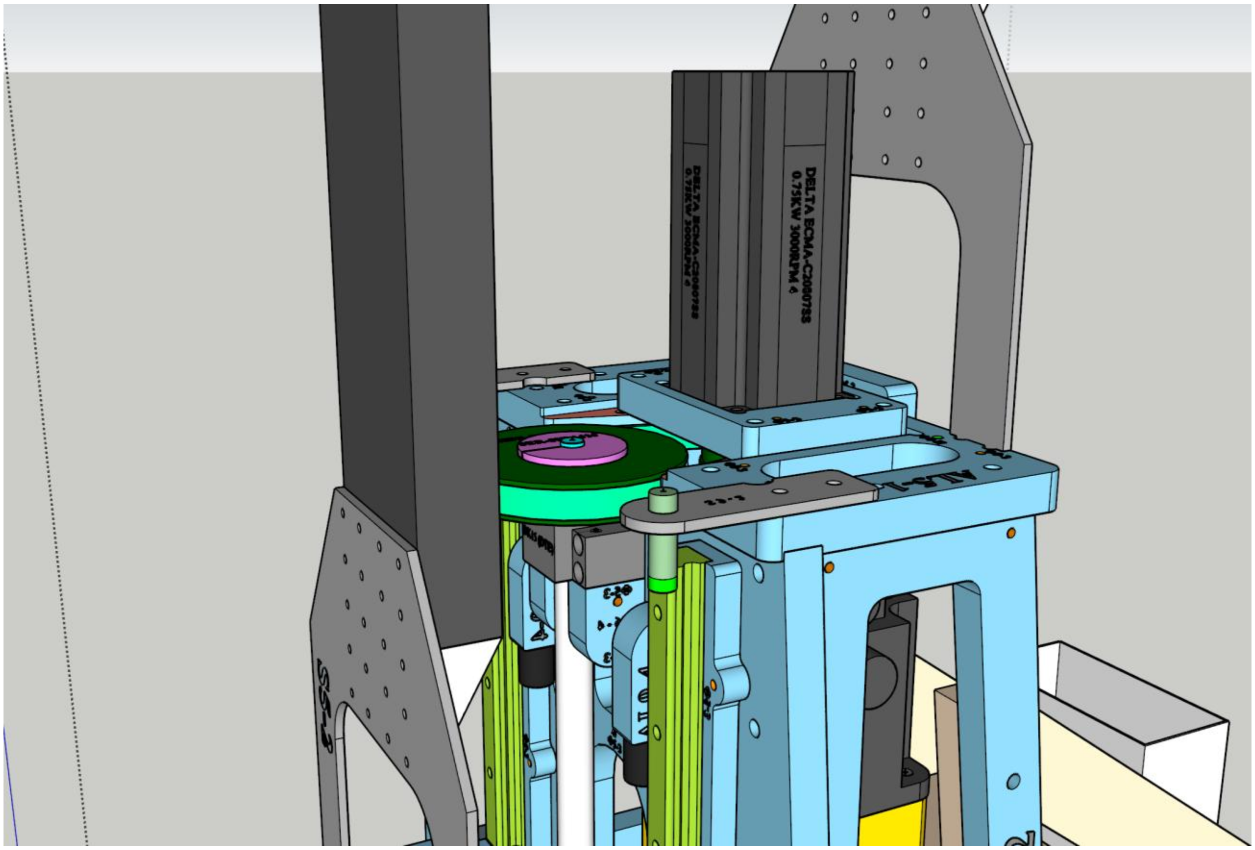


The CN-1 control connector of the driver is a 26-PIN type:





Z Axis Servo Motor – DELTA ECM-B3M-C20807RS1

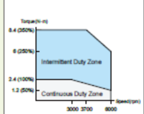
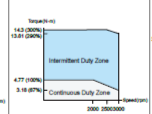
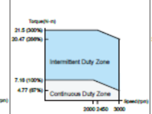
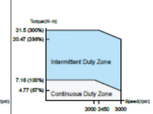


Parameter	Value
Power	0.75 kW
Torque	2.4 Nm (nominal) / 8.4 Nm (max)
Speed	3000 RPM (nominal) / 6000 RPM (max)
Current	4.27A (nominal) / 15.8A (max)
Encoder	24-bit incremental magnetic-optical (usable as single-turn absolute)
Features	Oil seal, 80 mm flange, 19 mm axis, **WITH BRAKE**

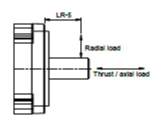
ECM-B3 Series Servo Motor Specifications

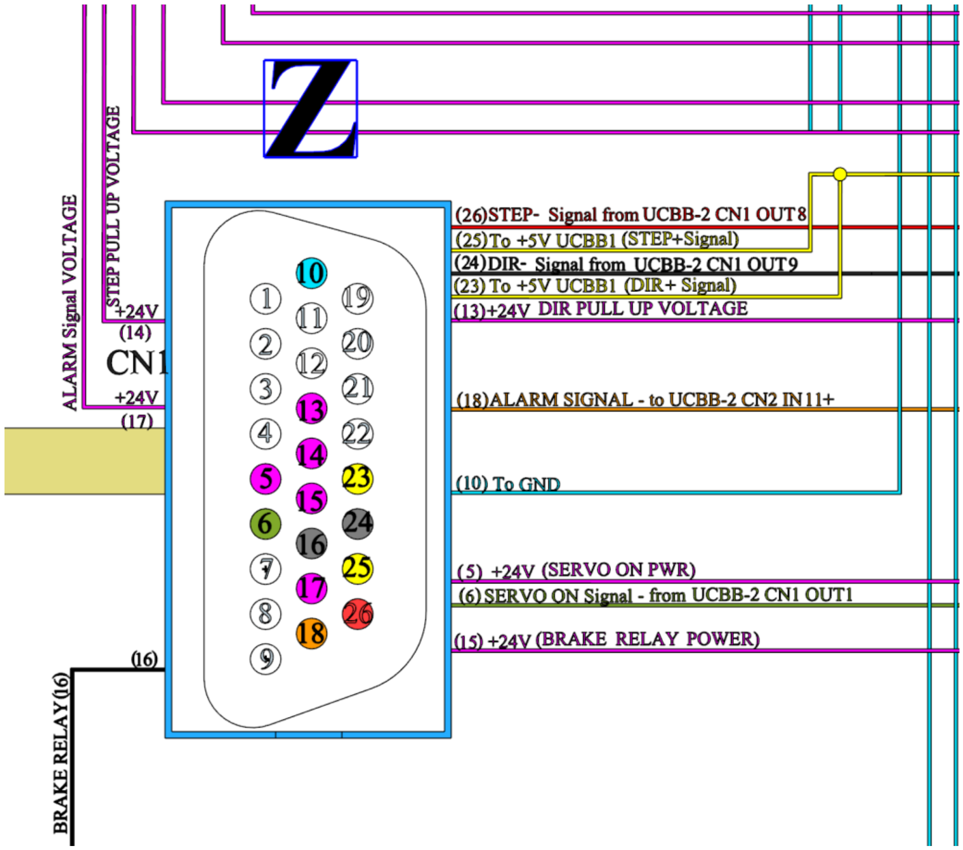
Electrical Specifications

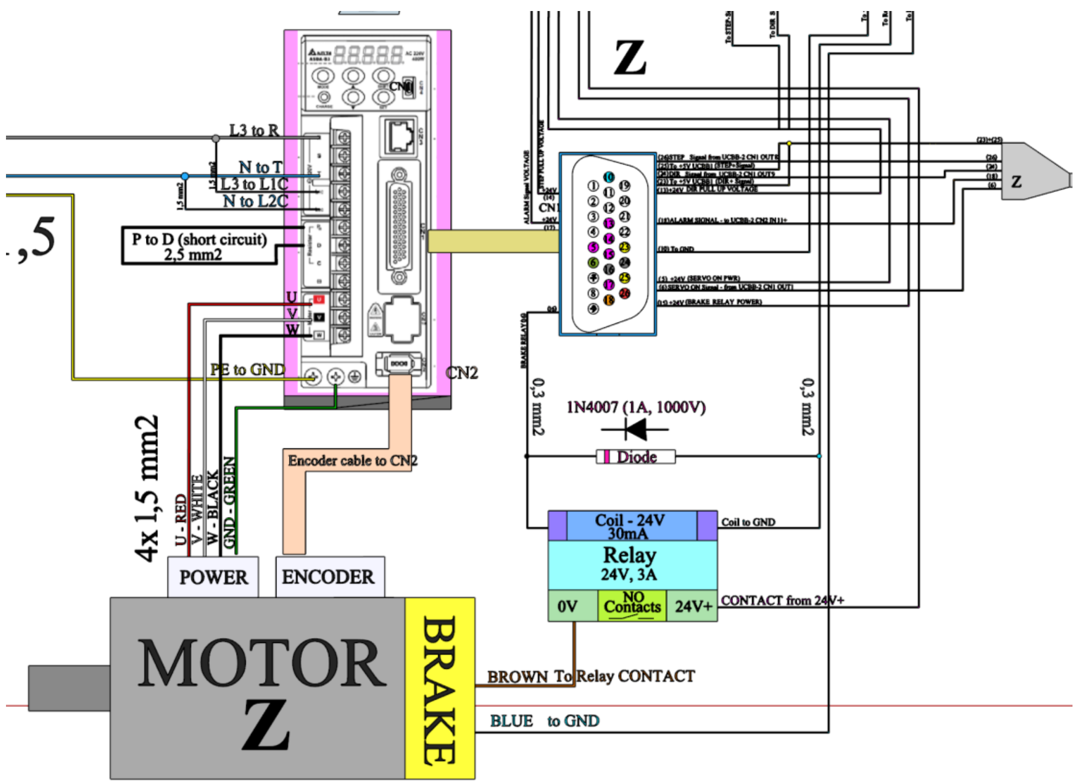
Medium Inertia Motor ECM-B3M Series

	ECM-B3-C 2 0807[*1]	ECM-B3-E 2 1310[*1]	ECM-B3-E 2 1315[*1]	ECM-B3-E 2 1320[*1]
Rated Power (kW)	0.75	1	1.5	2
Rated Torque (N-m)[*2]	2.4	4.77	7.16	9.55
Maximum Torque (N-m)	8.4	14.3	21.48	28.65
Rated Speed (rpm)	3000	2000		
Maximum Speed (rpm)	6000	3000		
Rated Current (Arms)	4.27	5.96	8.17	10.59
Max. Instantaneous Current (Arms)	15.8	19.9	26.82	34.20
Rated Power Rate (kW/s)[*3]	53.83(50.97)	29.21 (28.66)	45.69 (45.09)	62.25 (61.62)
Rotor Inertia ($\times 10^{-4}$kg.m^2)[*3]	1.07 (1.13)	7.79 (7.94)	11.22 (11.37)	14.65 (14.8)
Mechanical Time Constant (ms)[*3]	0.54 (0.57)	1.46 (1.49)	1.10 (1.12)	1.03 (1.04)
Torque Constant -KT (N-m/A)	0.56	0.80	0.88	0.90
Voltage Constant -KE (mV/(rpm))	20.17	29.30	31.69	32.70
Armature Resistance (Ohm)	0.55	0.419	0.260	0.198
Armature Inductance (mH)	2.81	4	2.81	2.18
Electrical Time Constant (ms)	5.11	9.55	10.81	11.01
Brake Holding Torque [Nt-m (min)] [*4]	2.5	10	10	10
Brake Power Consumption (at 20°C)[W]	8	21.5	21.5	21.5
Brake Release Time [ms (Max.)]	20	50	50	50
Brake Pull-In Time [ms (Max.)]	60	110	110	110
Max. Radial Loading (N)[*6]	392	490	686	980
Max. Axial Loading (N)[*6]	147	98	343	392
Weight (kg)[*3]	2.34 (3.15)	4.9 (6.3)	6.0 (7.4)	7 (8.5)
Derating (%) (with oil seal)	5	5	5	5
Torque Feature (T-N Curve)				
Insulation Class	Class A (UL), Class B (CE)			
Insulation Resistance	> 100 MΩ, DC 500V			
Insulation Strength	1.8 kVac, 1 sec			
Vibration Level (μm)	V15			
Operating Temperature	-20°C ~ 60°C*[5]			
Storage Temperature	-20°C ~ 80°C*[5]			
Storage & Operation Humidity	20 ~ 90%RH (non-condensing)			
Vibration Capacity	2.5 G			
IP Rating	IP67 (when using waterproof connections and when an oil seal is fitted to the rotating shaft (for an oil seal model))			
Certifications	CE c UL us			

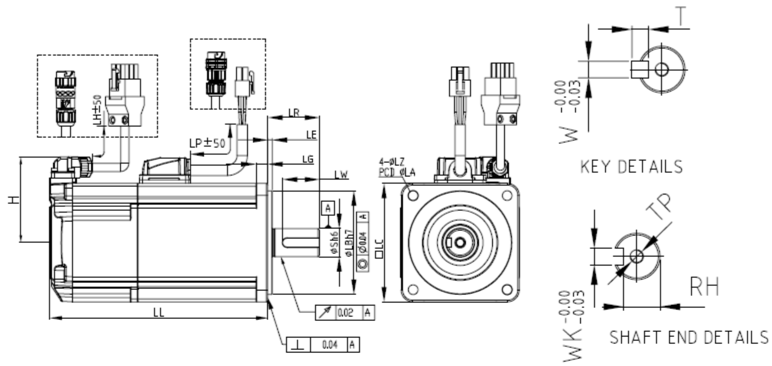
Notes:
1. In the servo motor model name, 1 represents the motor inertia and 2 represents the encoder type.
2. The rated torque is the continuous permissible torque between 0 to 40°C operating temperature which is suitable for the servo motor mounted with the following heat sink dimensions.
   F80: 250 mm x 250 mm x 6 mm
   F130: 400 mm x 400 mm x 20 mm
   Material: aluminum
3. ( ) = motor with brake
4. The built-in servo motor brake is only for keeping the object in a stopped state. Do not use it for deceleration or as a dynamic brake
5. If the operating temperature is over 40°C, refer to the power derating curves of B3 motors on page 27.
6. Please follow the max. tolerant loading of the motor shaft end listed below during operation




Z
ALARM Signal VOLTAGE
STEP PULL UP VOLTAGE
+24V
(14)
CN1
+24V
(17)
(16)
BRAKE RELAY(16)
(26)STEP- Signal from UCBB-2 CN1 OUT8
(25)To +5V UCBB1 (STEP+Signal)
(24)DIR- Signal from UCBB-2 CN1 OUT9
(23) To +5V UCBB1 (DIR+ Signal)
(13)+24V DIR PULL UP VOLTAGE
(18)ALARM SIGNAL - to UCBB-2 CN2 IN11+
(10) To GND
(5) +24V (SERVO ON PWR)
(6) SERVO ON Signal - from UCBB-2 CN1 OUT1
(15) +24V (BRAKE RELAY POWER)


Z
L3 to R
N to T
L3 to L1C
N to L2C
,5
P to D (short circuit)
2,5 mm2
U
V
W
PE to GND
CN2
Encoder cable to CN2
4x 1,5 mm2
U - RED
V - WHITE
W - BLACK
GND - GREEN
POWER
ENCODER
MOTOR
Z
BRAKE
0,3 mm2
1N4007 (1A, 1000V)
Diode
0,3 mm2
Coil - 24V
30mA
Coil to GND
Relay
24V, 3A
0V
NO
Contacts
24V+
CONTACT from 24V+
BROWN To Relay CONTACT
BLUE to GND

Dimensions of Motors with Frame Size of 80 mm or Below



Model	C $\boxed{2}$ 0401 $\boxed{3}\boxed{4}\boxed{5}$	C $\boxed{2}$ 0602 $\boxed{3}\boxed{4}\boxed{5}$	C $\boxed{2}$ 0604 $\boxed{3}\boxed{4}\boxed{5}$	C $\boxed{2}$ 0804 $\boxed{3}\boxed{4}\boxed{5}$	C $\boxed{2}$ 0807 $\boxed{3}\boxed{4}\boxed{5}$
LC	40	60	60	80	80
LZ	4.5	5.5	5.5	6.6	6.6
LA	46	70	70	90	90
S	$8(^{+0}_{-0.009})$	$14(^{+0}_{-0.011})$	$14(^{+0}_{-0.011})$	$14(^{+0}_{-0.011})$	$19(^{+0}_{-0.013})$
LB	$30(^{+0}_{-0.021})$	$50(^{+0}_{-0.025})$	$50(^{+0}_{-0.025})$	$70(^{+0}_{-0.030})$	$70(^{+0}_{-0.030})$
LL (w/o brake)	77.6	72.5	91	86.7	105.2
LL (with brake)	111.7	109.4	127.9	126.3	144.8
LH	300	300	300	300	300
LP	300	300	300	300	300
H	40	48.5	48.5	58.5	58.5
LR	25	30	30	30	35
LE	2.5	3	3	3	3
LG	5	7.5	7.5	8	8
LW	16	20	20	20	25
RH	6.2	11	11	11	15.5
WK	3	5	5	5	6
W	3	5	5	5	6
T	3	5	5	5	6
TP	M3 Depth 8	M4 Depth 15	M4 Depth 15	M4 Depth 15	M6 Depth 20

Notes:
1. In the servo motor model name, $\boxed{2}$ represents the encoder type, $\boxed{3}$ represents the brake or keyway / oil seal type, $\boxed{4}$ represents the shaft diameter and connector type, and $\boxed{5}$ represents the special code.
2. When $\boxed{4}$ in the servo motor model name is J or K, the connector is an IP67 waterproof connector.

Z Axis Servo Driver – DELTA ASD-B3A-0721-E

AC Servo drive, 0.75 kW, 220V, B3 series. Features 4 pulse inputs, analog voltage control, PR mode, RS-485 communication, and STO safety function. For more specs and data – see the tables/diagrams/pictures in the X/Y servo driver section above.

Servo Closed-Loop Operation

Although UCCNC sends Step/Direction commands similar to stepper motor control, the DELTA servo system operates as a closed-loop system at the driver level. The high-resolution encoder (17-bit on X/Y, 24-bit on Z) feeds real-time position and velocity information directly to the servo driver, which continuously compares commanded position against actual position. Any following error is corrected internally at thousands of iterations per second. This provides superior accuracy under varying loads, higher achievable speeds and accelerations, stall detection and alarm capability (reported back to UCCNC via the ALARM input), precise position holding at rest, and smooth motion even at very low speeds.

The Encoders Close the Loop *at the Servo Driver Level*

1. **Command Generation (UCCNC & UC300ETH):** UCCNC calculates the desired toolpath and determines how many steps each axis motor needs to take, and in which direction. It sends this information as high-frequency electrical pulses (Step signals) and a direction signal (Direction signal) via the UC300ETH controller and the UCBB breakout boards.
2. **Command Reception (Servo Driver):** The Delta Servo *Driver* (e.g., ASD-B3) receives these Step and Direction pulses from the controller. For the driver, each step pulse represents a tiny target increment of movement.
3. **Motor Actuation (Servo Driver):** Based on the received commands, the servo driver sends power (voltage and current) to the servo motor, instructing it to rotate.
4. **Feedback (Encoder -> Servo Driver): This is the critical part.** The high-resolution encoder (17-bit on X/Y, 24-bit on Z in your case) is directly connected to the *servo motor* and feeds the *actual* position and velocity information back *directly to the servo driver*.
5. **Error Correction (Servo Driver):** The servo driver *internally* and constantly compares:
    - The **commanded position** (derived from the incoming step pulses).
    - The **actual position** (read from the encoder).
6. **Closed-Loop Control (Within the Driver):** If there's any discrepancy between the commanded position and the actual position (this difference is called "following error"), the driver's internal logic (often a sophisticated DSP - Digital Signal Processor) instantly adjusts the power sent to the motor to correct the error. This happens extremely rapidly, thousands of times per second, ensuring the motor accurately follows the commands sent by UCCNC.

How Does This Help? (Benefits of Servo Encoders even without Controller Feedback):

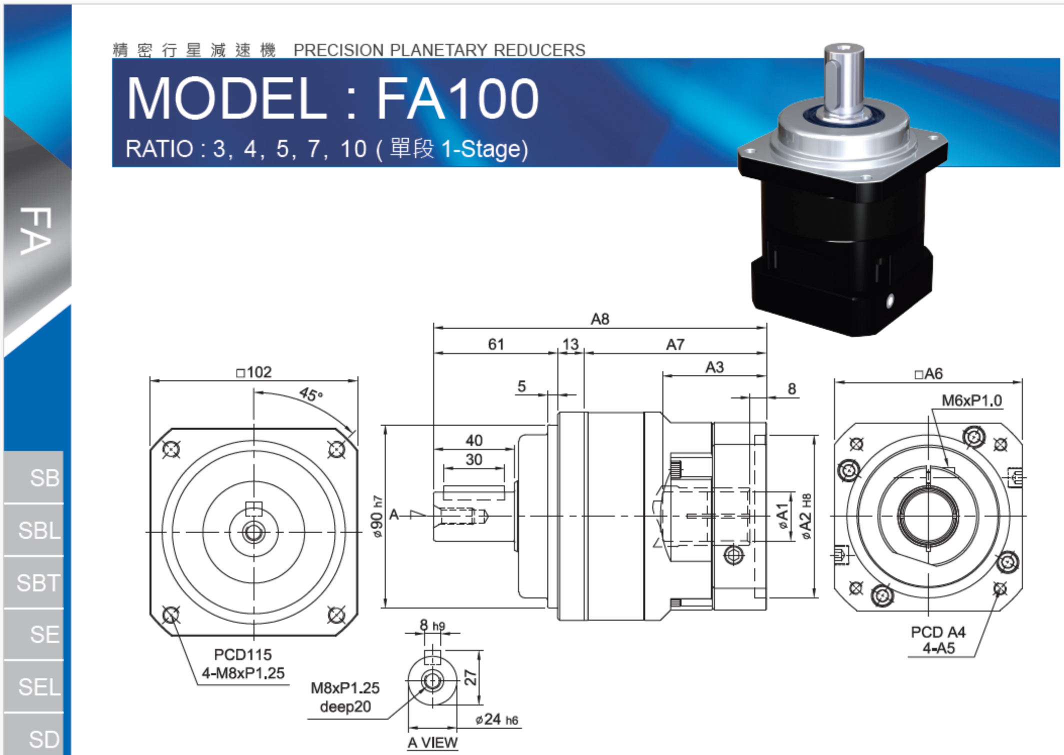
1. **Accuracy:** This internal closed loop ensures the motor *actually* reaches the commanded position. It automatically compensates for varying loads, friction, and minor mechanical imperfections that would cause an open-loop stepper system to lose steps or position inaccurately.

2. **Performance:** Because the driver knows the motor's exact position and speed, it can push the motor much harder (higher speeds and accelerations) than a stepper system without risking lost steps. It knows if the motor is lagging and can apply more power to catch up.
3. **Error Detection:** If the motor encounters an unexpected obstruction or the load is too high, the following error inside the driver will grow rapidly. The driver detects this large error and can trigger an *Alarm* state. This alarm signal *is* sent back to the UCBB/UCCNC (via the ALARM input pins you listed), allowing the main control system to halt the machine safely. Stepper systems generally don't know they've lost position until something goes visibly wrong.
4. **Stall Prevention/Detection:** Unlike steppers, servos don't inherently "lose steps." If they can't achieve the commanded position due to overload, they fault (alarm).
5. **Holding Torque & Stability:** Servos use the encoder feedback to actively hold their position precisely when stationary, resisting external forces better than steppers at rest.
6. **Smoothness:** Servo systems generally provide smoother motion, especially at very low speeds.

In Summary: While UCCNC/UC300ETH acts like it's controlling steppers by sending Step/Direction signals, the magic happens *within* the Delta Servo Drivers. The drivers use the high-resolution encoder feedback to create a fast, internal closed-loop system. This system ensures the motors precisely follow the Step/Direction commands, corrects for errors in real-time, and provides fault detection (via the Alarm signal back to UCCNC) if something goes wrong. The encoder is fundamental to the "servo" nature of the system, providing the accuracy, performance, and reliability advantages over open-loop stepper systems.

Gear Reducers and Transmission

X and Y Axes – Planetary Gear Reducers - Liming FA-100 - Planetary gear reducers with a reduction ratio of 1/7. Coupled with the servomotors.



入力法蘭尺寸表 Dimension of Input Flange

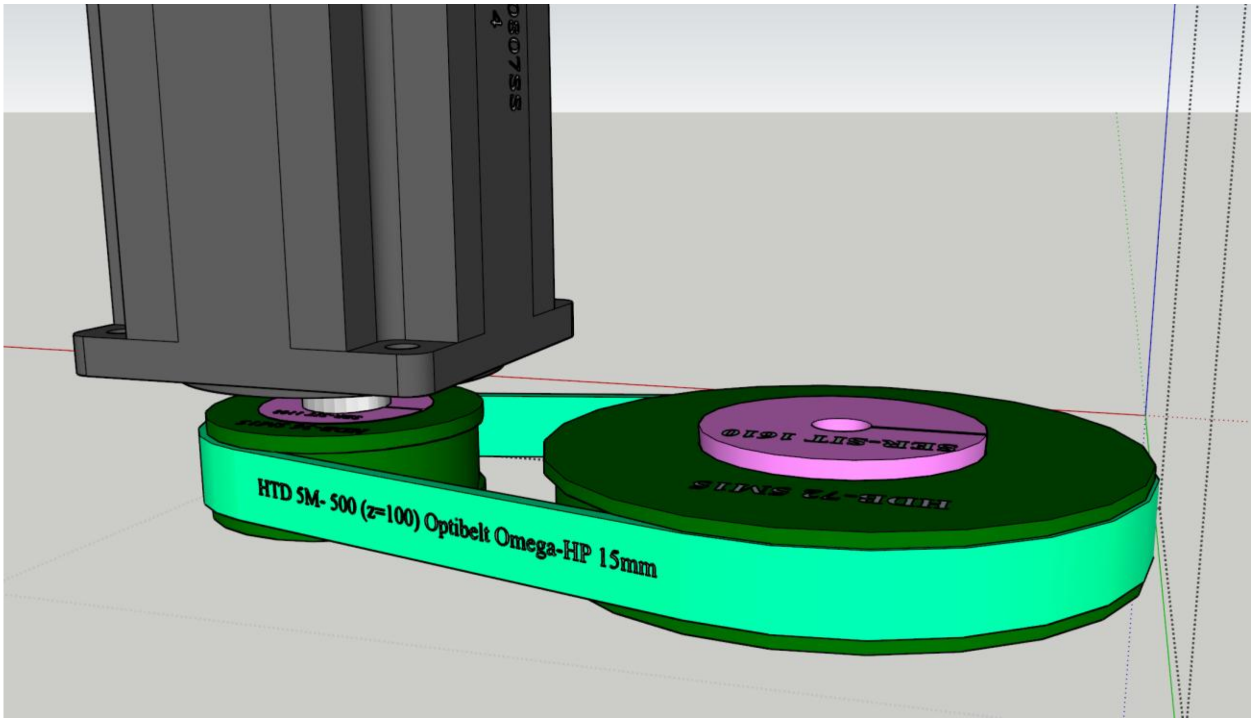
unit : mm

NO.	ØA1	ØA2	A3	A4	A5	A6	A7	A8
1	19、22、24	70	51	90	M6	92	89.5	163.5
2		80	51	100	M6	92	89.5	163.5
3		95	51	115	M6	110	89.5	163.5
4		95	51	115	M8	110	89.5	163.5
5		110	51	145	M8	130	89.5	163.5
6		110	65.5	145	M8	130	104	178

FA100-1 特性表

減速比 Ratio	代號	單位 Unit	3	4	5	7	10
額定輸出扭矩 / Rated Output Torque (Nominal Output Torque)	T_{2N}	Nm	165	146	160	149	141
最大加速扭矩 / Max. Acceleration Torque	T_{2B}	Nm	297	263	288	268	254
轉動慣量 / Mass Moments of Inertia		$kg.cm^2$	0.6	0.45	0.45	0.45	0.41
最大輸出扭矩 / Max. Output Torque 急停扭矩 / Emergency Stop Torque	T_{2NOT}	Nm	3 倍額定輸出扭矩 3 Times of Rated Output Torque				
額定輸入轉速 / Rated Input Speed	n_{1N}	rpm	3,000				
最大輸入轉速 / Max. Input Speed	n_{1B}	rpm	5,000				
扭轉剛性 / Torsional Rigidity		Nm/arcmin	14				
容許徑向力 / Max. Radial Force	F_{2rB}	N	3,200				
容許軸向力 / Max. Axial Force	F_{2aB}	N	1,600				
使用壽命 / Service Life	L_H	hr	S5 周期運轉 : >30,000 (S1 連續運轉 : >15,000 hrs) S5 Cycle Operation: >30,000 (S1 Continuous Operation: >15,000 hrs)				
效率 / Efficiency	η	%	≧97%				
使用溫度 / Operating Temperature		℃	-25℃~+90℃				
潤滑 / Lubrication			全合成潤滑油脂 Synthetic Grease				
防護等級 / Protection Class			IP 65				
安裝方向 / Mounting Position			任意 Any				
噪音值 / Noise Level		dB	≦65 dB				
重量 / Weight ±2%		Kg	3.76				

Z Axis – Timing Belt Reduction - high-quality timing belt **Optibelt Omega-HP 5M15** with **timing pulleys SIT TOP DRIVE HTD - HDB36** and **HDB72**. **Ratio1/2.**



Power Supplies for the control box

Supply	Model / Specifications	Function
24V DC	NES-350-24, 358W, 14.6A, 220VAC input	Powers both UCBB boards, servo drive logic circuits, all proximity sensors, and all solenoid valves
5V DC	MeanWell RS-15-5, 15W, 3A, 220VAC input	Powers the UC300ETH motion controller

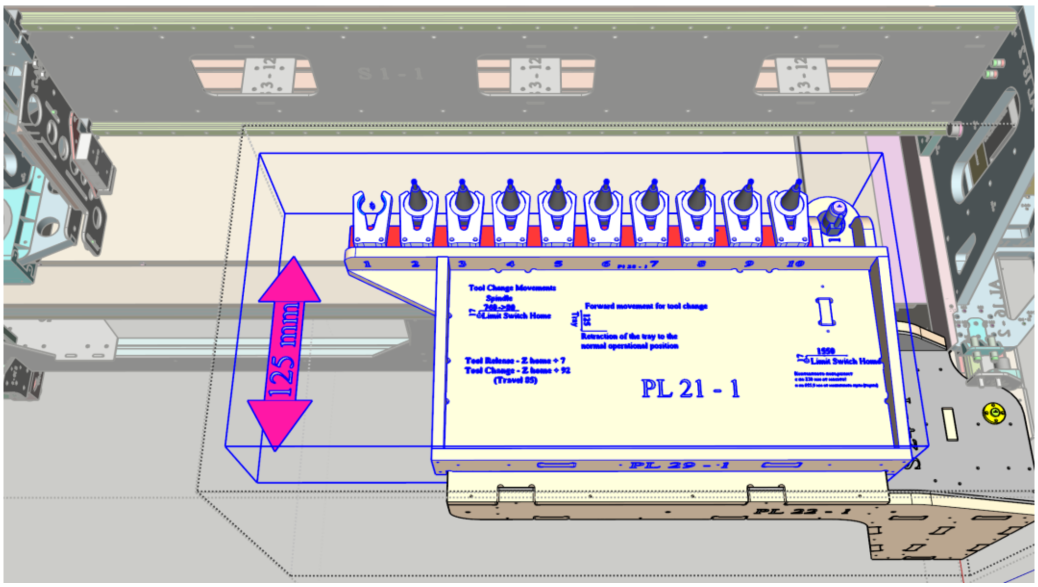
6. Machine Subsystems

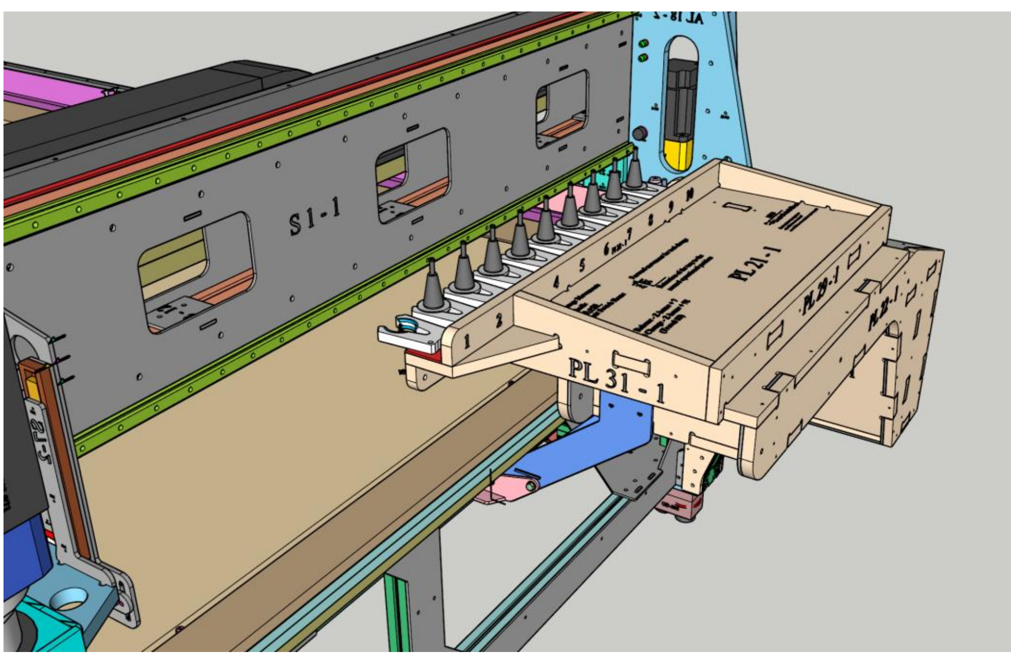
Retractable Tool Rack on Rotating Base

The tool rack is a retractable assembly holding 10 ISO30 tool holders with 80 mm spacing between clamp centers. The rack extends 125 mm in the Y-negative direction for tool change operations and retracts flush with the machine's front frame when not in use.

The extension/retraction of the tool rack is done automatically by the tool loading/offloading/changing macros M6 and M200006.

The rack can also be extended/retracted manually with the help of a dedicated UCCNC screen button and its macro M20710.



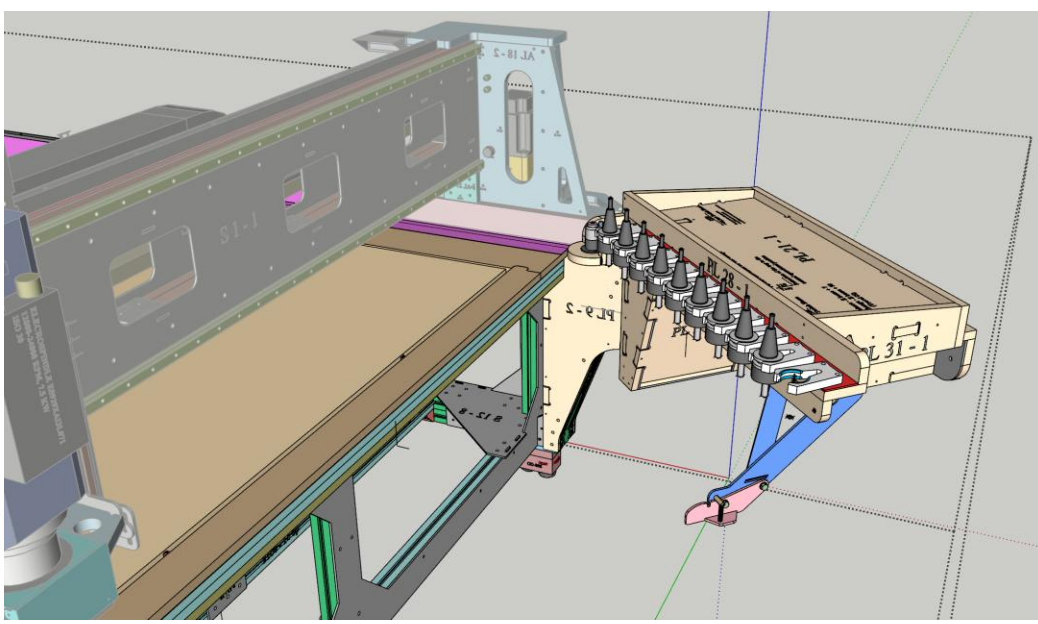


The Tool Rack has clamps for 10 tool holders ISO30. The spacing between the centers of the clamps is 80 mm. The rack is being extended/retracted by two double-acting cylinders TN16x125 with a stroke of 125 mm. The two cylinders are controlled by one SINGLE 24V 5/2 solenoid valve with its spring-loaded position corresponding to the Rack Retracted state (when the solenoid is activated with 24V power, the rack will extend). The Rack is oriented along the X axis and should be extended 125 mm in the Y negative direction for the tool changing routine to happen. When the rack is extended, it is breaching the working area of the machine. Special care should be taken not to operate the machine with the rack extended, or a collision might happen. For this purpose, a **PNP NC proximity sensor** is installed, which is being checked for triggered state (when the rack is retracted) during safety checks in the different macros. The sensor is positioned in such a way as to be:

- **Closed** (away from the tool rack) when the rack is **Retracted**
- **Open** (detecting the tool rack) when the rack is **Extended**

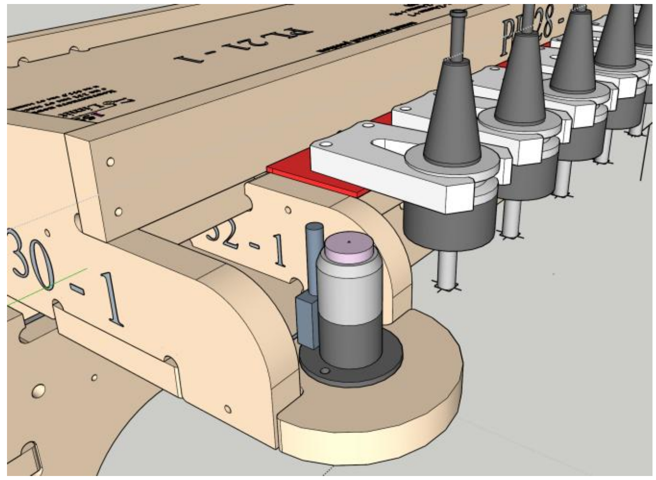
Rotating Base - the whole rack, together with the tool holder clamps and the tool probe, is mounted on a Rotating Base for the purpose of facilitating the tasks of loading/unloading the stock sheets being cut. The base is operated manually and has a spring-loaded locking mechanism for locking it firmly into the machine when in operating position. When rotated away from the machine 90 degrees, the whole front face of the working area becomes easily accessible. When rotated back to the machine, the Tool Rack is available to the machine for the purpose of changing the tools. For some of the macros of the software, it is essential to check and verify that the Tool Rack Probe is rotated towards the machine. For this purpose, a PNP NC proximity sensor is installed, which is being checked for triggered state (when the base is rotated towards the machine) during safety checks in the different macros. The sensor is positioned in such a way as to be:

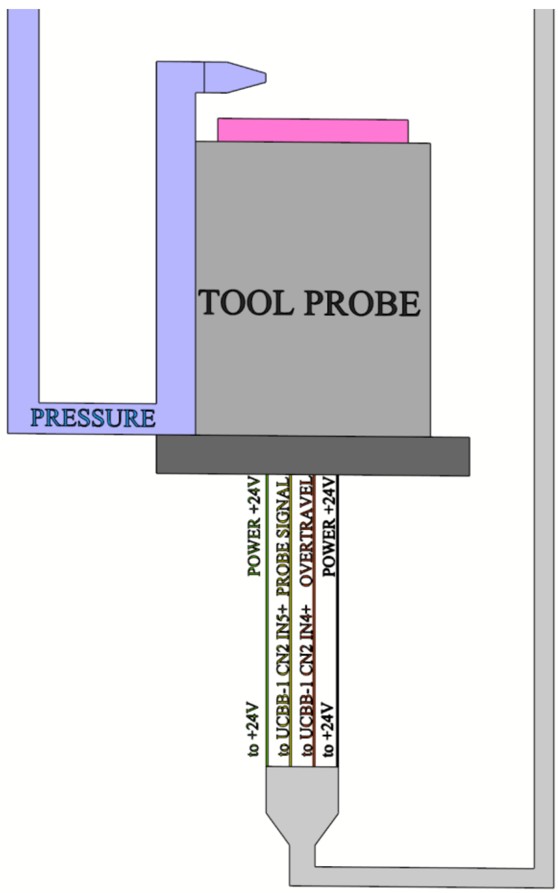
- **Closed** (away from the tool base) when the base is rotated **OUT** of the machine for loading/unloading
- **Open** (detecting the tool base) when the base is rotated **INTO** the machine for cutting



Tool Length Probe – mounted on the extendable Tool Rack

The tool length probe is a 24V plunger-type sensor mounted on the retractable tool rack. It uses Normally Closed logic: the internal switch is closed at rest and opens when the tool depresses the plunger past the actuation point.





The Probe is used for tool measurement automatically from the M6/M20006 tool loading/changing macros when the tool has no valid measurement data in the system registry. Once measured, the particular tool's offset (length) is stored in a registry and is used on all subsequent tool loadings (it is not necessary to measure the tool all the times it has been loaded).

The operator can manually use the tool probe for measuring any tool in the spindle (regardless of whether there is a measured offset for the tool in the registry) by using a dedicated UCCNC screen button and a tool measuring macro M20031.

The Rotating Tool Base should be rotated and locked INTO the machine, and the Tool Rack should be EXTENDED for the Tool Probe to be accessible for measuring.

The probe is equipped with an air nozzle for clearing the sensing surface from debris. The air nozzle is controlled by a 24V 2/2 Solenoid valve and is operated automatically when measuring is taking place.

The measured tool length is used to change the G54 Z coordinate (the working coordinates) so when G54 Z=0, the face of the instrument is at the work-piece upper surface. The exact distance of the probe's sensitive area from G53 Z0 (from the Machine Origin) and from G54 Z0 (from the upper surface of the workpiece) is not known. With the help of a special calibrating macro M20303, these distances are found. The probe being plunger type means the signal for Probe Z (the tool contact with the probe) is not sent exactly at the moment when the tool touches the probe, but rather a bit later, when the tool has travelled an additional small distance downward in order to push the plunger. When the plunger is pushed enough, an internal switch opens, and the signal for tool contact is sent. The so-called 'actuation distance' is typically between 0,025 and 0,1 mm.

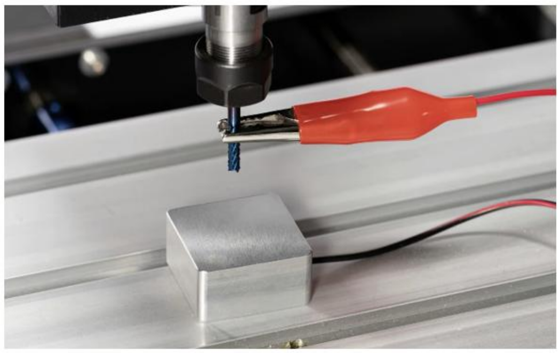
The probe has a second NC switch for sending an alarm signal for OVERTRAVEL if the tool has travelled an excessive distance and it is about to damage the probe. The signal is monitored by the UCCNC software, and the system is stopped to prevent any damage.

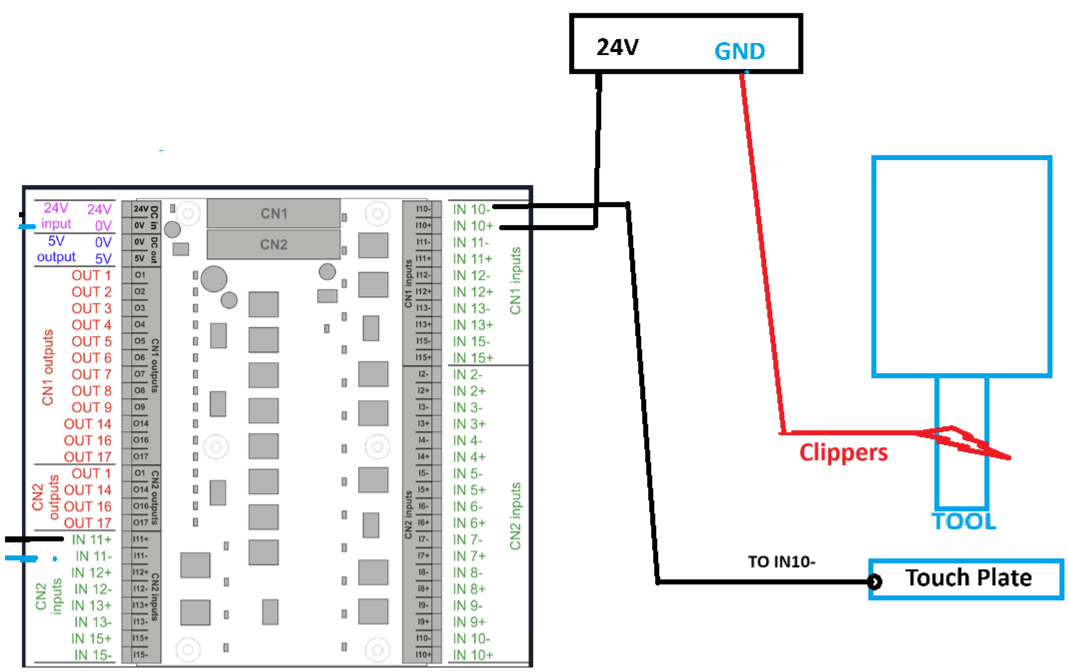
Note: This function is used as a non-standard Emergency Stop Button. If the operator is near the front–right corner of the machine and presses the plunger of the tool probe, the machine stops immediately and enters RESET mode.

Output mode	Normal Closed-NC
Stroke	5mm
Pre - travel	Almost Zero
Repeatability	0.001mm
Protection structure	IP67
Contact force	1.5N
Contact material	Alloy
Shell material	Grind 4S
Protective tube	1m nozzle small bending radius R25
Cable	3m oil resistant, 4 core diameter 5
Alarm signal when going out	NC (distance measurement signal about 2.5mm)
Contact rating	DC24V 20mA (Max) Recommended value 10mA (resistive load)

Touch Plate

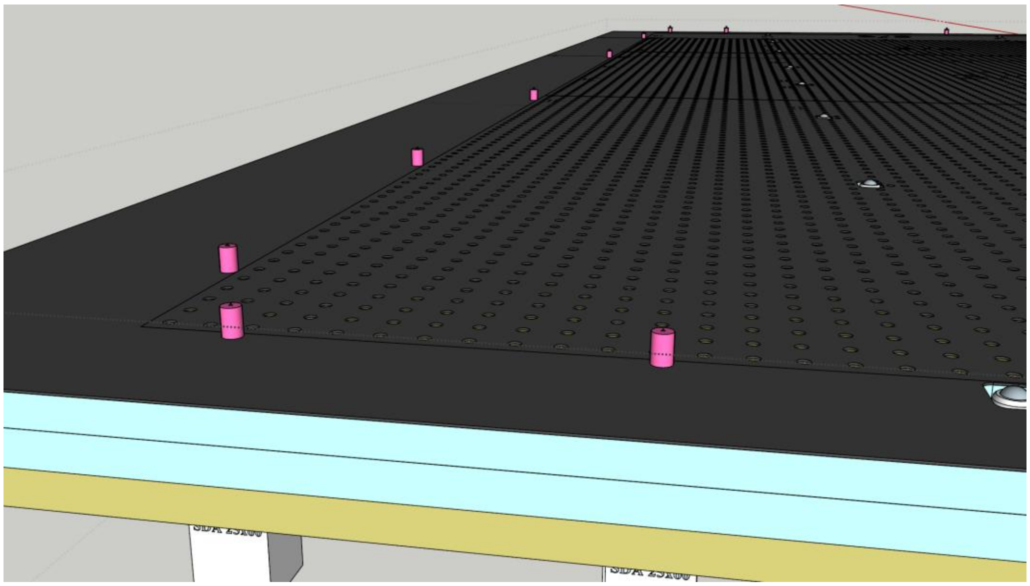
It is a solid rectangular (60 mm x 60 mm) aluminium plate with a known thickness of 20.0 mm, which is being used by a special measuring and calibrating macro M20303 for determining the G54 Z0 for the current tool (at G54 Z0, the tool face should be exactly at the upper surface of the workpiece), and for calibrating the Tool Probe. The Touch Plate is essentially a Normally Open Switch. It is being positioned MANUALLY by the operator on the front/left corner of the workpiece when the macro M20303 is used (it has a dedicated UCCNC screen button). The operator should also CONNECT the touch-plate clippers to the tool in the spindle. After the macro is finished, the Touch Plate is removed for the consecutive normal operations.





Pop-up Alignment Pins

Pneumatic pop-up pins are installed along the X and Y axes to enable rapid and repeatable workpiece registration. When extended, the pins protrude through the machine bed surface. Pushing a material sheet firmly against the pins aligns it precisely with G54 X0 and G54 Y0, positioning the front-left corner of the sheet at the G54 origin.



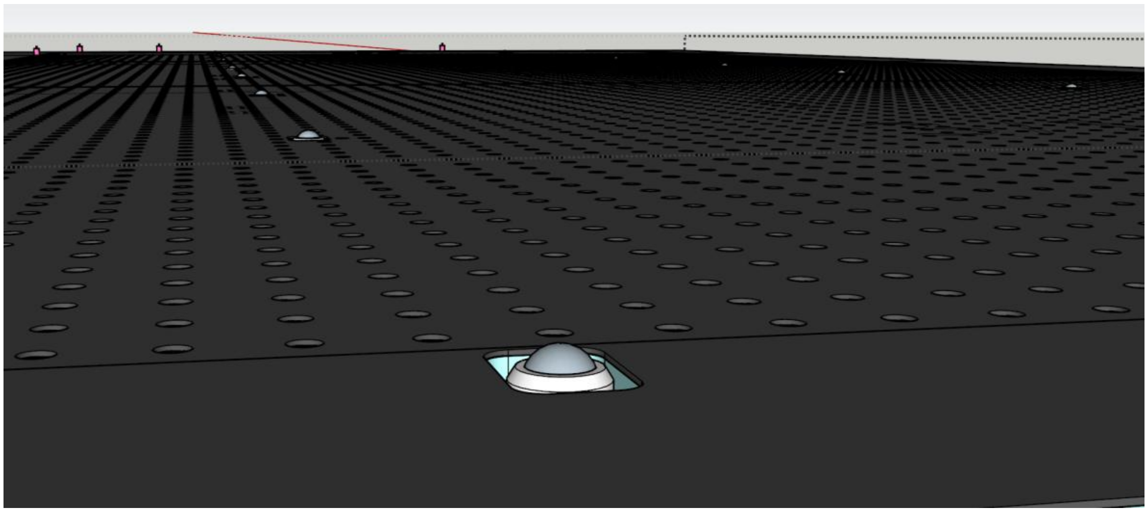
The Pins are extended/retracted by the operator when needed through a special macro M20730 with a dedicated button on UCCNC screen. When the pins are extended, they can interfere with the Spindle and its tool, so some special precautions are made for safety by checking whether the pins are retracted before operating the spindle.

The pins are being extended (pushed up through the machine bed)/retracted (hidden in the machine bed) by individual double-acting cylinders SDA25x60 with a stroke of 60 mm. All the cylinders are controlled by one 24V SINGLE 5/2 solenoid valve with its spring-loaded position corresponding to the retracted state (when the solenoid is activated with 24V power, the pins will extend to their UP position). A **PNP NC proximity sensor** is installed, which is being checked for triggered state (when the pins are extended) during safety checks in the different macros. The sensor is positioned in such a way as to be:

- **Closed** (away from the control pin) - when the pins are **Retracted** (hidden within the bed)
- **Open** (detecting the control pin) - when the pins are **Extended** (protruding through the bed)

Ball Bearings Transport System

Pneumatic ball bearings assist with loading and unloading material sheets. When extended, the ball transfer units protrude through the machine bed, allowing heavy sheets to be rolled into position with minimal effort.



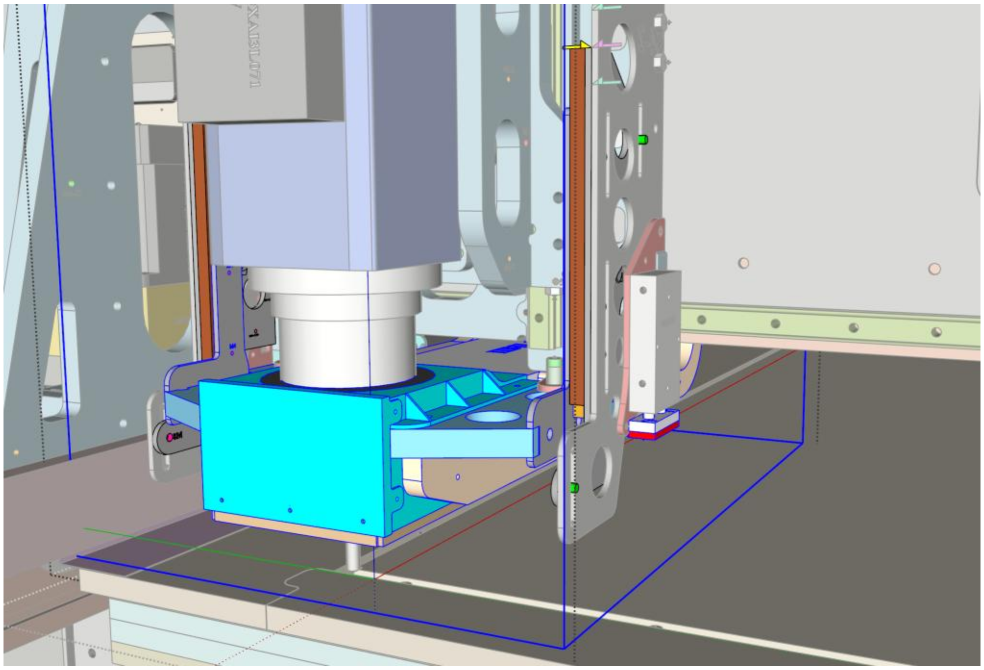
The ball bearings are extended/retracted by the operator when needed through a dedicated macro M20740 with an assigned button on the UCCNC screen. When the bearings are extended, they can interfere with the Spindle and its tool, so some special precautions are made for a safety check whether the bearings are retracted before operating the spindle.

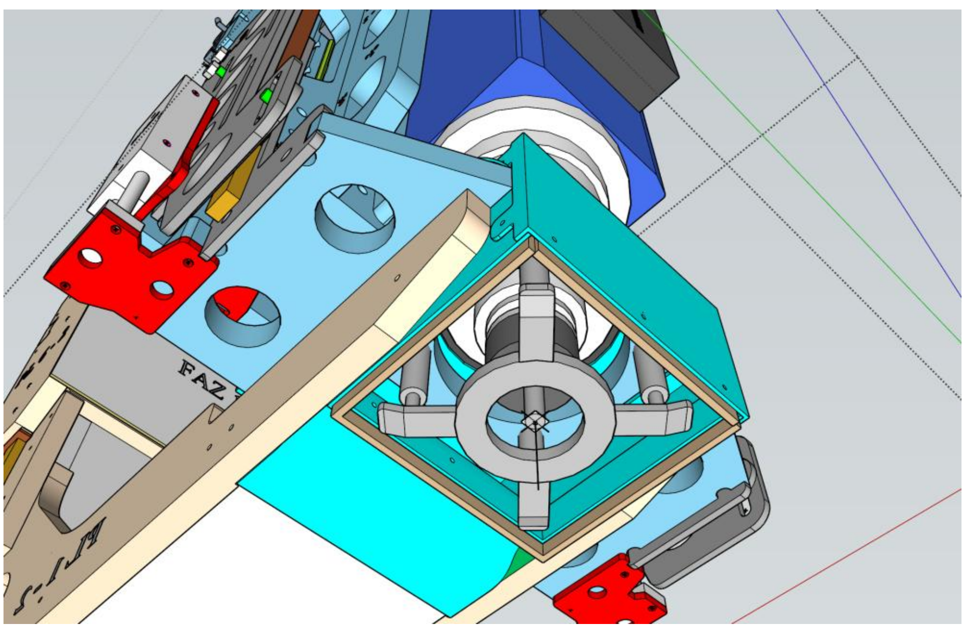
The ball bearings are being extended (pushed up through the machine bed)/retracted (hidden in the machine bed) by individual double-acting cylinders SDA25x25 with a stroke of 25 mm. All the cylinders are controlled by one 24V SINGLE 5/2 solenoid valve with its spring-loaded position corresponding to the retracted state (when the solenoid is activated with 24V power, the pins will extend to their UP position). A **PNP NC proximity sensor** is installed, which is being checked for triggered state (when the ball bearings are extended) during safety checks in the different macros. The sensor is positioned in such a way as to be:

- **Closed** (away from the control bearing) when the bearings are **Retracted** (hidden within the bed)
- **Open** (detecting the control bearing) when the bearings are **Extended** (protruding through the bed)

Retractable Dust Collector with Integrated Workpiece Pusher

The dust collector is a retractable assembly mounted around the spindle that captures debris during cutting operations. It is lowered into working position during machining and lifted clear during positioning moves, tool changes, and non-cutting operations.





The operation of the collector can be done automatically by the system through a special M9000 macro, which dynamically lowers/lifts the collector during the cutting cycles. The operator can manually operate the collector using a dedicated button on the UCCNC screen and the M20720 macro.

The Collector is being extended (pushed down)/retracted (lifted) by two double-acting cylinders TN16x50 with a stroke of 50 mm. The two cylinders are controlled by one 24V SINGLE 5/2 solenoid valve with its spring-loaded position corresponding to the Collector Lifted state (when the solenoid is activated with 24V power, the collector will extend to its down position). A **PNP NC** proximity sensor is installed, which is being checked for triggered state (when the collector is lifted) during safety checks in the different macros. The sensor is positioned in such a way as to be:

- **Closed** (away from the collector) when the collector is **Extended** (Lowered)
- **Open** (detecting the collector) when the collector is **Retracted** (Lifted)

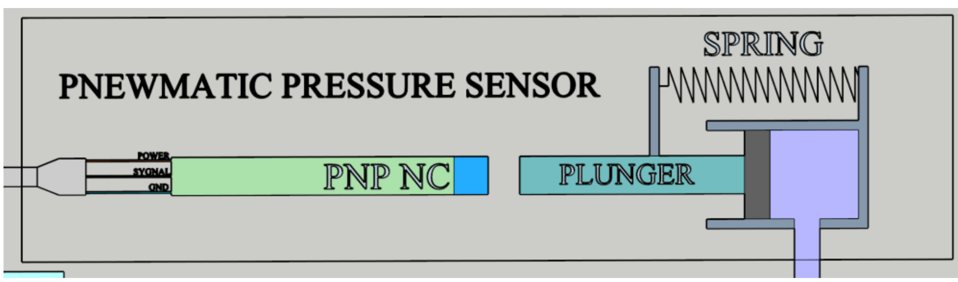
An integral part of the Dust Collector is the Spring-loaded Skid Type **Work-Piece Pusher**. When the collector is extended (pushed down) in working position close to the workpiece, the Pusher makes contact with the workpiece and effectively asserts pressure on it, keeping it in place mechanically while the tool is cutting it. The contact skid of the pusher has ball-type rollers on its lower surface, helping the pusher to move over the surface of the workpiece during the operation of the machine.

Pneumatic Pressure Sensor

A dedicated pneumatic pressure sensor continuously monitors the compressed air supply feeding the machine’s pneumatic subsystems. Adequate pressure is critical for the reliable operation of the spindle collet, tool rack, dust collector, alignment pins, and ball bearings.

The sensor is a simple mechanical plunger-type device that is spring-loaded. When the pressure in the pneumatic line is sufficient, the plunger is pushed out against the spring force. A PNP NC proximity sensor is installed, which is being checked continuously for triggered state (when the plunger is extended) from a dedicated macro M6000 running in a loop. The sensor is positioned in such a way as to be:

- **Closed** (away from the pressure plunger) when the plunger is **Retracted** from the sping when there is no pressure in the line
- **Open** (detecting the pressure plunger) when the plunger is **Extended** from the high pneumatic pressure against the spring force



Inductive Proximity Switches – LJ12A3-4-Z/AY

The machine is equipped with PNP NC (Normally Closed) inductive proximity switches on all axes. These switches serve triple duty as over-travel protection, homing references, and gantry squaring references.

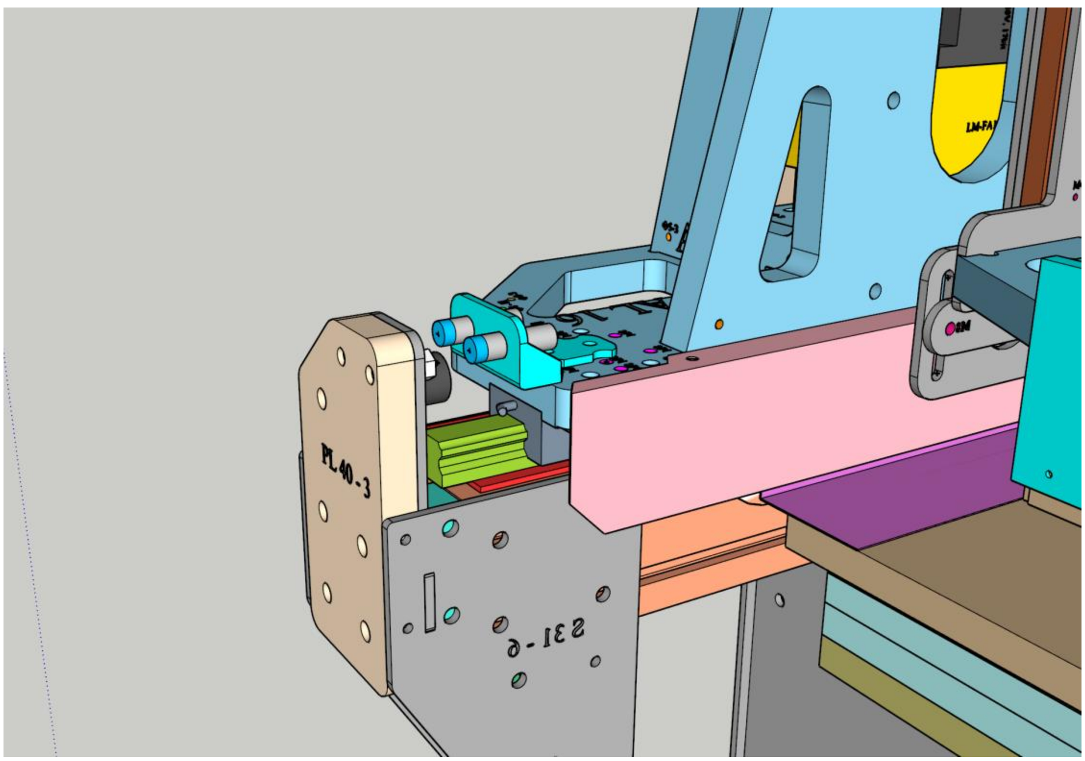
Model	LJ12A3-4-Z/BX	LJ12A3-4-Z/AX	LJ12A3-4-Z/BY	LJ12A3-4-Z/AY
Detecting Distance Max	4mm	4mm	4mm	4mm
Sensor Type	Proximity (Inductive)	Proximity (Inductive)	Proximity (Inductive)	Proximity (Inductive)
Sensor Output Switch Type	NPN – Normally open (NO)	NPN – Normally closed (NC)	PNP – Normally open (NO)	PNP – Normally closed (NC)
Supply Voltage Min	6Vdc	6Vdc	6Vdc	6Vdc
Supply Voltage Max	36Vdc	36Vdc	36Vdc	36Vdc
Current Consumption Max	300mA	300mA	300mA	300mA
Repeated precision	0.01mm	0.01mm	0.01mm	0.01mm
IP Rating	IP67	IP67	IP67	IP67
Operating Temperature	-25° ~ 75°C	-25° ~ 75°C	-25° ~ 75°C	-25° ~ 75°C
Material	Metal	Metal	Metal	Metal
SKU	1176001	1176002	1176003	1176004
Price	2.80 €	2.80 €	2.80 €	2.80 €

Pictures, dimensions,table data and technical drawings of this product are made with reasonable care and liability is excluded for the accuracy and correctness of this data.

Limit switch installation points:

Location	Axis / Direction	Function
Front-Left Corner of Table	Y Negative (Master)	Y-axis home/limit and gantry squaring
Front-Right Corner of Table	Y Negative (Slave/A)	A-axis home/limit and gantry squaring
Aft-Left Corner of Table	Y Positive	Y-axis positive travel limit
Left Side of Gantry	X Negative	X-axis home/limit
Right Side of Gantry	X Positive	X-axis positive travel limit
Upper Side of Spindle Assembly	Z Positive	Z-axis home/limit (upper)
Lower Side of Spindle Assembly	Z Negative	Z-axis lower travel limit

In addition, the same type and model of proximity switches are used in the Pressure Sensor, and for determining the positions of the Rotating Tool Base, the Tool Rack, the Positioning Pins, the Ball Bearings, and the Dust Collector.

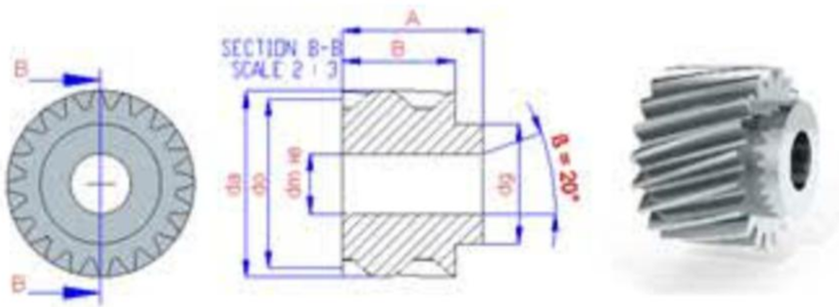


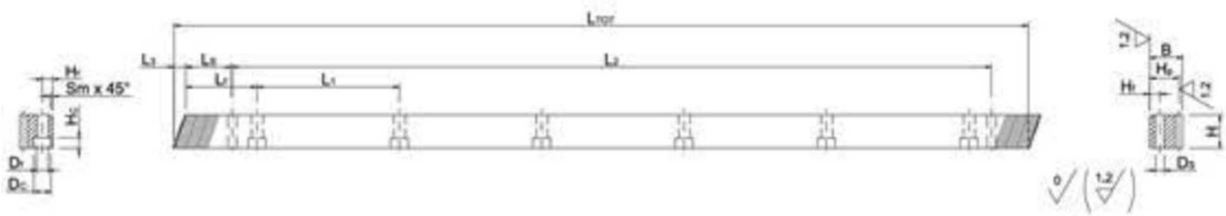
7. Mechanical Systems

Helical Rack GAMBINI and Pinion Drives DOGUS CALIP

Used for the Y LEFT, Y RIGHT, and X axis**.**

LEFT THREAD, MOD 2.0, Z:25 (number of teeth on the pinions)





CODICE ITEM CODE	M	L TOT	L3	Lf	L1	Ls	H	B	Hf	Ds	Sm	Df	Dc	Hc	Hp	Z	L2	N° fori holes	PESO kg
1HR29R010050H	1	500	7,27	62,5	125	30,29	20,5	20	8	5,7	2	7	11	7	19	150	439,4	4	1,48
1HR29R010100H	1	1000	7,27	62,5	125	30,29	20,5	20	8	5,7	2	7	11	7	19	300	939,4	8	2,97
1HR29R010150H	1	1500	7,27	62,5	125	30,29	20,5	20	8	5,7	2	7	11	7	19	450	1439,4	12	4,45
1HR29R015050H	1,5	500	7,27	62,5	125	30,29	20,5	20	8	5,7	2	7	11	7	18,5	100	439,4	4	1,44
1HR29R015100H	1,5	1000	7,27	62,5	125	30,29	20,5	20	8	5,7	2	7	11	7	18,5	200	939,4	8	2,88
1HR29R015150H	1,5	1500	7,27	62,5	125	30,29	20,5	20	8	5,7	2	7	11	7	18,5	300	1439,4	12	4,32
1HR29R020050H	2	500	8,86	62,5	125	31,71	25	24	8	5,7	2	7	11	7	22	75	436,6	4	2,09
1HR29R020100H	2	1000	8,86	62,5	125	31,71	25	24	8	5,7	2	7	11	7	22	150	936,6	8	4,20
1HR29R020150H	2	1500	8,86	62,5	125	31,71	25	24	8	5,7	2	7	11	7	22	225	1436,6	12	6,30
1HR29R020200H	2	2000	8,86	62,5	125	31,71	25	24	8	5,7	2	7	11	7	22	300	1936,6	16	8,41

Ball Screw (Z-Axis) TBI 20x5 C7 with Nut TBI SFU20x5

Nominal Model Code of Nut

G SFU R 025 05 T4 D + N3

G	SFU	R	025	05	T4	D	+	N3
①	②	③	④	⑤	⑥	⑦		⑧

①

Product Code

②

Nominal Model	
S	S : Single nut
	D : Double nut
F	F : With flange
	C : Without flange
U	NI : NI type nut
	NU : NU type nut
	H : H type nut
	NH : NH nut (A solution for slide table)
	Y : Y type nut
	V : V type nut
	U : DIN nut
	M : M type nut
	K : K type nut

③

Threading Direction
R : Right
L : Left

④

Nominal Diameter
Unit : mm

⑤

Lead
Unit : mm

⑥

Number of Turns (Turn·Row)
Turn : T : 1
A : 1.5 (or 1.7/1.8)
B : 2.5/2.8
C : 3.5
D : 4.8
ex : (2.5 × 2 = B2)

⑦

Flange Type
N : Not cutting
S : Single cutting
D : Double cutting

⑧

Nut Surface Treatment
S : Standard
B1 : Black Oxidation
N1 : Hard Chrome Plating
P : Phosphating
N3 : Nickel Plating
N4 : Raydent
N5 : Chrome Plating

C Ball Screw

■ 2-3-4 Preload of Rolled Ball Screw

The standard preloading for Rolled Ball Screw is P0. If P1 preloading is required, please contact ***TBI***

C
Ball Screw

2-1 Nominal Model Code of Ball Screw

SFU	R	025	05	T4	D	G	C5	-	600	-	P1	-	B2	+	N3	N3
①	②	③	④	⑤	⑥	⑦	⑧		⑨		⑩		⑪		⑫	⑬

①

Nominal Model	
S	S : Single nut
	D : Double nut
	O : OFF set double nut
F	F : With flange
	C : Without flange
U	NI : NI type nut
	NU : NU type nut
	H : H type nut
	NH : NH nut (A solution for slide table)
	Y : Y type nut
	V : V type nut
	U : DIN nut
	M : M type nut
	K : K type nut

②

Threading Direction
R : Right
L : Left

③

Nominal Diameter
Unit : mm

④

Lead
Unit : mm

⑤

Number of Turns (Turn·Row)
Turn : T : 1
A : 1.5 (or 1.7/1.8)
B : 2.5/2.8
C : 3.5
D : 4.8
ex : (2.5 × 2 = B2)

⑥

Flange Type
N : Not cutting
S : Single cutting
D : Double cutting

⑦

Product Code
G : Ground
F : Rolled

⑧

Accuracy Grade
C0, C1, C2, C3, C5, C7, C10

⑨

Overall Length of Shaft
Unit : mm

⑩

Axial Clearance and Preload Value
P0, P1, P2, P3, P4

⑪

Number of Nut
(Leave blank if only one nut is required) Ex : Install two nuts on a shaft B2

⑫

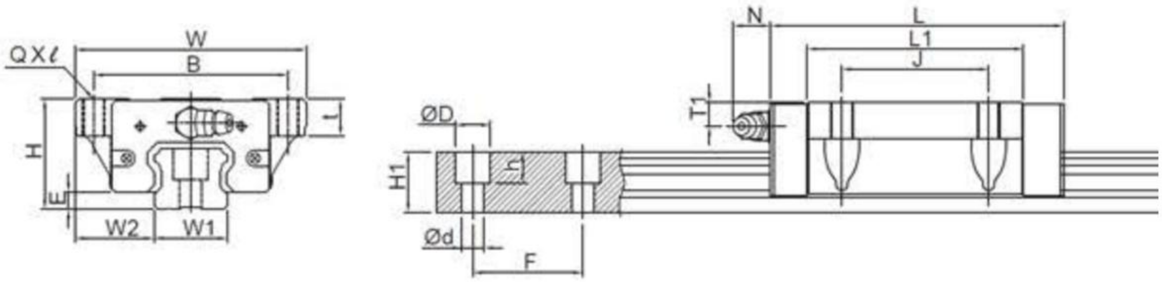
Nut Surface Treatment
S : Standard
B1 : Black Oxidation
N1 : Hard Chrome Plating
P : Phosphating
N3 : Nickel Plating
N4 : Raydent
N5 : Chrome Plating

⑬

Shaft Surface Treatment
S : Standard
B1 : Black Oxidation
N1 : Hard Chrome Plating
P : Phosphating
N3 : Nickel Plating
N4 : Raydent
N5 : Chrome Plating

※No symbol required when plating is not needed.
※An inspection report is provided for ground ball screws with an accuracy higher than C5.

Linear Guide Rails TBI TR25 with GUIDE BLOCKS TBI TRS25FN-H-Z1 – Used for all the axes of the machine.



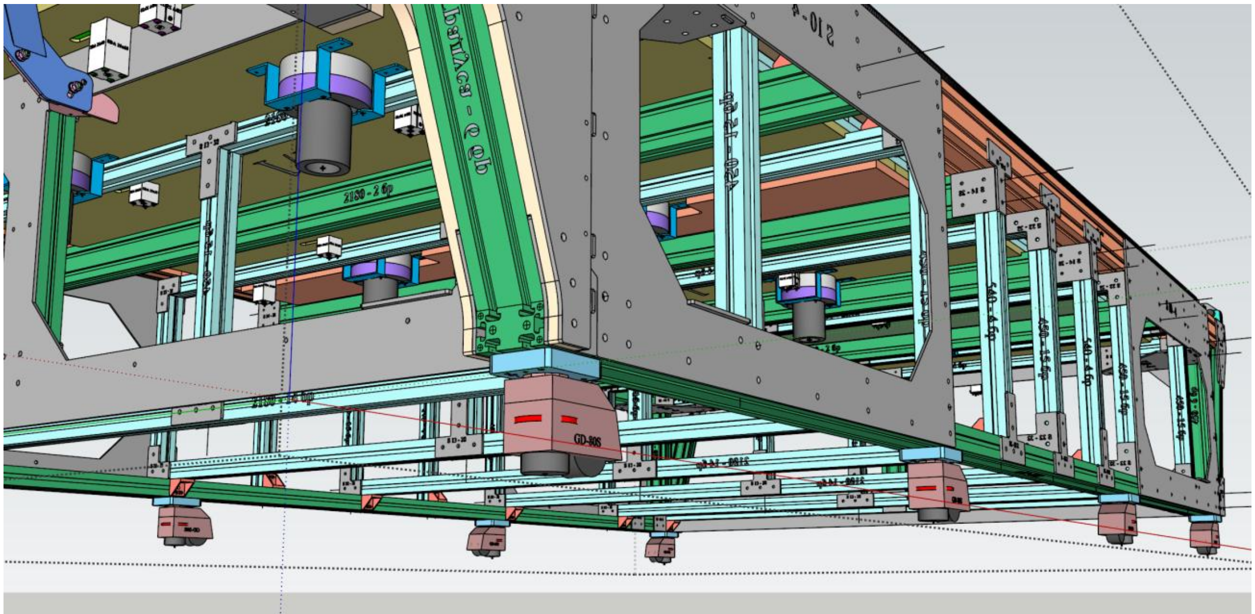
Model No.	Assembly (mm)			Block Dimension (mm)										Rail (mm)					
	H	W2	E	W	B	J	t	L	L1	QXℓ	T1	Oil Hole	N	W1	H1	ØD	h	Ød	F
TRH15FN	24	16	3.2	47	38	30	8	56.9	39.5	M5X8	5.5	M4X0.7	7	15	13	7.5	6	4.5	60
TRH15FL								65.4	48										
TRH20FN	30	21.5	4.6	63	53	40	10	75.6	54	M6X10	6.5	M6X1	14	20	16.5	9.5	8.5	6	60
TRH20FE								99.6	78										
TRH25FN	36	23.5	5.8	70	57	45	12	81	59	M8X12	7.5	M6X1	14	23	20	11	9	7	60
TRH25FE								110	88										

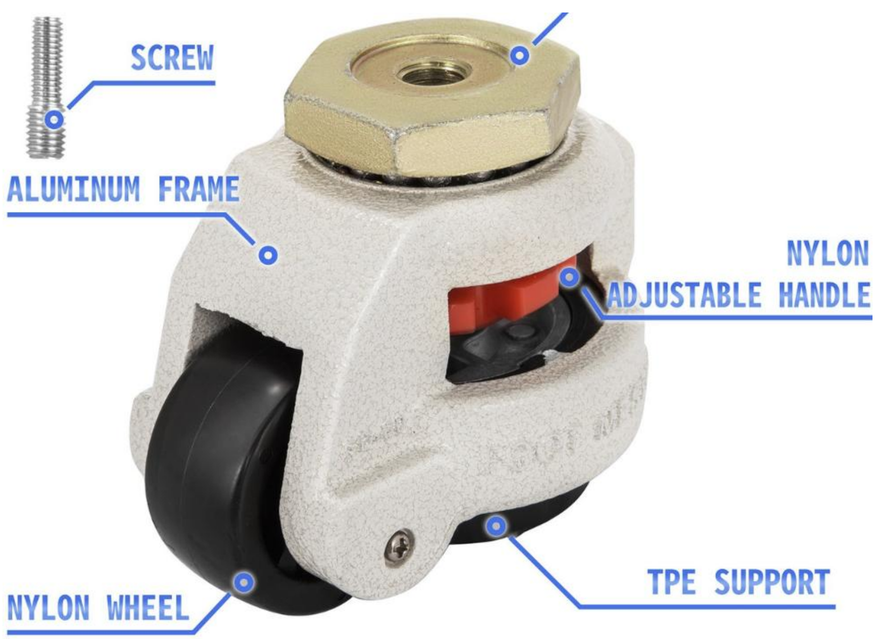
8. Adjustment Capabilities

A distinguishing feature of the StratoCNC is its comprehensive built-in adjustment provisions for all three axes. These adjustment capabilities allow the machine to be precisely aligned during initial commissioning and maintained to specification throughout its operational life.

Machine Levelling

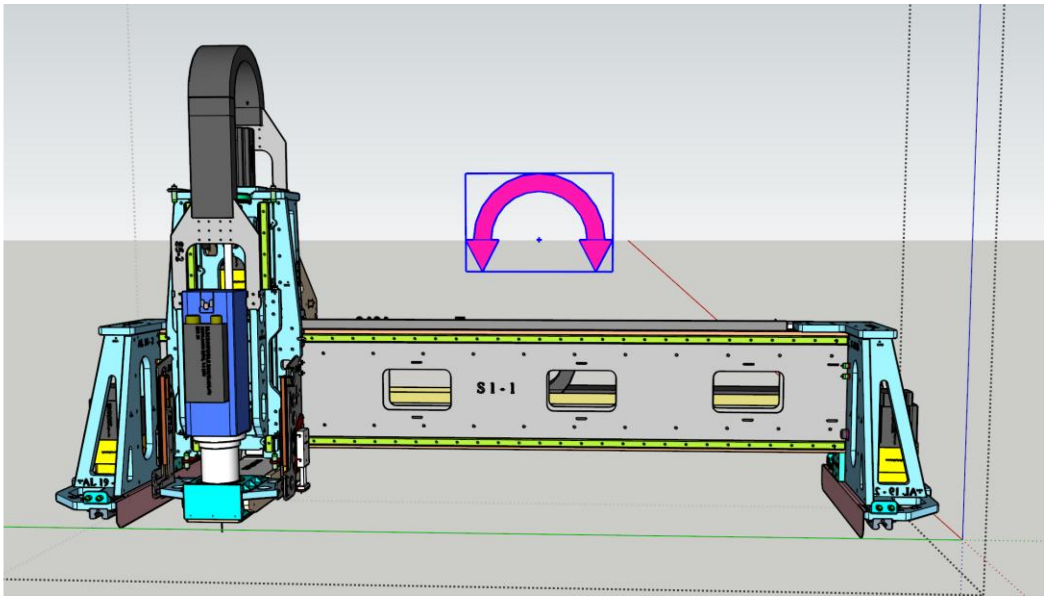
The whole machine structure is sitting on 8 heavy-duty industrial leveling casters **GD-80-S** with up to 15 mm height adjustment capability and 500 kg max loading each.

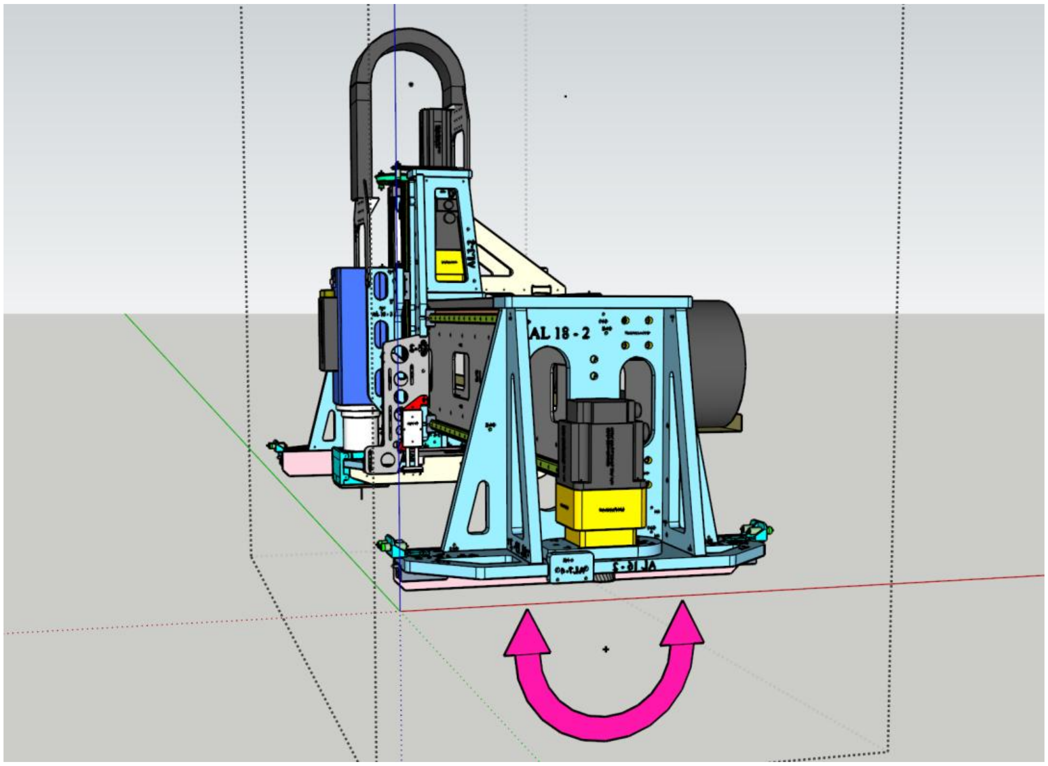


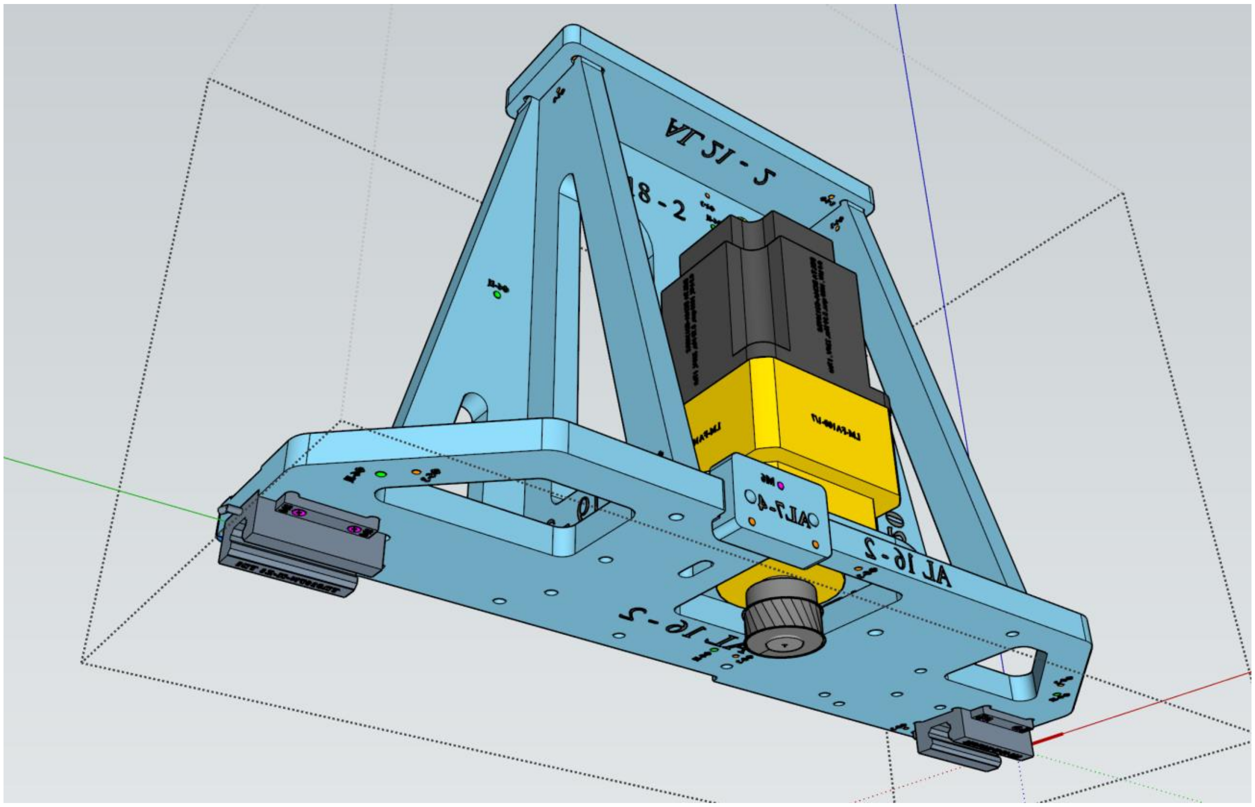


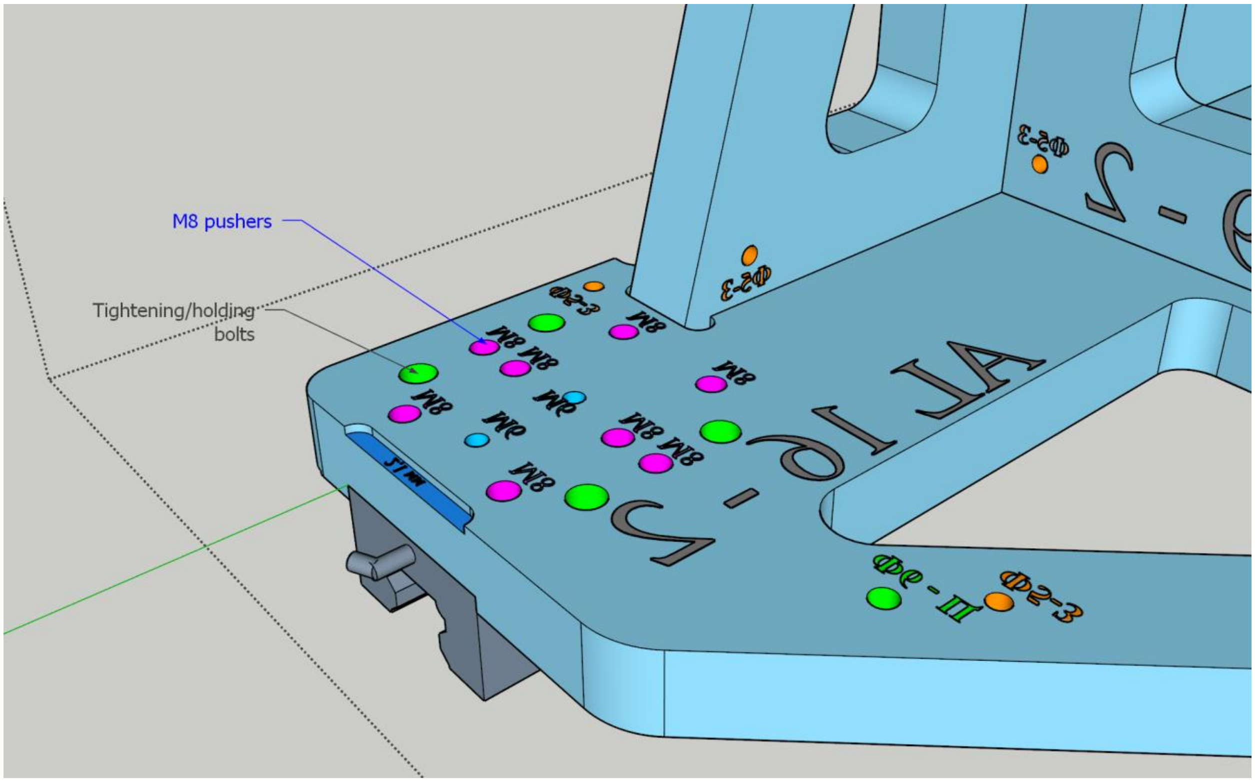
Gantry Assembly Levelling

The gantry assembly rides on eight linear guide blocks. Each guide block is secured to its aluminium side cart with 4 bolts, and each cart includes 8 M8 pusher bolts per block for height adjustment. By selectively tightening the pusher bolts while loosening the holding bolts, each of the eight mounting points can be raised or lowered independently by up to 5 mm. This provides full control over gantry levelling in both the X and Y directions.



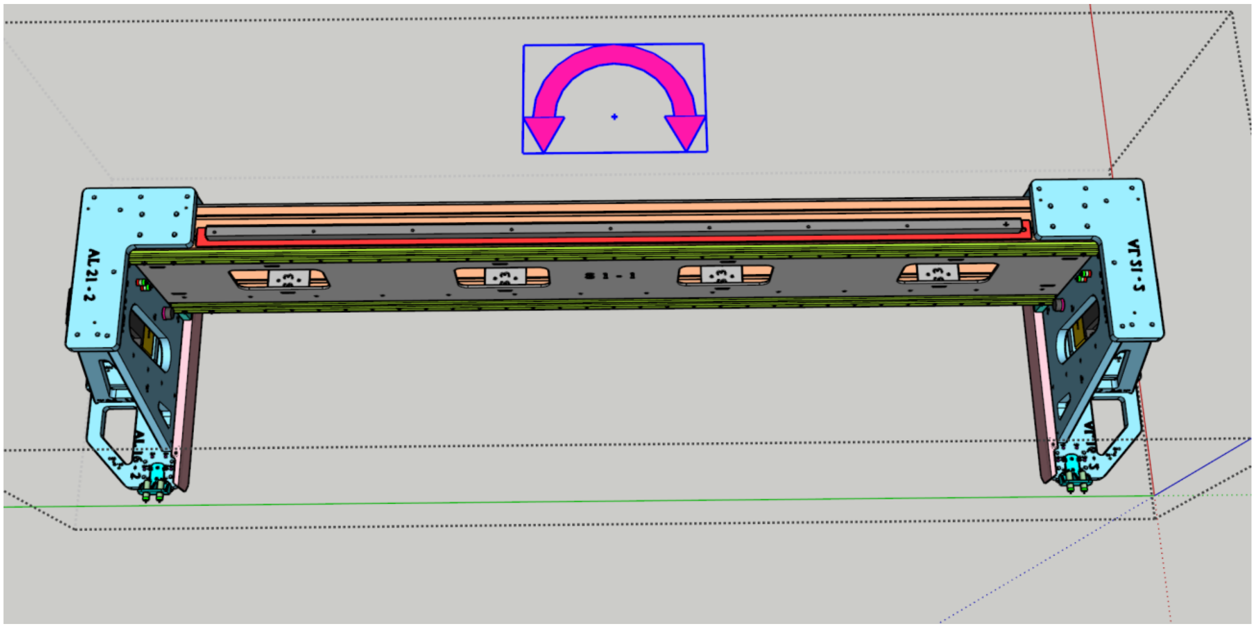




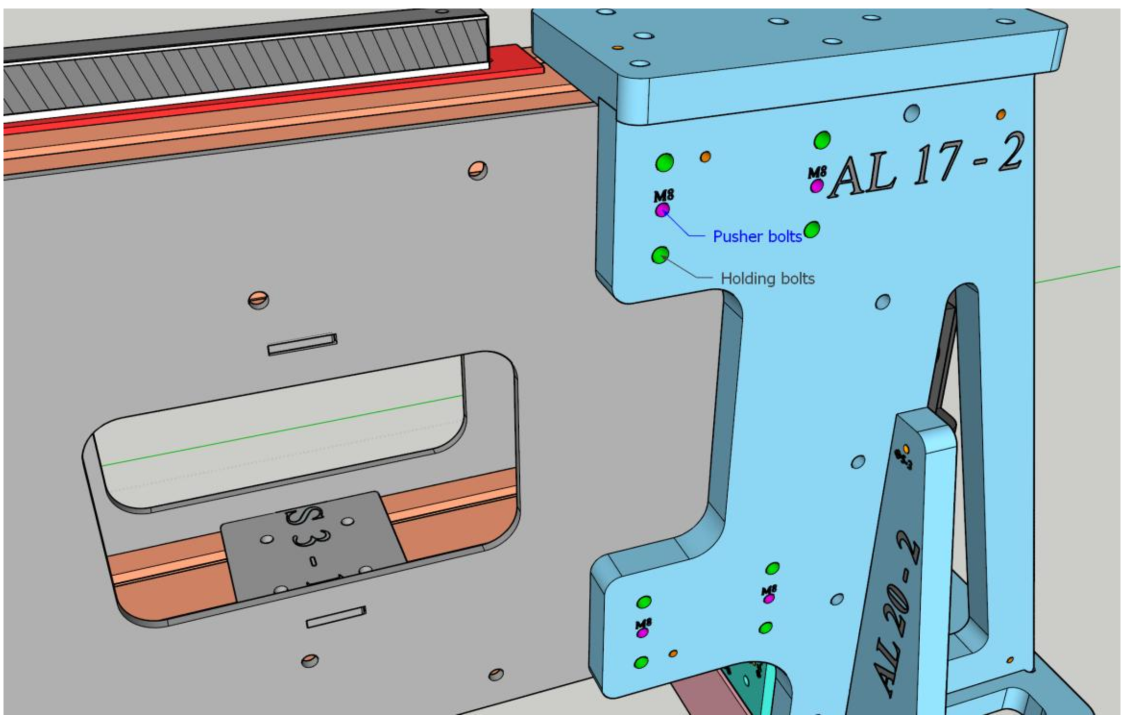

M8 pushers
Tightening/holding bolts

Gantry Squaring

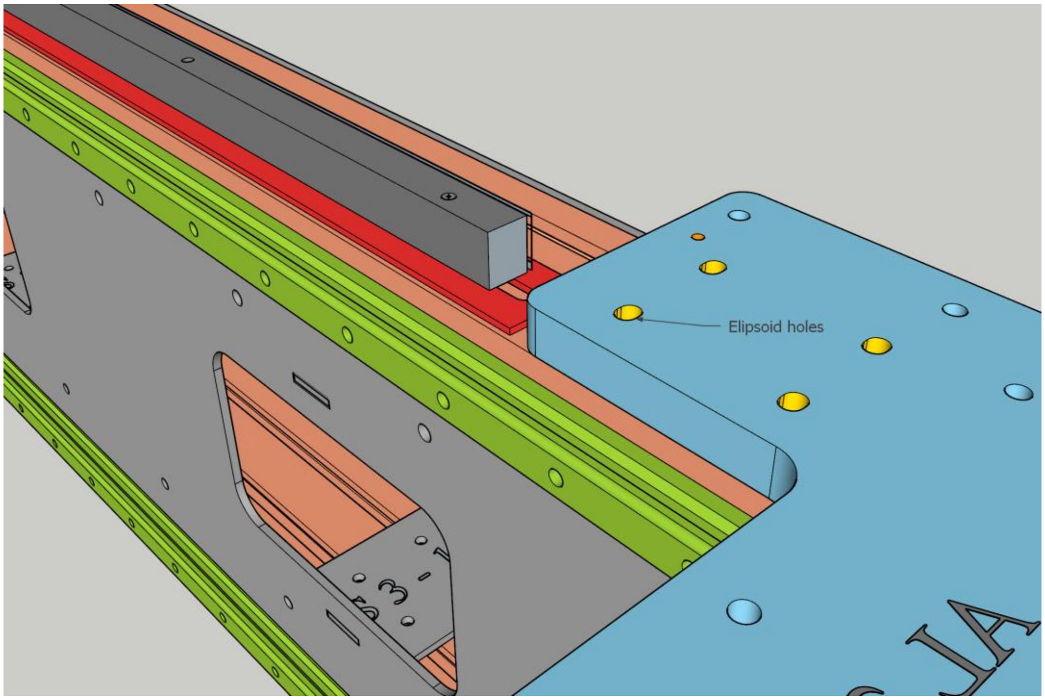
Ensuring that the gantry beam is precisely perpendicular to the side guide rails (X-axis perpendicular to Y-axis) is one of the most critical alignment tasks. The StratoCNC provides mechanical provisions for this adjustment on both side carts.



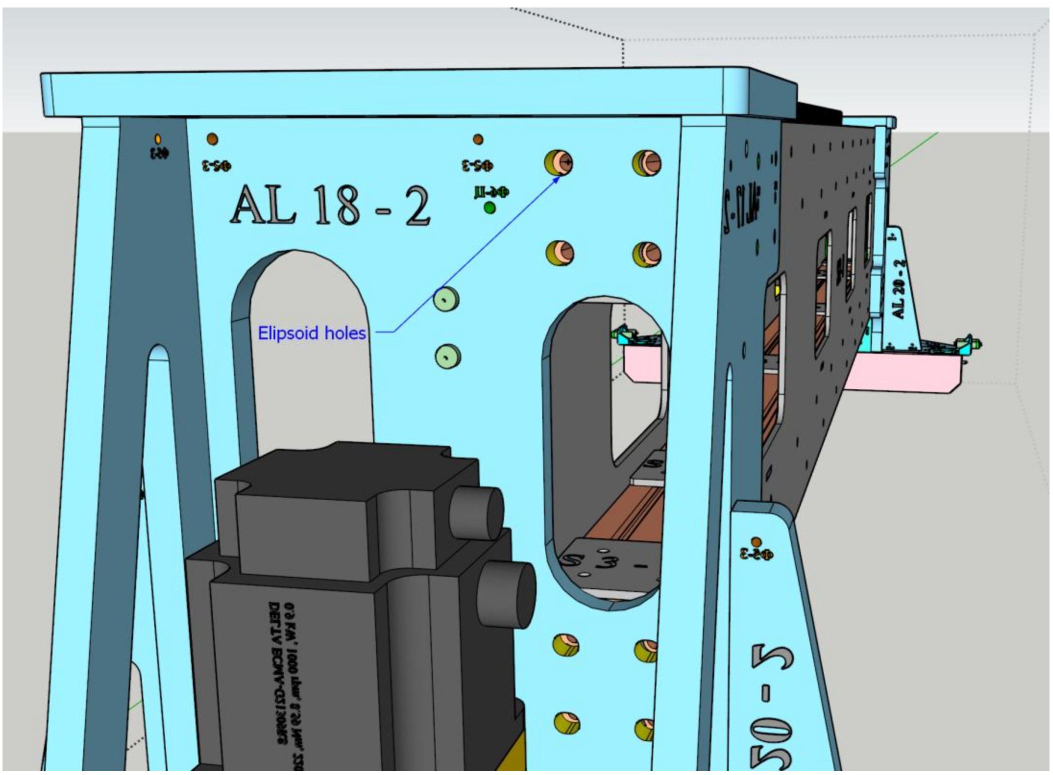
In StratoCNC, the gantry itself is mounted to each of the two carts with 20 bolts M8. Eight of these bolts are positioned on the back side of the cart. Alongside these, four PUSHER M8 bolts are used to push the gantry away from the back of the cart. The beam beds in the carts are 2 mm wider than the beam, effectively allowing for 2 mm play of each of the gantry sides.

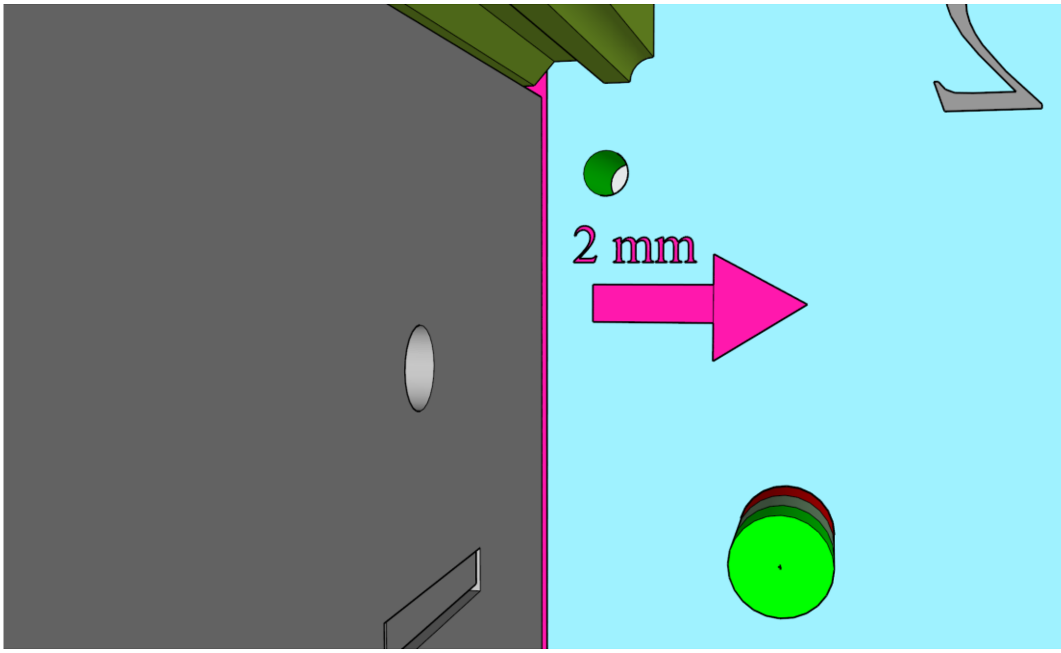


The holes for the four holding bolts on the top of the cart has ellipsoidal shape, allowing for the bolts and the gantry itself to move forward/backward.



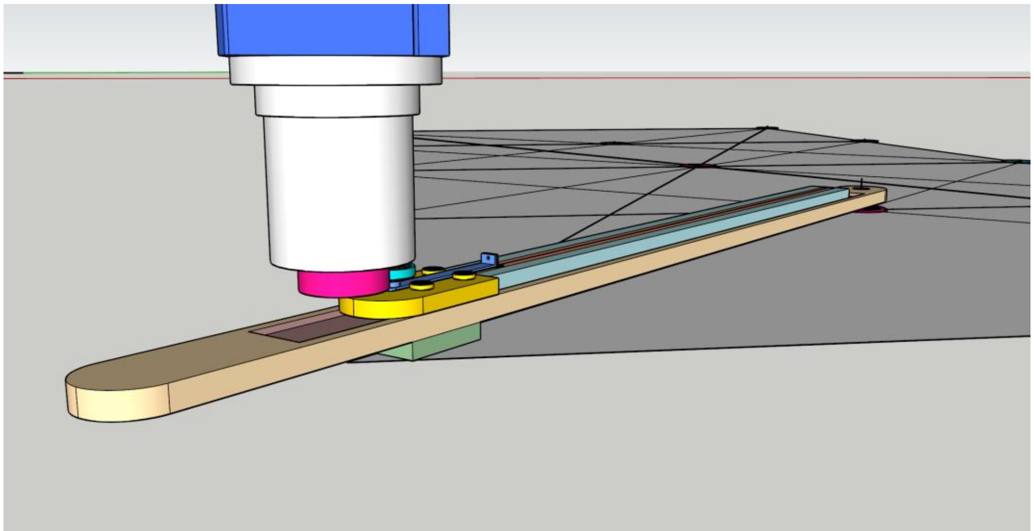
The holes for the holding bolts on the side of the cart has ellipsoidal shape, allowing for the bolts and the gantry itself to move forward/backward.





The **squaring procedure** – untighten all the 20 holding bolts of both sides of the gantry, tighten the 8 pusher bolts on the lagging side to move it the required distance (max 2 mm possible), then tighten all the bolts on both sides.

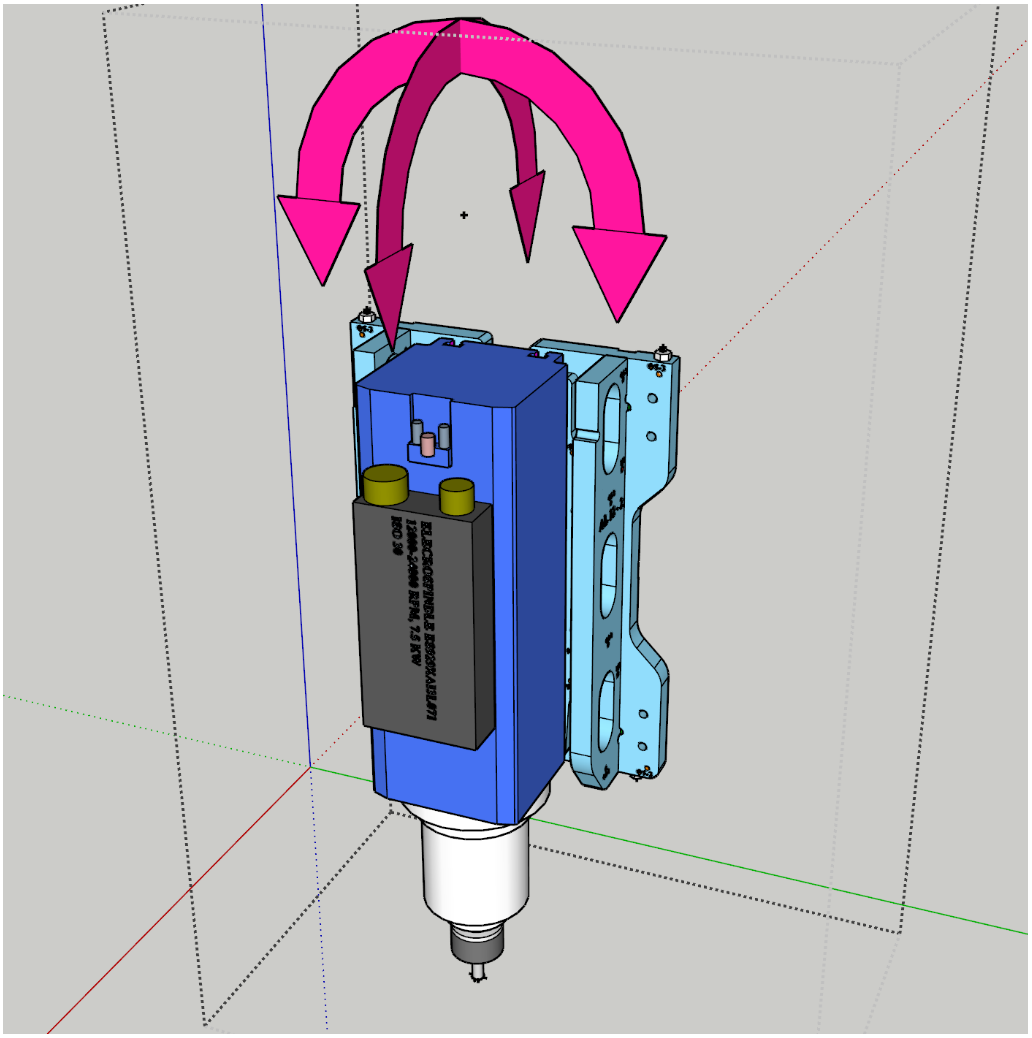
To determine if the gantry needs squaring and how much exactly, SFERRI LTD has developed a unique and proprietary measuring routine called **THE HALF-DIAGONAL BEAM COMPASS METHOD**.



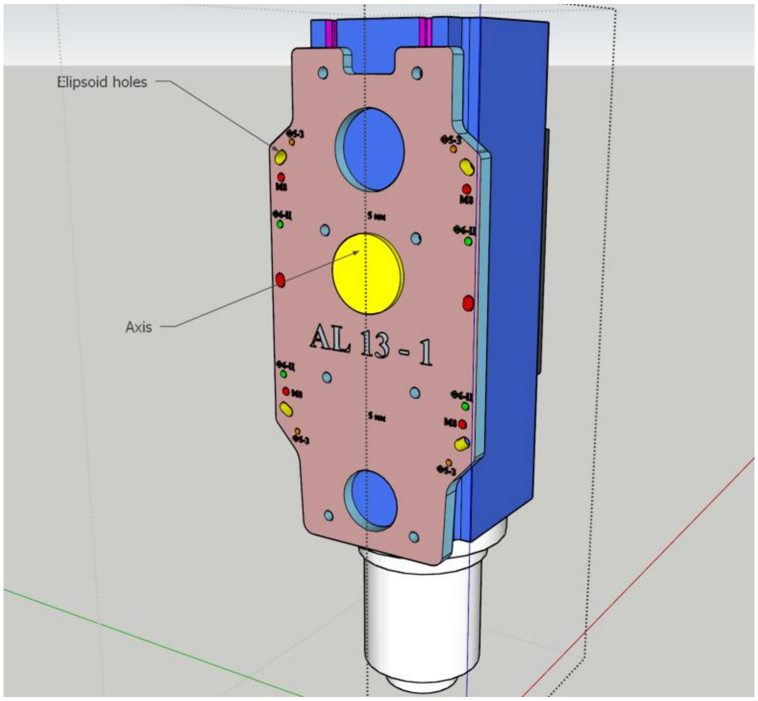
The combination of measuring and adjusting ensures the mechanical perpendicularity of the gantry. During the daily operations, the StratoCNC uses a special homing and squaring macro M20301 with a dedicated UCCNC screen button, which can determine small deviations from the perpendicularity and fine-tune it with the help of the servo motors, the homing proximity switches, and the UCCNC. The system will not allow normal operations unless the homing and squaring procedure is completed.

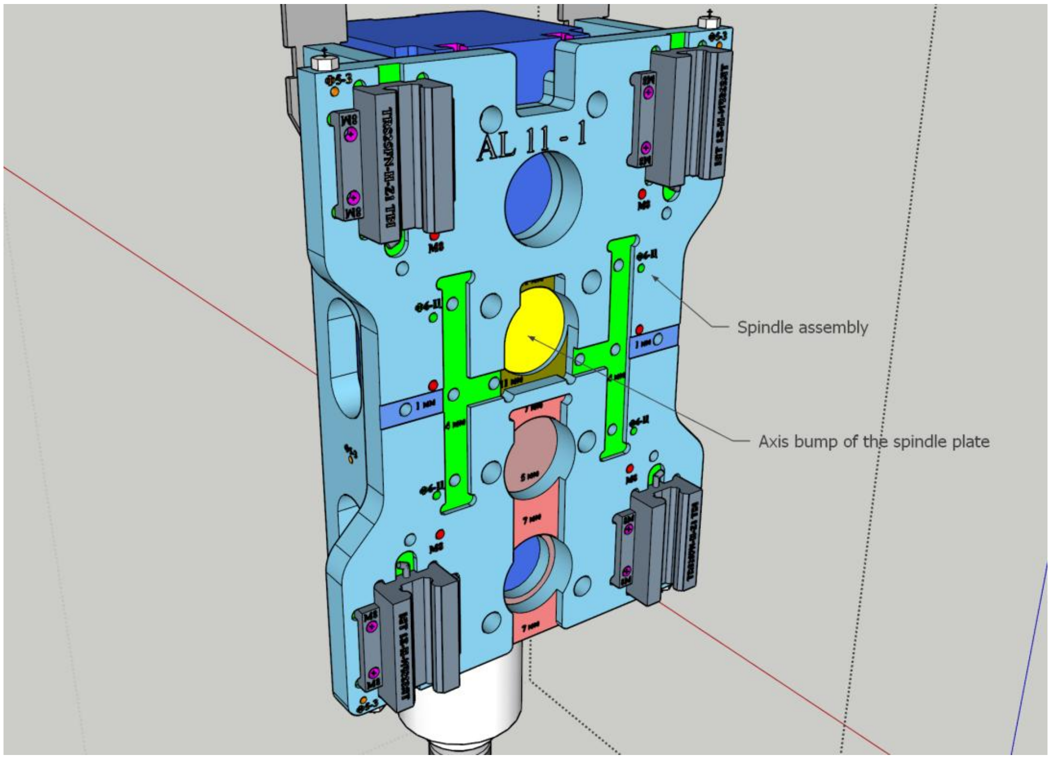
Spindle Tramming

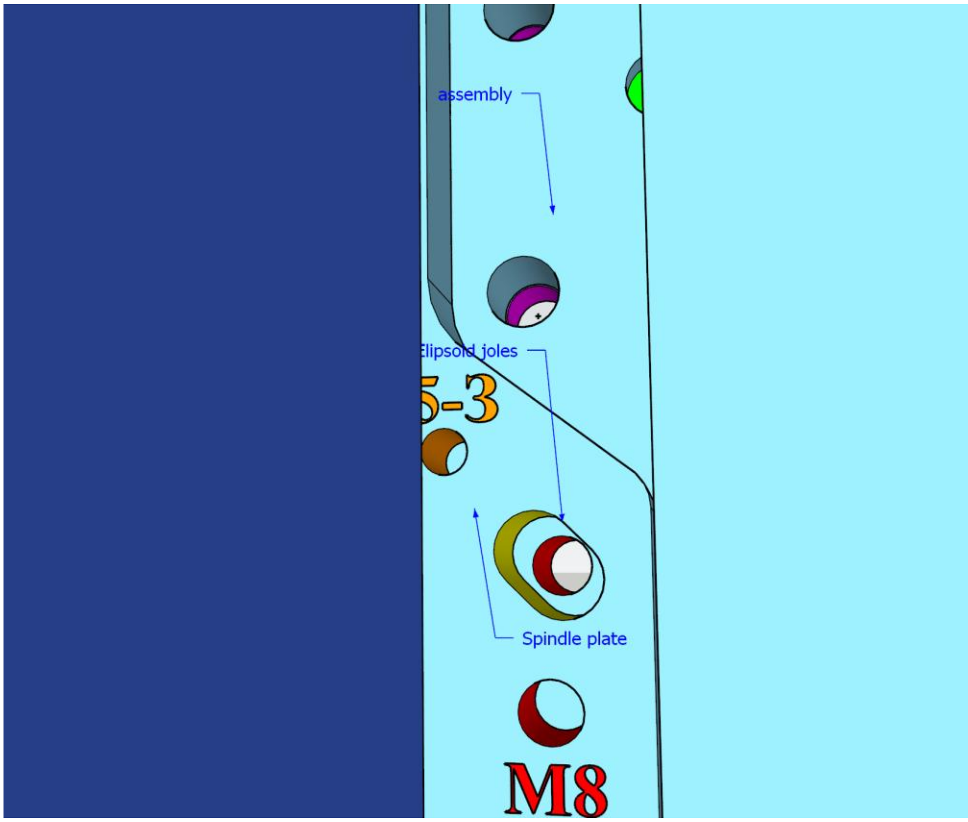
The spindle assembly includes provisions for adjusting spindle perpendicularity to the work surface in both the X-axis (side-to-side tilt) and Y-axis (front-to-back tilt) planes.



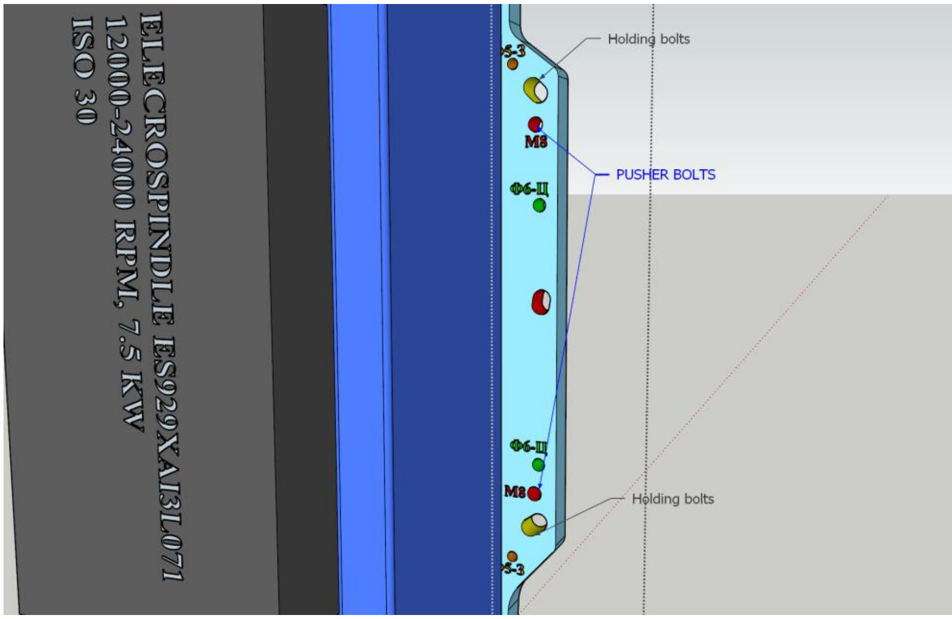
For rotating the spindle **sideways**, there is an axis bump machined on the back of the mounting aluminum plate. The holding bolts of the plate are ellipsoids and are positioned in such a way that the plate and spindle can rotate around the axis bump. The axis bump is inserted in a circular bed in the back of the spindle assembly. So the procedure is to to untitghten the four holding bolts in the ellipsoid holes and rotate the spindle with the plate around the axis bump, then tighten the bolts again.







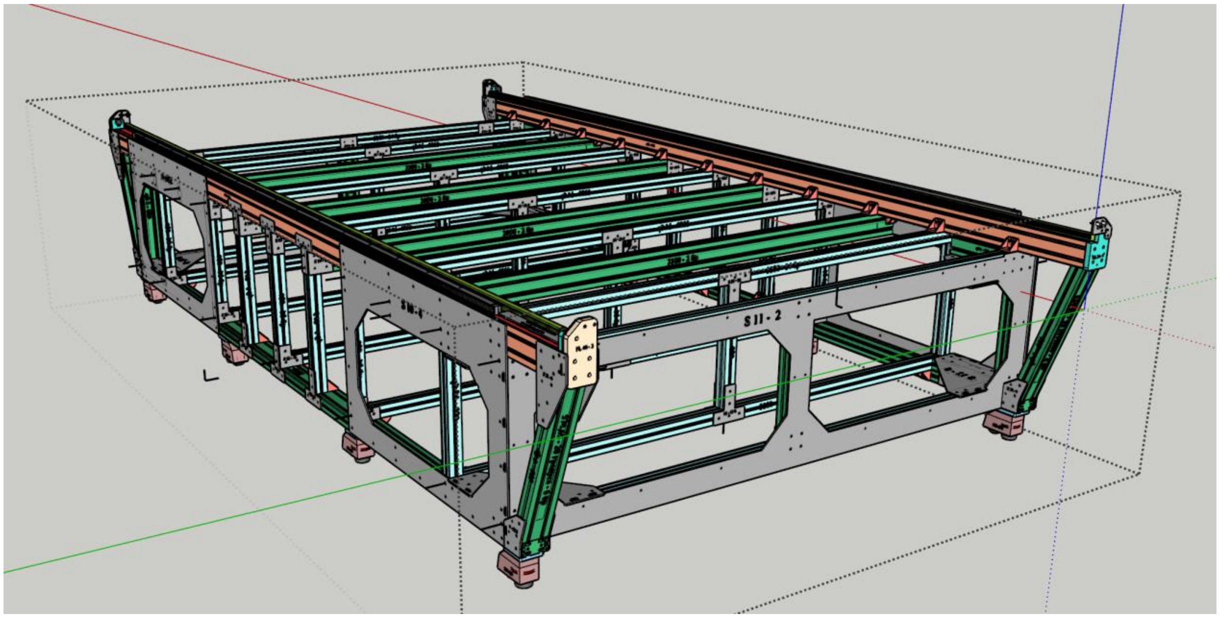
For rotating the spindle **forward/backward**, there are four M8 PUSHER bolts on the four corners of the spindle back-plate. By untightening the four holding bolts in the ellipsoid holes and tightening the two upper or the two lower pusher bolt the spindle can be adjusted vertically. All the bolts are tightened afterwards.

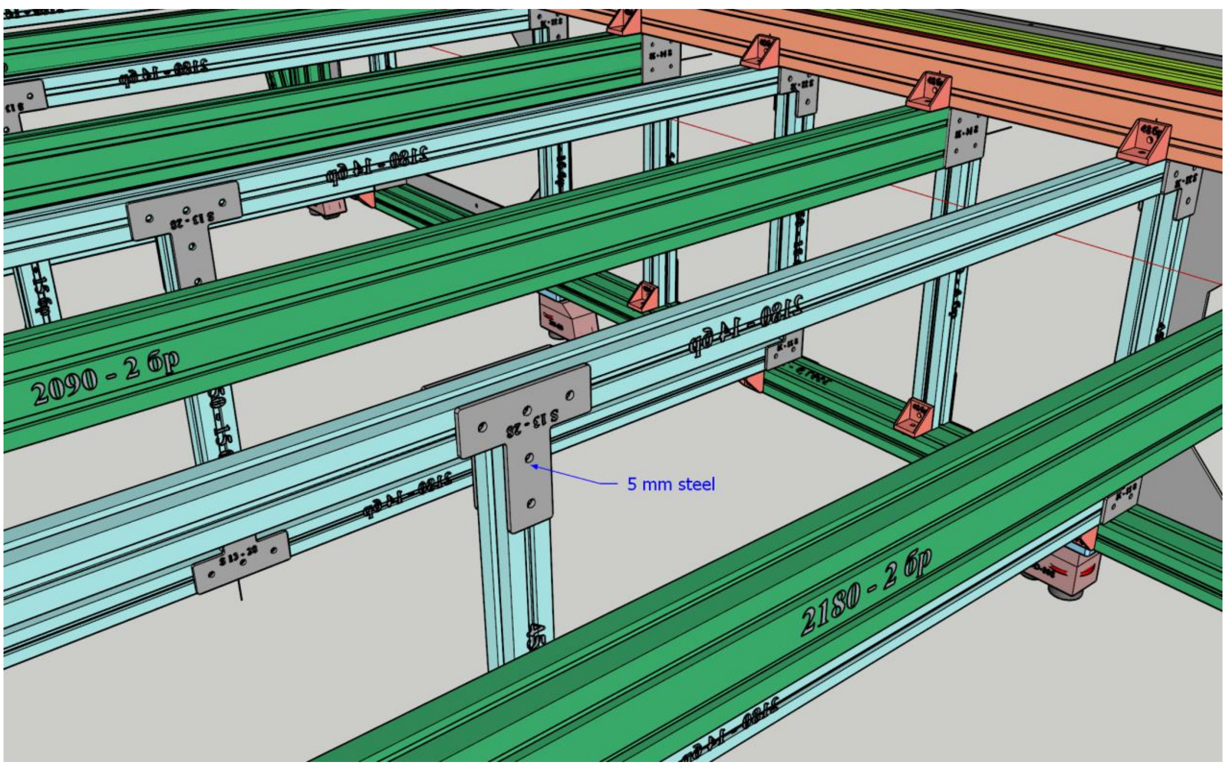


9. Structural Design

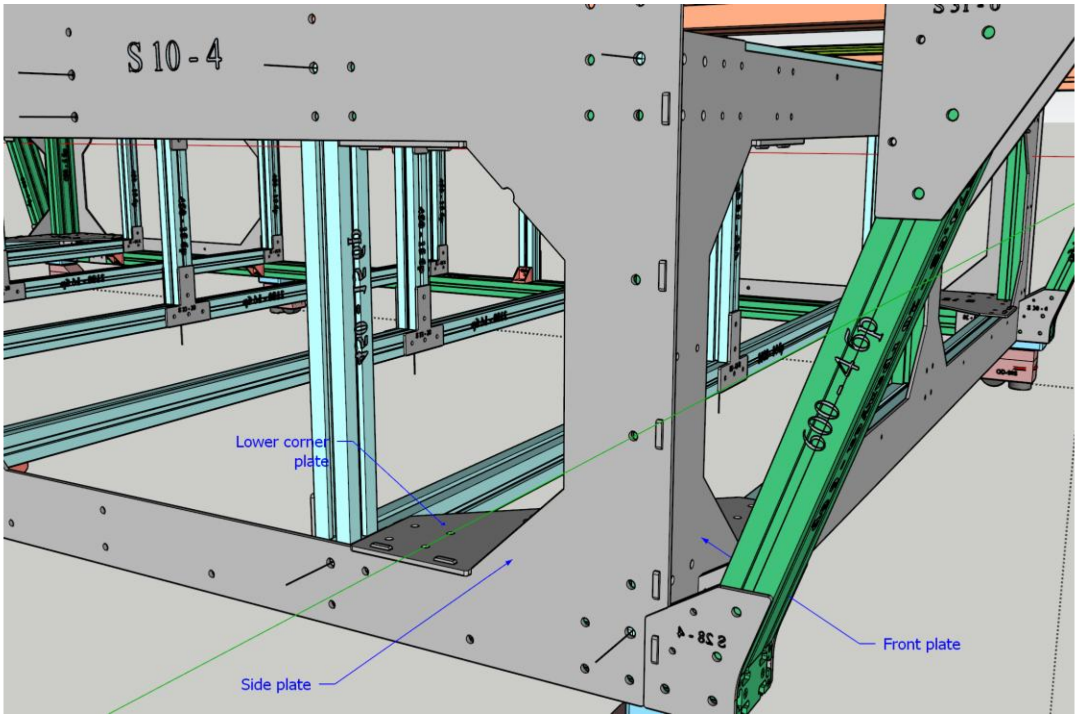
Main Structure

The main structure of the machine is made with aluminium extrusions 90x90, 90x45, 45x45, and steel plates, and planks with 5 mm thickness. All the 90x90 and 90x45 extrusions are filled with epoxy granite for vibration dampening and added stiffness of the structure.



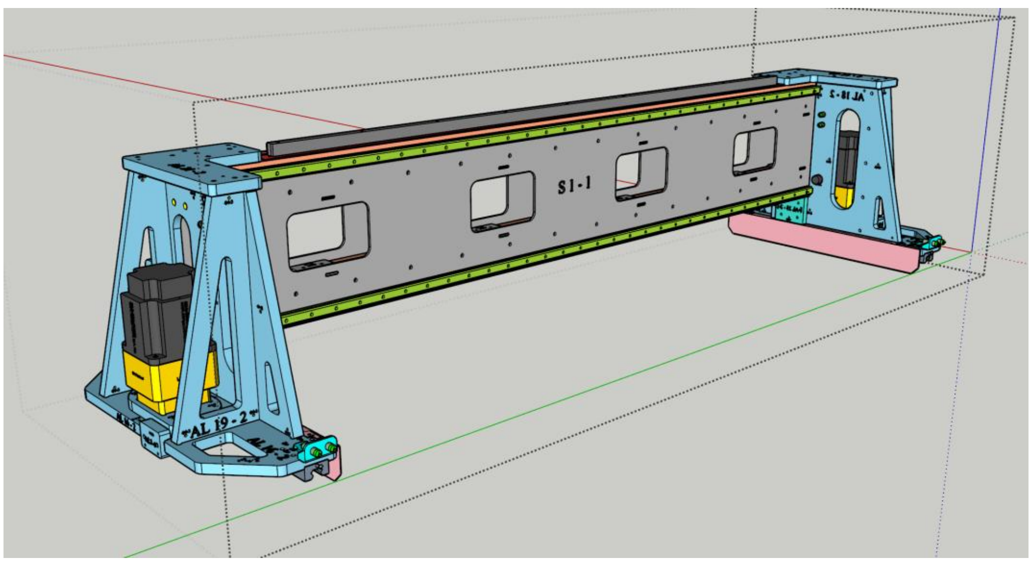


The steel plates are laser-cut and are for ensuring three-dimensional precision and stiffness of the overall structure. In each of the 8 corners of the three-dimensional parallelepiped, the steel plates form a box-like shape with thongs and corresponding positioning grooves in the mating elements for precise positioning and assembly.

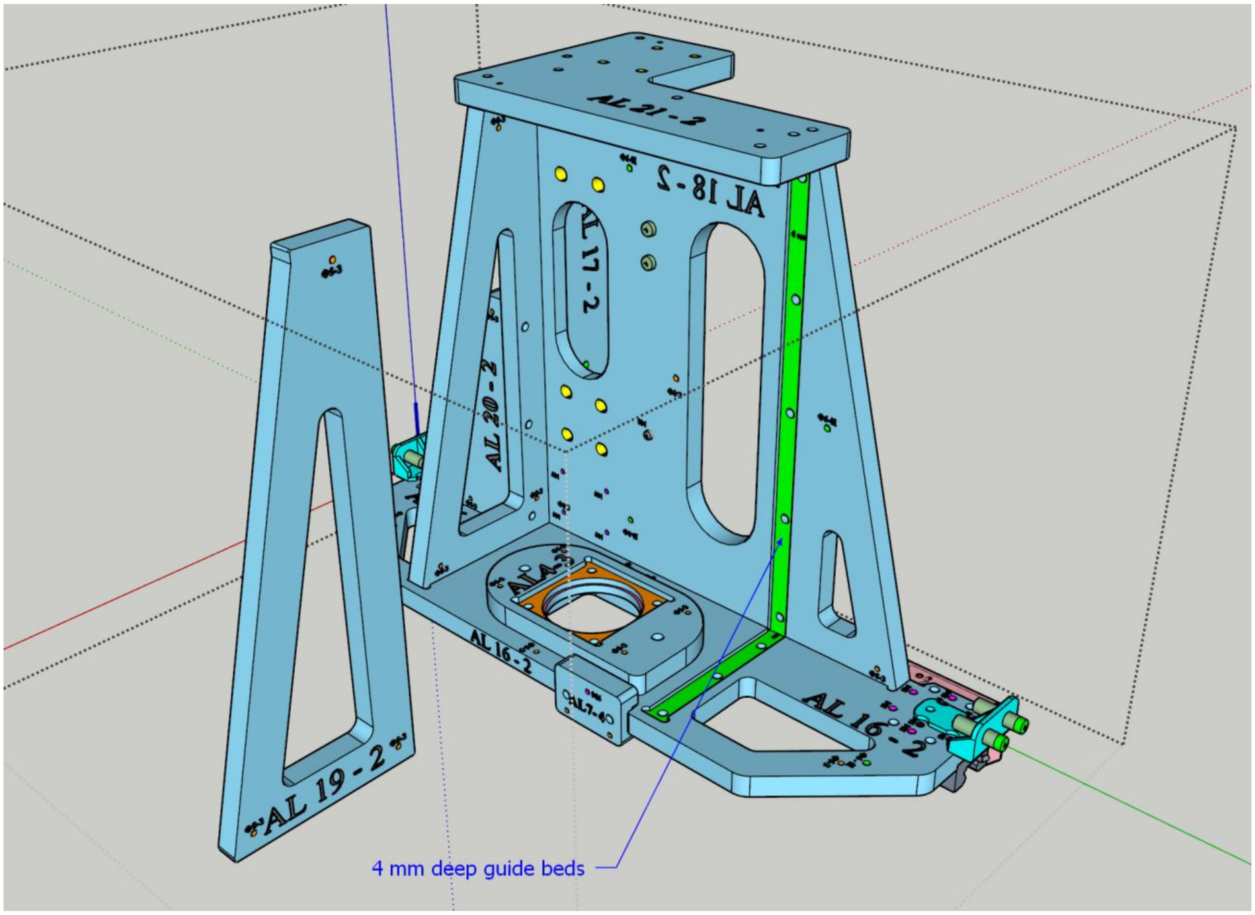


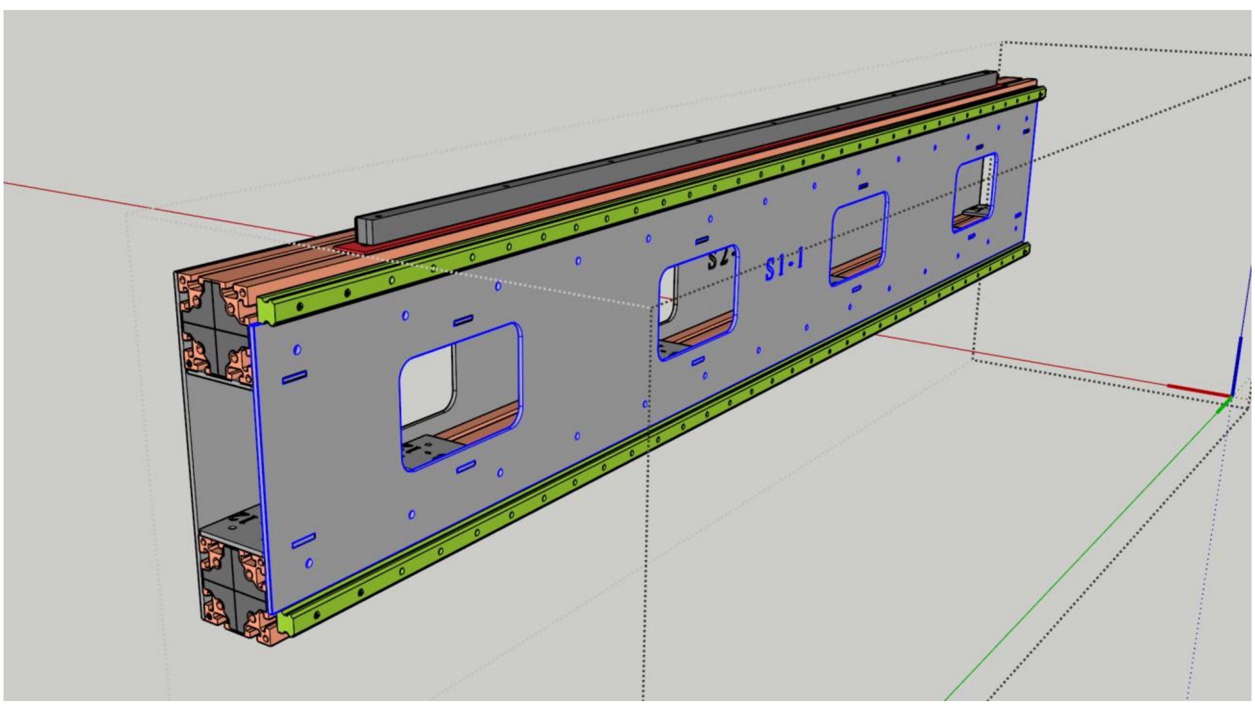
Gantry Assembly

The gantry assembly consists of two aluminium side carts and a gantry beam. Each cart is fabricated from 20 mm-thick aluminium elements forming a box structure, with machined guide beds in mating surfaces to ensure assembly precision and structural rigidity.



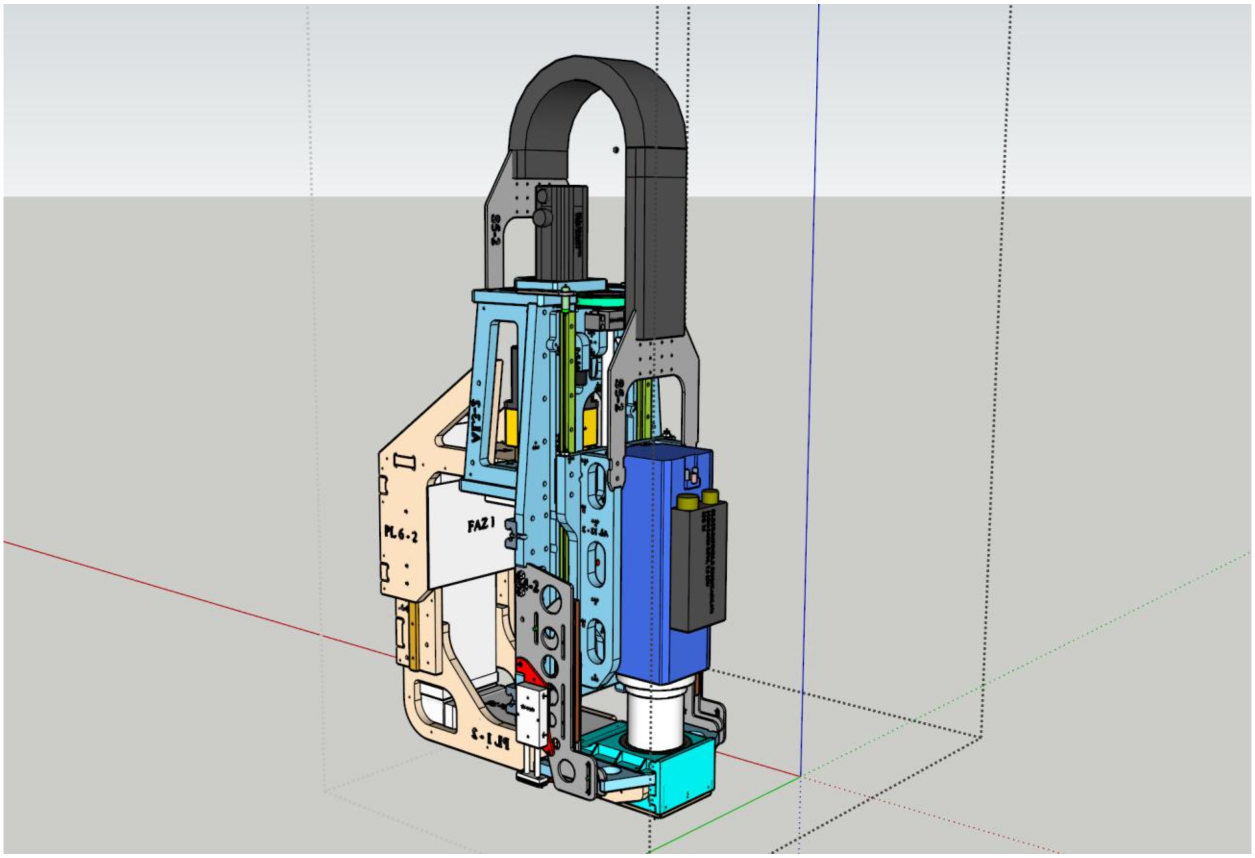
The gantry beam is made from two 90x90 extrusions filled with epoxy granite and two 5mm-thick steel plates (back and front), and some additional positioning steel planks.





Spindle Assembly

The spindle assembly follows the same construction philosophy as the side carts, using 20 mm-thick aluminium elements in a box configuration with machined guide beds. Some additional 5 mm steel elements provide supplementary rigidity. The assembly houses the HSD spindle, the Z-axis servo motor and timing belt transmission, the dust collector mounting, and the spindle tramming adjustment hardware.

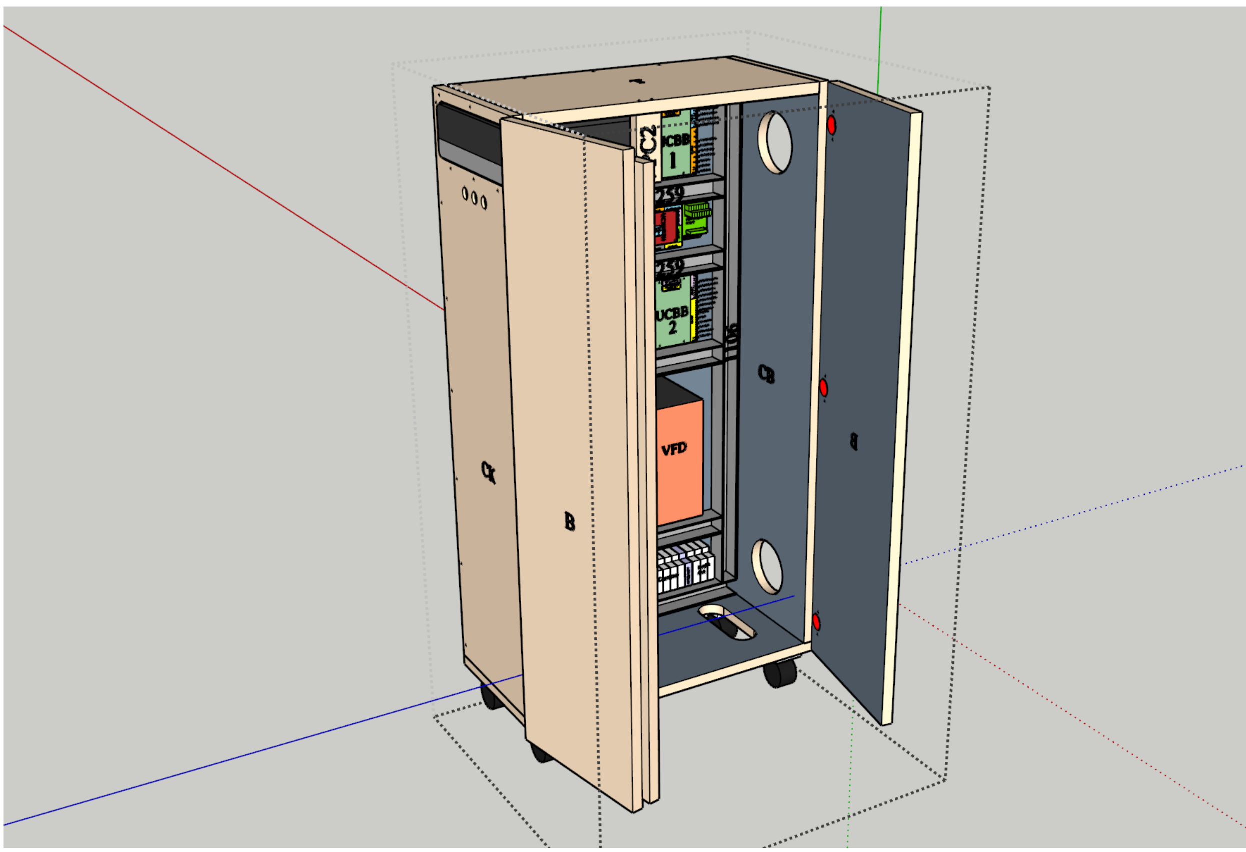


10. CONTROL CABINET

All the electronic components (the main PC, the Servo drives, the VFD, the UC300ETH controller, the break-out boards) together with the Circuit breakers, EMI filters, cables, busses, 24VDC power supply, and 5VDC power supply are housed in a standalone cabinet.

The cabinet is wide and spacious for easy installation, good airflow between the components, and is fitted with two powerful fans and a filter for providing a constant volume of purified cooling air.

All the wiring in the cabinet is made with high-quality SHIELDED cables and the extensive use of FERRITES. The wiring management was carefully designed not to put the sensitive signal cables close to the notoriously noise-emitting VFD and power cables.



11. STANDARD OPERATING PROCEDURES

“Start of the Day” Routine – Powering On, Homing, Squaring, Warming Up

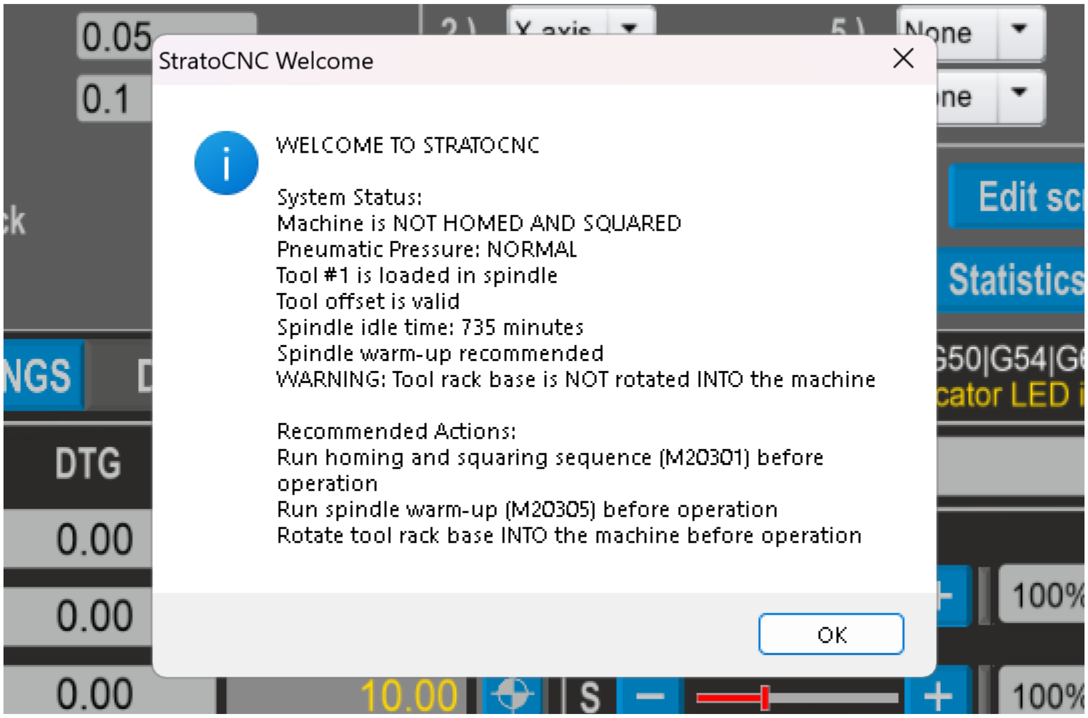
This section describes the recommended daily startup sequence for the StratoCNC. Following this procedure ensures the machine is properly initialized, all safety systems are active, and the spindle is ready for production work.

Step 1 – Power On the System

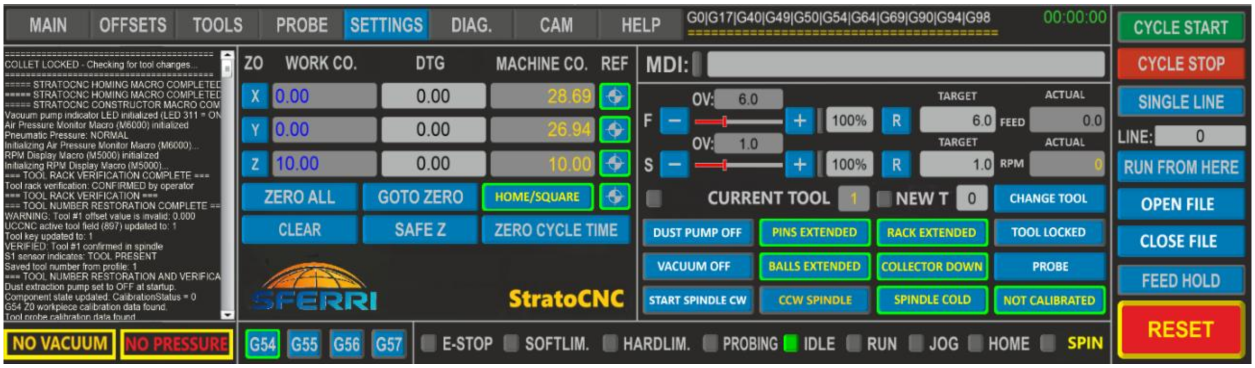
Turn on the main power to the control cabinet using the GREEN BUTTON on the side of the cabinet. Verify that the air compressor is running and has reached adequate operating pressure (check the pressure regulator and ensure the gauge reads at least 6 bar). Turn on the Control PC and allow it to boot fully. Wait at least 3 minutes after power-on for the UC300ETH motion controller to initialize its communication network and establish a stable Ethernet link with the PC. You can verify the controller is ready by checking the green power LED and the blue communication LED on the UC300ETH board - both should be lit.

Step 2 – Launch UCCNC

Start the UCCNC software. On startup, the Constructor Macro M99998 runs automatically. It initializes all system states, synchronizes the on-screen button indicators with the actual physical positions of all pneumatic components (tool rack, dust collector, collet, pins, ball bearings), and starts all background monitoring loops. A welcome message will appear in the status bar summarizing the machine status and recommending the next steps.



After the M99998 is done and you press OK on its dialog box, you will see many flashing buttons indicating some procedures should be done and many static buttons in yellow/green indicating the physical state of the different machine components (Tool Rack, Positioning Pins, Transport Ball Bearings, Dust Collector, etc.). You cannot start normal operations until ALL the blinking buttons have been extinguished and ALL the machine components have been retracted. There are also indications about the lack of vacuum and the lack of pneumatic pressure in the lower left corner of the screen.



The big yellow **[RESET]** button in the lower right corner is also flashing in red. To continue with the sequence, press the **[RESET]** button and exit the reset state of the machine, which is the default one on start-up. The button would change to its non-flashing yellow state, and you would be able to continue with the next step:

Step 3 – Home and Square the Machine

Press the flashing **[HOME/SQUARE]** button on the UCCNC screen. This triggers macro M20301, which performs the complete homing and gantry squaring sequence. The system first checks that the machine is idle, the spindle is stopped, and all pneumatic components are in safe positions. It then homes the Z axis first (moving upward), followed by the X axis (moving left), and finally the Y axis (moving toward the front of the machine). Each axis uses a two-stage homing process: a fast approach to the limit switch, a short back-off, and then a slow precision approach for accurate home position. After all three axes are homed, the system performs gantry squaring by independently driving each side of the Y axis to the front limit switches, ensuring the gantry beam is precisely perpendicular to the guide rails. The Home/Square button will stop flashing and will change its state to a white **[HOMED/SQUARED]** when the procedure completes successfully.

WARNING: The system will not allow normal operations (cycle start, tool changes, or probing) until the homing and squaring procedure is completed. Always home the machine FIRST before attempting any other operations with the machine.

Step 4 – Warm Up the Spindle

Press the flashing **[SPINDLE COLD]** button on the UCCNC screen. This triggers macro M20305, which runs a controlled multi-stage warm-up routine, gradually increasing the spindle RPM over several stages. The warm-up sequence follows the HSD spindle manufacturer's recommendations for extending bearing life and ensuring thermal stability. If the spindle was recently warmed (the system tracks warm-up timestamps), the macro will skip the warm-up and inform you accordingly. When the warm-up procedure is completed, the button changes to its white **[SPINDLE WARM]** state and stops flashing to indicate the spindle is warm.

Note: The system keeps track of when the spindle is running and for how long. If a period of 1 hour of idle time is detected, the system flags the spindle as COLD and changes the button back to its flashing **[SPINDLE COLD]** state. Now you have to run the warm-up cycle again by pressing the button before starting using it. Running a cold spindle under heavy load can reduce bearing life.

Step 5 – Calibrate G54 Z0 and the Tool Probe (if needed)

If you are working with a new workpiece thickness or if the workpiece surface Z0 needs to be re-established, press the **[CALIBRATED/NOT CALIBRATED BUTTON]** to run macro M20303. This macro performs two tasks in a single streamlined workflow: first, it measures the G54 Z0 (the top surface of the workpiece) using the touch plate and Tool N1 (Tool N1 is the only tool allowed to be used with the M20303 macro - it is the REFERENCE Tool). If another tool is in the spindle – change it to Tool 1 and continue with the M20303.

Second - if needed, it calibrates the Tool Length Probe by moving to the probe location and measuring its exact Z position relative to the machine and workpiece zeros. Once the tool probe has been calibrated, the button changes to its white **[CALIBRATED]** state and stops flashing, indicating calibration has been completed and the probe is good for measuring the other tools.

Step 6 – Retract all the machine components before starting a cycle

Press the green/yellow **[RACK EXTENDED]** button to retract the Tool Rack. When completed, the button will change to its white state **[RACK RETRACTED]**.

Press the green/yellow **[PINS EXTENDED]** button to retract the Positioning Pins. When completed, the button will change to its white state **[PINS RETRACTED]**.

Press the green/yellow **[BALLS EXTENDED]** button to retract the Transport Ball Bearings. When completed, the button will change to its white state **[BALLS RETRACTED]**.

Press the green/yellow **[COLLECTOR DOWN]** button to lift the Dust Collector together with the integrated Spring Pusher. When completed, the button will change to its white state **[COLLECTOR UP]**.

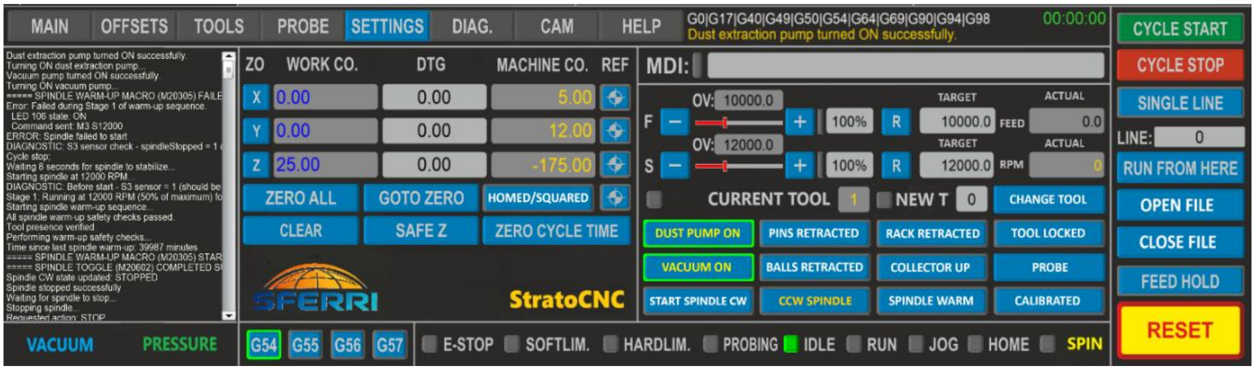
Step 7 – Start the Vacuum Pump, the Dust Extraction Pump, and the Compressor

Press the white **[VACUUM OFF]** button to switch on the VACUUM pumps. When completed, the button will change to its green/yellow state **[VACCUM ON]**, and the yellow warning lamp NO VACCUM will change to its blue VACUUM state.

Press the white **[DUST PUMP OFF]** button to switch on the Dust Extraction Pump. When completed, the button will change to its green/yellow state **[DUST PUMP ON]**.

Switch on the compressor and open the main pneumatic shut-off valve. The red warning lamp NO PRESSURE will change to its green PRESSURE state.

And NOW the system is initialized and READY TO GO. The screen should look like this for normal operations:



The **[RESET]** button is yellow and not flashing – the system is out of RESET mode and ready.

All the buttons are in their white and non-flashing states, indicating all the procedures have been completed and all the components have been retracted.

The warning lamps in the lower left corner are blue VACCUM and green PRESSURE, indicating there is vacuum and pneumatic pressure available.

There are two yellow/green buttons – **[DUST PUMP ON]** and **[VACUUM ON],** indicating the respective pumps are running and ready for operations.

The work coordinates in the DRO are X0 Y0 Z25 – the machine is positioned with the center of the tool's face exactly at 25 mm over the left front upper corner of the loaded sheet of material.

(At this position, the machine coordinates should indicate X5 Y12 and some negative value for Z)

Loading a Sheet on the Table

The StratoCNC is designed for efficient handling of large sheet materials (up to 1525 x 1525 mm plywood sheets). The integrated ball bearing transport system and pop-up alignment pins make loading quick and precise.

Step 1 – Extend the Ball Bearings

Press the white **[BALLS RETRACTED]** button on the UCCNC screen (macro M20740) to extend the pneumatic ball bearing rollers. The rollers rise above the table surface, allowing sheets to be slid easily across the table with minimal effort. Verify the button changes its state to the yellow/green **[BALLS EXTENDED]**.

Step 2 – Extend the Positioning Pins

Press the white **[PINS RETRACTED]** button on the UCCNC screen (macro M20730) to extend the positioning pins. These pins pop up at the G54 X0/Y0 corner position, providing a precise physical reference for the front-left corner of the workpiece. Verify the button changes its state to the yellow/green **[PINS EXTENDED]**.

Step 3 – Slide the Sheet onto the Table

Slide the sheet onto the table surface using the ball bearing rollers. Push the sheet until it contacts the positioning pins on both the X and Y edges. This ensures the sheet is precisely aligned with the G54 work coordinate origin.

Step 4 – Retract the Pins and Bearings

Once the sheet is positioned, retract the positioning pins by pressing the Pop-up Pins button again, then retract the ball bearings by pressing the Ball Bearings button again. The sheet now rests flat on the vacuum table surface. Verify the buttons change to their white states.

Step 5 – Engage the Vacuum

Press the white **[VACUUM OFF]** button on the UCCNC screen (macro M20760) to turn on the vacuum system. The 8-zone decentralized vacuum table, with its individual vacuum motors and perforated rubber surface, provides secure and even holding of the workpiece. Verify that the sheet is firmly held before proceeding with any machining operations.

Note: the UCCNC button is switching ON all 8 vacuum motors. There is a vacuum control box with 8 individual ON/OFF buttons for the motors and 8 potentiometers for adjusting the vacuum force individually. If some of the motors are not needed, they can be switched off.

Loading a G-code File and Starting, Stopping, Pausing a Cycle

Loading a File

Click the white **[LOAD FILE]** button in the UCCNC interface to open the file browser. Navigate to your G-code file (.nc, .tap, .gcode, or .txt) and select it. The file will be loaded, and the toolpath will be displayed in the 3D viewer. Review the toolpath visually to confirm it matches your expected cutting pattern and that all tool paths are within the machine working area.

Starting a Cycle

Press the green **[CYCLE START]** button on the UCCNC screen. Before execution begins, the M8000 General Safety Monitor performs a comprehensive set of pre-run safety checks. These checks verify that the machine is homed and squared, Z0 has been calibrated, a tool is present in the spindle with a valid measured offset, the spindle is stopped, and the collet is locked, all pneumatic components are in safe positions (rack retracted, pins retracted, bearings retracted, dust collector lifted), and air pressure is adequate. If the spindle is cold (idle for too long), the system will prompt you to run the warm-up routine first. If all checks pass, the system will optionally ask whether you want the dust extraction pump turned on, and then the G-code program begins executing.

During a cutting cycle, the system automatically manages the spindle (starting, stopping, and setting RPM as directed by the G-code), automatically performs tool changes (M6 commands trigger the full automated tool change sequence, including measurement of unmeasured tools), and dynamically controls the dust collector position based on the Z-axis height via the M9000 macro.

Note: There is a green cycle timer next to the **[CYCLE START]** button, which measures the elapsed time during the running of the cycle.

Pausing a Cycle

Use this for: Clearing chips, checking a dimension, or answering the phone.

- Press the button **[FEED HOLD]**
- What Happens:
    - Motion decelerates smoothly to a stop, but the **machine is NOT losing its physical machine position** (coordinates). The machine knows exactly where it is
    - **The Spindle stays ON** (This is a safety feature: if the spindle stopped while the tool was touching material, the tool would break)
    - The **Cycle Timer continues running**. This is normal. The timer measures "Wall Clock Time" (total time elapsed since the job started), not just "Cutting Time." If you pause for 2 minutes, your total cycle time will be 2 minutes longer.
    - The button **[FEED HOLD]** flashes in green; the button **[CYCLE START]** is NOT active; the button **[CYCLE STOP]** is active and can be used to stop the cycle entirely
- To resume the cycle - press the flashing button **[FEED HOLD]**. The cycle immediately resumes from where it was stopped, and the button stops flashing.

Stopping a Cycle

Use this for: Aborting a job completely (e.g., broken tool, wrong setup, or end of shift).

- Press the red button **[Cycle Stop]**.
- **What Happens:**
    - Motion decelerates to a stop.
    - **The Spindle usually stays ON** (for safety) until you manually turn it off

    - The system "forgets" the active run state. It is no longer "in the middle of a cut"—it is now "Stopped."
    - The **Cycle Timer stops.**
- **The Danger:**
    - **DO NOT press the green button [CYCLE START] immediately after a Cycle Stop.**
    - If you do, the machine may crash or may try to plunge into the material without restarting the spindle (because the "Spindle ON" command was at the beginning of the file, which the machine has already passed).

Note: If you need to stop the machine immediately in an **emergency**, use one of the Physical **Emergency Stop Buttons,** OR press the yellow screen button **[RESET],** OR press the plunger of the Tool Probe. See the Abnormal Operating Procedures chapter for details.

How to SAFELY Resume after a "Cycle Stop"

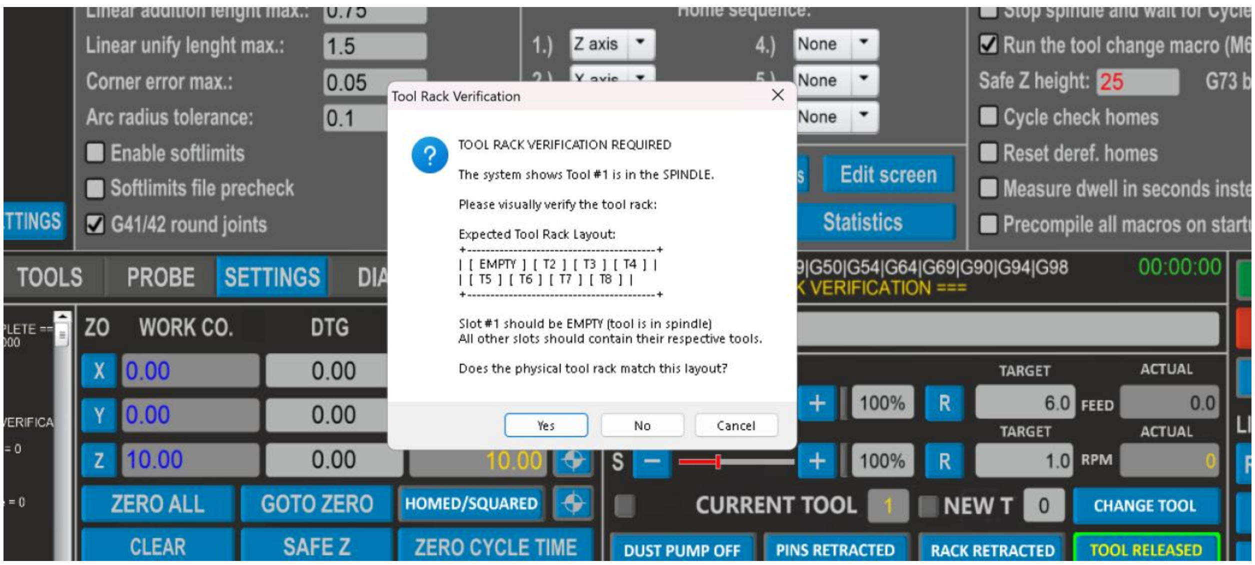
If you stopped the machine (Cycle Stop) and need to finish the part, you must use the **"Run From Here"** procedure:

1. **Locate Position:** Scroll through the G-code on the screen to find the line number close to where you stopped (start a few lines early to be safe).
2. **Select:** Click on that line of code.
3. **Run From Here:** Press the **[RUN FROM HERE]** button.
4. **Verify:** UCCNC will scan the file up to that point to "learn" the correct spindle speed and feed rate.
5. **Start Spindle:** A dialog will appear prompting you to start the spindle. Ensure it is running.
6. **Resume:** Press **[CYCLE START]**. The machine will move to the position and resume cutting safely.

Tool Manipulation – Loading, Offloading, Changing, Measuring

Tool State Initialization

The UCCNC maintains a tool registry of up to 20 tools with their number and their individual offsets (basically, how many mm they extend down from the spindle nose). The system is keeping track of which tool number is being loaded and if that tool has a valid offset. During initialization, the StratoCNC constructor macro M99998 checks the system registry for the current tool number in the spindle and prompts the operator to visually confirm that the slot of the loaded tool assumed by the system is actually empty (*I assume Tool 1 is loaded – can you confirm I'm right?).* This step is specifically designed to make sure the system registry is in sync with the real-world state of the tool rack, and prevents system crashes if the tool rack has been manually manipulated (changing broken or worn tools, or placing/unplacing tools at tool rack).



The current tool in the spindle is shown in yellow in the **CURRENT TOOL** field. If there is no tool (the spindle is empty), the current tool number is 0. The **CURRENT TOOL** field is not directly operated by the operator but rather by the system, and is displaying the assumed tool by the system in the spindle.

The operator uses the **NEW T** (new tool) field or the MDI to enter the desired new tool number if he intends to use the automated tool loading/offloading/changing procedures – the tool changing macro M6 or the button **[CHANGE TOOL],** which is linked to the tool changing macro M20006.

Tool Loading

So, you have just started the StratoCNC for the first time, and there is no tool loaded in the spindle yet. How do you load one if there is no [LOAD TOOL] button? The answer is Tool Change – for StratoCNC, the tool loading is just an operation of tool changing – Change from Tool 0 (no tool) to Tool 1.

The M6 (MDI and G-code) and M20006 (screen button) tool changing macros have inbuilt logic that checks the current vs the new tool number and decides intelligently how to proceed:

A. Current tool 0 - New tool 1 **-->** *I need to **LOAD** Tool 1 in the empty spindle*

B. Current tool 1 - New tool 2 **-->** *I need to **CHANGE** Tool 1 with Tool 2*

C. Current tool 1 - New tool 0 **-->** *I need to **OFFLOAD** Tool 1 from the spindle to the tool rack*

D. Current tool 1 - New tool 1 **-->** *Hmm, I don't need to do anything*

E. Current tool 1 - New tool 13 **-->** ***Error!*** *I have only 10 slots. Check your new tool N, please!*

Having that in mind, the logic above, there are at least 8 ways to load a tool in the empty spindle:

1. **From a G-code file** - Write M6 T1 in the G-code file. Load the file and start the cycle – the StratoCNC will load the tool and update the **CURRENT TOOL** field. Since this is the first tool loading, the tool has no valid offset. The system will recognize this and will start an automated tool measurement routine. Once measured, the tool's offset will be loaded for use during the cutting, and it will also be saved in the system registry for subsequent usage (there will be no need for measuring the tool every time it is being used).
2. **From the MDI only** - Write M6 T1 in the MDI and press ENTER – the system will load the tool and do exactly the same things as with Method 1.
3. **With the NEW T field and the MDI** – Write the tool number in the **NEW T** field and press ENTER, then write M6 in the MDI and press ENTER. The system will load the tool and do exactly the same things as with Method 1.
4. **With the NEW T field and the [CHANGE TOOL] button** - Write the tool number in the **NEW T** field and press the **[CHANGE TOOL]** button. The system will load the tool and do exactly the same things as with Method 1.
5. **With the MDI and the [CHANGE TOOL] button** – Write T1 in the MDI and press ENTER, then press the **[CHANGE TOOL]** button. The system will load the tool and do exactly the same things as with Method 1.
6. **With the [TOOL LOCKED/RELEASED] button** – press the button and the system (M20770 macro) first makes some safety checks (machine idle, spindle stopped). Since you are in front of the control terminal and you play with the UCCNC screen buttons, the macro starts a 10-second countdown to give you time to walk to the spindle. The collet then opens for approximately 2 seconds, during which you insert the tool by hand. The collet automatically closes after the brief window. The system detects the presence of the tool in the spindle and prompts you to register the tool number. Then the system updates the **CURRENT TOOL** field and checks for a valid tool offset in the registry. If there is one, it is being loaded and used. If there is not (the tool hasn't been measured yet), the system reminds you with a prompt that the tool needs to be measured before using it.

7. **With the physical button 'A' on the spindle's face** – press the button and the system (M8001 macro) opens the collet for a period of 3 seconds during which you insert the tool. When the 3 seconds have elapsed, the collet is closed automatically. The system then checks and verifies that there is a tool now in the spindle and prompts you to register its number. Then the system updates the **CURRENT TOOL** field and checks for a valid offset. It then prompts you to measure the tool if there is none.
8. **With the physical red override button on the collet solenoid** – the collet solenoid is mounted on the right top side of the spindle's assembly, right behind the spindle. On the solenoid itself, there is a red override button. When pressed, the solenoid activates and opens the collet immediately, regardless of the software and the electrical signals sent to it. There is no countdown, and there are no safety checks – the collet stays open until the button is pressed. So you open the collet with the button, insert the tool, and close the collet. Now the M4000 Collet Override Monitor Macro kicks in and detects that a tool has been inserted. It prompts you to enter the tool's number, checks for a valid offset, and if no offset exists, prompts you to measure the tool. The **CURRENT TOOL** field is also updated. This method should only be used when the software-controlled methods are not available.

Tool Offloading

The logic is absolutely the same as for the tool loading – there is no [OFFLOAD TOOL] button, and the procedure is simply a tool change from the current tool to Tool 0 (no tool).

As with the loading, there are many ways to do it:

1. **From a G-code file** - Write M6 T0 in the G-code file. Load the file and start the cycle – the StratoCNC will offload the current tool to its parcing position on the tool rack and will reset the **CURRENT TOOL** field to 0.
2. **From the MDI only** - Write M6 T0 in the MDI and press ENTER – the system will offload the current tool and do exactly the same things as with Method 1.
3. **With the NEW T field and the MDI** – Write 0 in the **NEW T** field and press ENTER, then write M6 in the MDI and press ENTER. The system will offload the current tool and do exactly the same things as with Method 1.
4. **With the NEW T field and the [CHANGE TOOL] button** - Write 0 in the **NEW T** field and press the **[CHANGE TOOL]** button. The system will offload the current tool and do exactly the same things as with Method 1.
5. **With the MDI and the [CHANGE TOOL] button** – Write T0 in the MDI and press ENTER, then press the **[CHANGE TOOL]** button. The system will offload the current tool and do exactly the same things as with Method 1.
6. **With the [TOOL LOCKED/RELEASED] button** – press the button and the system (M20770 macro) first makes some safety checks (machine idle, spindle stopped). Since you are in front of the control terminal and you play with the UCCNC screen buttons, the macro starts a 10-second countdown to give you time to walk to the spindle. The collet then opens for approximately 2 seconds, during which you take out the tool by hand. The

collet automatically closes after the brief window. The system detects the lack of a tool in the spindle and automatically resets the **CURRENT TOOL** field to 0.

7. **With the physical button 'A' on the spindle's face –** press the button and the system (M8001 macro) opens the collet for a period of 3 seconds, during which you take out the tool. When the 3 seconds have elapsed, the collet is closed automatically. The system then checks and verifies that there is no tool now in the spindle and resets the **CURRENT TOOL** field to 0.

8. **With the physical red override button on the collet solenoid** – press the button to open the collet and take out the tool. The system then detects the lack of the tool and resets the **CURRENT TOOL** field to 0. This method should only be used when the software-controlled methods are not available.

Tool Changing

As with the loading and offloading, there are 8 ways to do a tool change:

1. **From a G-code file (recommended)** - The primary method for tool changes is the automated macro M6. It is used with the parameter T (the desired new tool number). In the G-code file, just write 'M6 T1' if you want Tool Number 1 to be used for a toolpath X, and then 'M6 T3' for toolpath Y, and then'M6 T4' for the next one, and so on, and so on. Once you load the file and start the cycle, the Strato CNC will automatically perform a complete tool change sequence every time it needs a new tool for the upcoming toolpath or operation in the file. The idea is to start the cycle and forget. If the g-code calls for Tool'N' as the first tool, but there is no tool in the spindle, the system will load the right tool, then it will change it whenever a new toolpath with a new tool has to be cut. Every time a new tool is loaded, the **CURRENT TOOL** field is updated, and the system also checks in the UCCNC Tool Registry for its stored offset, loads it, and uses it. If there is no valid offset for the loaded tool (it hasn't been measured up to this moment), the system automatically measures the tool with the help of the tool probe, updates the system registry with the newly measured offset, and uses it.
2. **From the MDI only** - Write M6 T1 in the MDI and press ENTER – the system will change the current tool to Tool 1 and do exactly the same things as with Method 1.
3. **With the NEW T field and the MDI** – Write 1 in the **NEW T** field and press ENTER, then write M6 in the MDI and press ENTER - the system will change the current tool to Tool 1 and do exactly the same things as with Method 1.
4. **With the NEW T field and the [CHANGE TOOL] button** - Write 1 in the **NEW T** field and press the **[CHANGE TOOL]** button - the system will change the current tool to Tool 1 and do exactly the same things as with Method 1.

5. **With the MDI and the [CHANGE TOOL] button** – Write T1 in the MDI and press ENTER, then press the **[CHANGE TOOL]** button - the system will change the current tool to Tool 1 and do exactly the same things as with Method 1.
6. **With the [TOOL LOCKED/RELEASED] button** – press the button and the system (M20770 macro) first makes some safety checks (machine idle, spindle stopped). Since you are in front of the control terminal and you play with the UCCNC screen buttons, the macro starts a 10-second countdown to give you time to walk to the spindle. The collet then opens for approximately 3 seconds, during which you take out the current tool by hand and insert the new one. The collet automatically closes after 3 seconds have elapsed. The system detects that there is still a tool in the spindle (there was one before the opening of the collet) and asks you if a tool change took place. If the answer is YES, it prompts you to enter the tool number, and then it checks for a valid offset for this tool. If there is none, it prompts you to measure the tool before use. The **CURRENT TOOL** field is also updated with the new tool number.
7. **With the physical button 'A' on the spindle's face** – press the button and the system (M8001 macro) opens the collet for a period of 3 seconds, during which you take out the old tool and then insert the new one. When the 3 seconds have elapsed, the collet is closed automatically. The system detects that there is still a tool in the spindle (there was one before the opening of the collet) and asks you if a tool change took place. If the answer is YES, it prompts you to enter the tool number, and then it checks for a valid offset for this tool. If there is none, it prompts you to measure the tool before use. The **CURRENT TOOL** field is also updated with the new tool number.
8. **With the physical red override button on the collet solenoid** – press the button to open the collet and take out the old tool and insert the new one. Release the button and the collet locks. The system detects that there is still a tool in the spindle (there was one before the opening of the collet) and asks you if a tool change took place. If the answer is YES, it prompts you to enter the tool number, and then it checks for a valid offset for this tool. If there is none, it prompts you to measure the tool before use. The **CURRENT TOOL** field is also updated with the new tool number. This method should only be used when the software-controlled methods are not available.

Tool Measurement

Press the white **[PROBE]** button on the UCCNC screen (macro M20031) to manually measure any tool currently in the spindle (or write M20031 in the MDI and press ENTER). The system extends the tool rack, lowers the dust collector for access, cleans the probe surface with an air blast, and performs a two-stage probing sequence (fast approach followed by a slow precision probe). The measured tool length offset is stored in the system registry and applied to the UCCNC tool offset fields. The tool does not need to be re-measured on subsequent loadings unless you want to verify the offset.

Note: The automatic tool changing macros M6 and M20006 have a provision for automatic measurement of any tool that has no valid offset in the system registry. The goal is for the operator not to be needed once the cycle is started.

Note: The system would not allow for any tool measurement (both directly via the button or during the automated tool changing in a cycle) if the Tool Probe has not been calibrated. The calibration is done with the M20303 macro and its **[NOT CALIBRATED]** button.

Touch Plate Operation – Finding G54 Z0 and Calibrating the Tool Probe

The touch plate is used to establish the exact G54 Z0 position (the top surface of your workpiece) and to calibrate the tool length probe. Both operations are performed by a single macro M20303, triggered by the **[CALIBRATED] / [NOT CALIBRATED]** button. **The macro is designed to work with ONLY TOOL N1 as a reference tool and will terminate itself if another tool is loaded.**

Procedure

Place the touch plate on the surface of the workpiece, directly below the spindle, and attach the alligator clips to the spindle. The touch plate has a known thickness that is stored in the system configuration. Press the **[CALIBRATED] / [NOT CALIBRATED]** button. The macro will prompt you to confirm the touch plate is in position. The spindle then descends in a two-stage probing motion (fast approach, retract, slow precision probe) until the tool contacts the touch plate. The system calculates the exact G54 Z0 offset by subtracting the known touch plate thickness from the probed position. Then the system urges the operator to remove the touch plate and to unclip the Alligator clips from the spindle. Stage 1 is complete!

Stage 2 for calibrating the Tool Probe is optional. After the first stage is done and the touch plate/alligator clips are removed, the macro asks the operator whether to proceed with calibrating the tool probe. If NOT – the spindle is returned to its safe position and the macro is complete. If the operator answers YES, the macro moves the spindle to the probe location, extends the rack, lowers the dust collector, cleans the probe with an air blast, and performs a precision probe to determine the probe's upper surface Z position relative to both the machine origin and the workpiece surface. The probe is flagged as CALIBRATED in the system registry and is cleared to measure all the other tools if needed. The button changes its state to white **[CALIBRATED]**, which is an indication to the operator that the probe is calibrated and good for measuring.

Note: The calibration status of the probe is a static one (the probe can not move up and down in space, so once calibrated, it can be assumed it will stay calibrated). That means on the subsequent system initializations, the M99998 constructor macro will read in the registry that the probe has been calibrated before and will set the button to its white **[CALIBRATED]** state. That being said, it is still a good operational practice to run the tool probe calibration routine at the beginning of the day or at least once per week.

Note: You need to run the Stage 1 G54 Z0 calibration whenever you change the workpiece thickness or whenever you suspect the Z0 reference may have shifted. The **[CALIBRATED]** status of the button only indicates that the tool probe has been calibrated and has nothing to do with the decision whether and when to run the touch probe stage 1 routine.

Tool Probe Operation

The plunger-type tool length probe is mounted on the retractable tool rack. It uses a Normally Closed switch that opens when the tool depresses the plunger. The probe has a typical actuation distance of 0.025 to 0.1 mm.

During automated tool changes (M6 / M20006), tools are automatically measured if they do not have a stored offset. You can also measure any tool manually by pressing the white **[PROBE]** button (macro M20031) or by writing M20031 in the MDI and pressing ENTER. The probe must be calibrated (via M20303, the touch plate, and Tool 1) before measurements are valid.

During the probing routine, the probe's surface is cleaned by an airstream to prevent dust or particles from interfering with the measurement.

The probe also has an overtravel alarm switch that triggers an immediate machine stop if the plunger is depressed beyond the safe limit, protecting both the probe and the tool.

This function is used to turn the probe into an Emergency Stop Button when the operator is near the probe (near the front right corner of the machine).

Spindle Operation – Protections, Indications, Starting CW/CCW, Stopping, Overriding the RPM

The 7.5 KW ATC Digital HSD Spindle is by far the most expensive and the most important component of StratoCNC. Therefore, very special care and effort have been taken during the development to make sure the spindle will be thoroughly protected and properly utilized.

There are no physical buttons for starting/stopping the spindle. All the operations are done through the software by using UCCNC screen buttons or macro commands from the MDI or from within the G-code files.

The StratoCNC system features an intelligent thermal monitoring routine that tracks spindle idle time to protect internal bearings and ensure machining precision. The system (via the M8000 safety monitoring macro) automatically flags the spindle as "Cold" if it has been stationary for more than 60 minutes (or upon initial startup – by the constructor macro M99998), indicated by a flashing yellow/green [SPINDLE COLD] button. When this indicator is active, operators must run the automated Warm-up Cycle by pressing the flashing button to properly lubricate and stabilize the spindle before high-load cutting; however, if machining resumes within the 60-minute window of the last stop, the system retains its "Warm" status and requires no additional warm-up. For more information about the logic and architecture of the spindle warm-up, check the separate

'SPINDLE WARM UP TRACKING SYSTEM' manual, which is part of the standard StratoCNC documentation suite.

The spindle management and control macro system also includes strict hardware safety interlocks to prevent equipment damage during operation. Most critically, the system performs a Tool Presence Check (via Sensor S1) before executing any start command (M3/M4/M20601/M20602); if no tool holder is detected in the collet, rotation is blocked to prevent damage to the drawbar assembly. Additionally, the control logic enforces mutual exclusivity between Clockwise (CW) and Counter-Clockwise (CCW) modes, requiring a complete spindle stop (M5) before a change in direction is permitted. Manual control is handled via dedicated toggle macros (M20601/M20602), which verify the shaft has physically stopped spinning before allowing new commands, ensuring safe transitions between operational states.

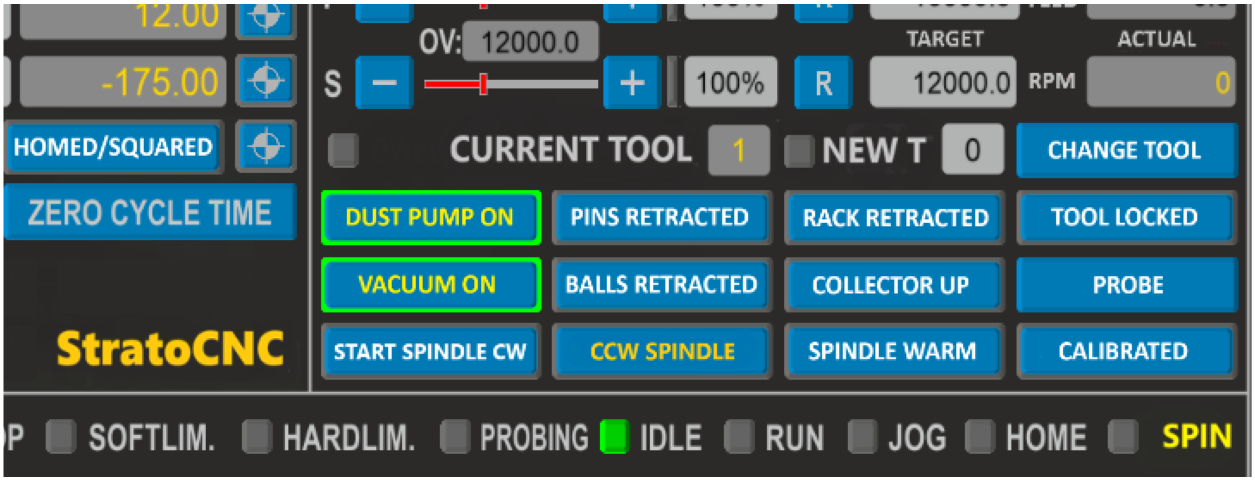
To ensure long-term reliability, the machine utilizes a dedicated background monitoring loop (M10002) that continuously scans spindle telemetry for faults, including thermal overload, cooling fan failure, and spindle electronics malfunctions. If any critical fault is detected during operation, the system automatically triggers an Emergency Safety Sequence: the current G-code cycle is immediately halted, spindle rotation is forced off, and the Z-axis retracts to a safe height to prevent workpiece damage. A proper warning message is displayed together with a dedicated red warning lamp SPINDLE.

A monitoring macro is reading the control voltage provided by the VFD and is constantly calculating and displaying the actual RPMs of the spindle in a dedicated DRO field.

To indicate reliably, undeniably, and constantly whether the spindle is running or not, the system uses a myriad of visual cues to the operator:

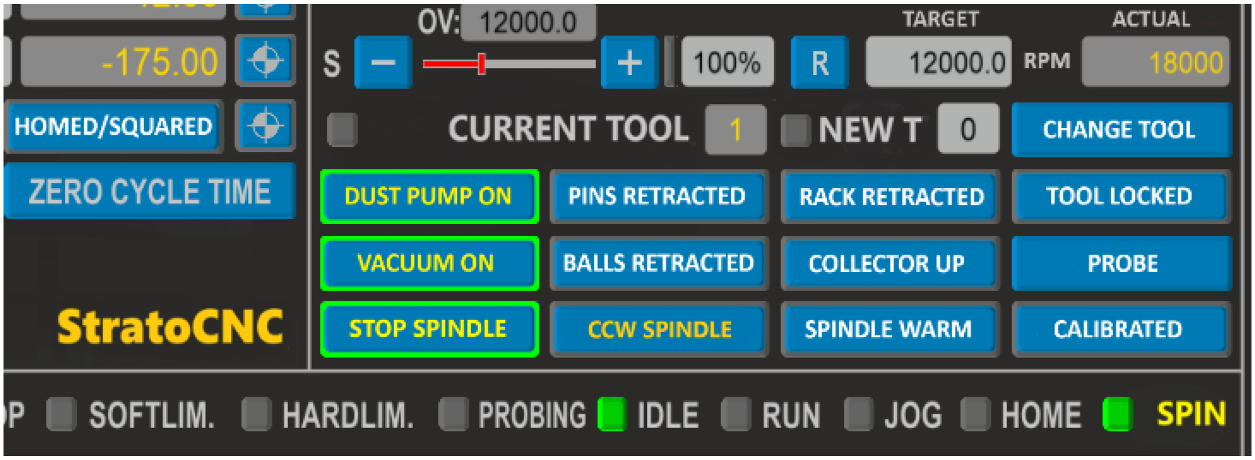
1) When the spindle is **IDLE:**

- The two screen buttons for starting/stopping the spindle are in their inactive states – white **[START SPINDLE CW]** and golden **[CCW SPINDLE]**
- The screen LED with the yellow sign SPIN is **OFF** (grey)
- The yellow DRO field for the actual RPM shows 0



2) When the spindle is **RUNNING:**

- Depending on whether the spindle is running clockwise or counterclockwise ONE of the two screen buttons for starting/stopping is showing its active yellow/green state **[STOP SPINDLE]**, indicating it is ready to be used for stopping the spindle. The other button is still in its inactive state.
- The screen LED with the yellow sign SPIN is ON (green)
- The yellow DRO field for the actual RPM shows some RPM



Every time the spindle is started/stopped by a button, or by an MDI command, or by a G-code file command during the cycle running – the three different indications will be dynamically synced with the spindle's state and will be showing the operator what is going on with the spindle, whether it is rotating, with what RPM, and in which direction.

With all that made clear, let's now see how to operate this baby!

Starting the Spindle Clockwise (Normal Operation)

The spindle can be STARTED clockwise in 5 different ways:

1. **From a G-code file** - Write M3 S15000 in the G-code file. Load the file and start the cycle – the system will start the spindle clockwise with 15000 RPM.
2. **From the MDI only** - Write M3 S15000 in the MDI and press ENTER – the system will start the spindle clockwise with 15000 RPM.
3. **With the TARGET RPM field and the MDI** - Write 15000 in the **TARGET RPM** field and press ENTER, then write M3 in the MDI and press ENTER – the system will start the spindle clockwise with 15000 RPM.
4. **With the TARGET RPM field and the [START SPINDLE CW] button** - Write 15000 in the **TARGET RPM** field and press ENTER, then press the white **[START SPINDLE CW]** button – the system will start the spindle clockwise with 15000 RPM.

5. **With the MDI and the [START SPINDLE CW] button** – Write S15000 in the MDI and press ENTER, then press the white **[START SPINDLE CW]** button – the system will start the spindle clockwise with 15000 RPM.

Starting the Spindle Counter-Clockwise (NON STANDARD OPERATION)

Because counterclockwise rotation is not a standard operation, to start it with the golden **[SPINDLE CCW]** screen button requires a two-step confirmation (both confirmation dialogs default to NO for maximum safety). In an effort to make the inadvertent starting of the spindle in the wrong direction unlikely, we designed the two buttons with different colors and different signs, and also added a two-step confirmation (*You want me to run the spindle counterclockwise! Are you sure?!*). All other safety checks are the same as for CW operation.

The spindle can be STARTED counterclockwise in 5 different ways:

1. **From a G-code file** - Write M4 S15000 in the G-code file. Load the file and start the cycle – the system will start the spindle counterclockwise with 15000 RPM.
2. **From the MDI only** - Write M4 S15000 in the MDI and press ENTER – the system will start the spindle counterclockwise with 15000 RPM.
3. **With the TARGET RPM field and the MDI** - Write 15000 in the **TARGET RPM** field and press ENTER, then write M4 in the MDI and press ENTER – the system will start the spindle counterclockwise with 15000 RPM.
4. **With the TARGET RPM field and the [SPINDLE CCW] button** - Write 15000 in the **TARGET RPM** field and press ENTER, then press the golden **[SPINDLE CCW]** button, then press OK on the first dialog box, then press OK on the second dialog box – the system will start the spindle counterclockwise with 15000 RPM.
5. **With the MDI and the [SPINDLE CCW] button** – Write S15000 in the MDI and press ENTER, then press the golden **[SPINDLE CCW]** button, then press OK on the first dialog box, then press OK on the second dialog box – the system will start the spindle counterclockwise with 15000 RPM.

Stopping the Spindle

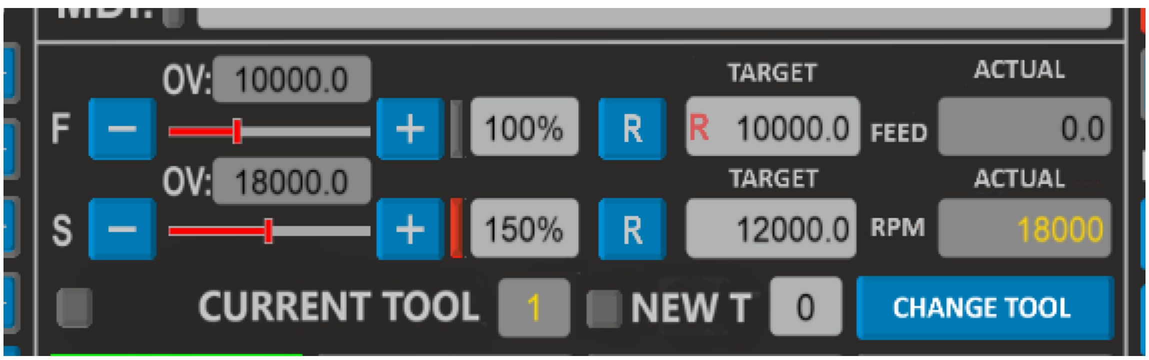
The spindle can be STOPPED in 3 different ways:

1. **From a G-code file** - Write M5 the G-code file
2. **From the MDI** - Write M5 in the MDI and press ENTER
3. **With the active yellow/green [STOP SPINDLE] button** – Depending on the direction of the rotation, one of the two start/stop buttons will be in its active yellow/green **[STOP SPINDLE**] state – press that button to stop the rotation of the spindle.
4. With the power button – Just stop the power to the machine…………….... I'M JOKING! I wanted to wake up! ☺ PLEASE NEVER DO THIS!

Overriding the Spindle speed

By using the +/- buttons on the Spindle Override Slider marked **S,** you can always increase or decrease the target RPM dynamically during the running cycle. You can also write a percentage directly into the active field next to the slider and press ENTER. The calculated new target speed will be displayed over the slider. The max allowed speed can never be higher than the absolute maximum spindle speed setting, which in our case is 24000 RPM. If your desired percentage is higher than 100%, a red LED will be illuminated next to it. If the percentage is less than 100% (you are decreasing the RPM), the LED will be green.

In the picture bellow the target RPM have set to 12000 in the TARGET RPM field. Then the operator entered 150 in the percentage field and pressed ENTER – the calculated new RPMs are 18000 and are displayed above the slider. The spindle reached that speed, which is shown by the golden actual RPM field. The red LED shows that the target speed has been increased:



Dust Collector Operation – Dynamic and Manual

Dynamic Operation (During Cutting)

When a Cycle is started, the operator is presented with a dialog box, and he has to choose whether to use dynamic or manual operation of the Dust Collector for this particular cycle.

If he chooses the Dynamic option, the M9000 Dynamic Dust Collector Control macro automatically manages the dust collector position based on the Z-axis height. When the Z axis descends below a configurable threshold (default: 5.0 mm above the workpiece), the dust collector lowers. When the Z axis rises above the threshold (during rapid positioning moves), the collector is lifted.

The reason for this behavior is that an integral part of the dust collector is the Spring-loaded Workpiece Pusher. Its purpose is to mechanically exert pressing force on the workpiece being cut in order to augment the pulling vacuum force and ensure the workpiece is held in place securely. This is especially important when a small element is being cut, and there is simply not enough gripping area for the vacuum table to provide adequate holding force. The idea of the pusher is to act as the holding hand of a person holding the cutting tool with one hand and the element being cut with the other.

And why do we need to lift the pusher during the positioning moves? Technically, we don't, but these moves are generally more rapid, and we felt it would be better if we lifted the pusher during these to extend its life and to prevent unnecessary mechanical fiddling with the workpiece at high speeds.

Manual Operation

If the operator chooses a manual operation during the cycle start, the M9000 dynamic macro is disabled for the duration of that particular cycle. Then the operator has the choice to lift or lower the collector together with the pusher whenever or however he determines pertinent.

Press the white **[COLLECTOR UP]** / yellow/green **[COLLECTOR DOWN]** button on the UCCNC screen (macro M20720) to manually toggle the dust collector between its raised and lowered positions. A proximity sensor verifies the collector position, and the button state indicates the current state.

Tool Rack Operation – Extending and Retracting

Press the white **[RACK RETRACTED]** / yellow/green **[RACK EXTENDED]** button on the UCCNC screen (macro M20710) to manually toggle the tool rack between its retracted and extended positions. The rack extends 125 mm in the Y-negative direction, breaching the working area. A proximity sensor confirms the rack position, and the button state indicates the current state.

The tool rack is automatically extended and retracted by the tool change macros (M6 / M20006), the tool measuring macro (M20031), and the tool probe calibrating macro (M20303), so manual operation is normally only needed for maintenance or setup tasks.

WARNING: Never start a cutting cycle with the tool rack extended. The M8000 safety monitor checks rack position before allowing cycle start, but always verify visually as a good practice.

Rotating Tool Base Operation – Rotating, Locking, Unlocking

The entire tool rack assembly, including the tool holder clamps and the tool length probe, is mounted on a manually operated rotating base. When rotated 90 degrees away from the machine, the full front face of the working area becomes accessible for loading and unloading sheet material. When rotated back into the machine, the rack is available for automated tool changes and tool length measurements/tool probe calibration.

To rotate the base, release the spring-loaded locking mechanism by pushing down the locking handle, swing the base out (or in), and allow the locking mechanism to engage when the base reaches its end position. A proximity sensor detects whether the base is rotated into the machine, and several macros check this sensor before performing operations that require the tool rack or probe.

WARNING: Always ensure the rotating base is fully locked into the machine before running any tool change, tool measurement, or probe calibration operations.

Ball Bearings Operation – Extending and Retracting

Press the white **[BALLS RETRACTED]** / yellow/green **[BALLS EXTENDED]** button on the UCCNC screen (macro M20740) to toggle the pneumatic ball bearing transport rollers. When extended, the rollers rise above the table surface, allowing sheets to be slid easily for loading and unloading. When retracted, the rollers drop below the surface, and the sheet rests flat on the vacuum table. A proximity sensor confirms the bearing position, and the button state indicates the current state of the ball bearings.

Note: Always retract the ball bearings before engaging the vacuum or starting a cutting cycle. The M8000 safety monitor checks bearing position before allowing cycle start.

Positioning Pins Operation – Extending and Retracting

Press the white **[PINS RETRACTED]** / yellow/green **[PINS EXTENDED]** button on the UCCNC screen (macro M20730) to toggle the pop-up positioning pins. When extended, the pins rise above the table surface at the G54 X0/Y0 corner position, providing a precise physical reference for positioning the workpiece. When retracted, the pins are flush with or below the table surface. A proximity sensor confirms pin position, and the button LED indicates the current state.

Note: Retract the pins after the workpiece is positioned and before starting a cutting cycle. The M8000 safety monitor checks the pin position before allowing cycle start.

Vacuum Table Operation

Press the Vacuum On/Off button on the UCCNC screen (macro M20760) to toggle the vacuum system. The StratoCNC uses a decentralized 8-zone vacuum table, where each zone has its own vacuum motor. This design provides uniform suction across the table, is low-maintenance, and eliminates the need for a single large (and expensive) vacuum pump. The upper surface of the table is covered with 1 mm perforated rubber for improved holding characteristics. The button LED indicates whether the vacuum is currently active.

Tip: Always ensure the ball bearings are retracted, and the sheet is lying flat before engaging the vacuum. For small workpieces that do not cover all zones, the uncovered zones still operate but with reduced effect, which is normal.

Dust Suction Pump Operation

Press the Dust Pump On/Off button on the UCCNC screen (macro M20750) to toggle the dust extraction pump. This pump provides suction through the retractable dust collector housing around the spindle, removing chips and dust during cutting operations. The pump can be turned on manually at any time, and it is also offered as an option during the M8000 pre-cycle safety checks before a cutting file starts. The button LED indicates whether the pump is currently active.

WARNING: After an E-Stop event, always visually inspect the machine, workpiece, and tool for damage before re-homing and resuming operations.

12. ABNORMAL OPERATING PROCEDURES

General Safety Rules and Precautions

The StratoCNC incorporates multiple layers of safety protection through its macro system, hardware interlocks, and monitoring loops. However, situations may arise that require the operator to understand how the machine responds to faults and how to recover from abnormal conditions. This section describes the procedures for handling the most common abnormal situations.

The general principle of the StratoCNC safety system is to detect faults early, halt operations immediately, inform the operator with clear diagnostic messages, and prevent further operations until the fault is resolved. All health monitoring macros (M10001, M10002, M10003) use a standardized two-state machine architecture: State 0 performs normal monitoring at regular intervals, and State 1 manages the alarm-active condition with automatic recovery when conditions clear.

General Error Handling and Management

When the StratoCNC macro system detects an error during any operation, it follows a consistent pattern. The running operation is stopped immediately. An error message is displayed on the UCCNC status bar describing the specific issue. The operator is typically offered options: fix the issue and retry, or abort the operation. In critical situations (such as during a tool change), a SafetyRecovery function attempts to automatically stabilize the machine by retracting the Z axis to a safe height, stopping the spindle, retracting all pneumatic components, and locking the collet.

If you encounter an error message, read it carefully. The macro system provides specific, actionable information about what went wrong and what you should do. Common error types include precondition failures (something was not set up correctly before the operation), sensor verification failures (a physical condition does not match expectations), and hardware faults (a component is not responding as expected).

Tip: Monitor the UCCNC status messages continuously during operation. The macro system provides detailed feedback about every significant action and any detected anomalies.

Emergency Stop Button Operations

The StratoCNC has Emergency Stop (E-Stop) buttons that immediately halt all machine motion and disable the servo drives when pressed. The E-Stop circuit uses a Normally Closed configuration, meaning the circuit is complete during normal operation and is broken when the button is pressed. This fail-safe design ensures that a wiring fault also triggers the emergency stop.

Location of the E-stop Buttons

The StratoCNC has 4 E-stop buttons in total. They are mounted roughly on the 4 corners of the machine, covering as much as possible of the working area and giving the operator the ability to stop the machine from almost all the possible angles and in most of the possible working situations.

One E-stop button is mounted on the face of the control cabinet, which is situated close to the front left corner.

Two individual buttons are mounted on each of the back corners.

The front right corner has an improvised E-Stop button – the tool-measuring probe. The probe has an overtravel protection circuit. The signal from this circuit is wired and set in such a way in UCCNC that when the plunger is pressed, the circuit activates and the UCCNC enters the RESET mode, just like if an E-stop button were pressed. So when the operator is around the front right corner, and an emergency arises, he has to use the plunger of the probe as an E-stop Button.

When to Use the E-Stop

Use the Emergency Stop in any situation where immediate machine halt is necessary: an impending collision, unexpected machine behavior, a tool breakage event, a fire or smoke condition, any situation where personal safety is at risk, or when you observe the machine moving in an unexpected direction or at an unexpected speed.

What Happens When E-Stop is Pressed

All axis motion stops immediately. The servo drives are disabled (the motors lose holding torque, but the Z axis has a mechanical brake that engages automatically when power is removed). The UCCNC software enters the RESET state (see below). The spindle continues running - the VFD does not receive an explicit stop command via E-Stop.

Recovery After E-Stop

Release the E-Stop button by twisting it clockwise (or pulling it, depending on the button type). Click the RESET button in UCCNC to clear the RESET state and re-enable communication with the controller. The machine has now lost its homing reference. You must re-home the machine by pressing the Home/Square button (M20301) before any further operations. After homing, if you were in the middle of a cutting job, you will need to re-establish G54 Z0 if the workpiece or tool has moved, and reload and restart the G-code file. You may use the **[RUN FROM HERE]** to continue working from the point of disturbance. It is always a good idea to start several lines before the actual line of stop.

RESET State

UCCNC enters the RESET state if:

- An axis Proximity Limit Switch triggers – the machine has reached some of its axes' limit switches. In normal conditions, this should never happen as there are software limits set in UCCNC that should prevent the machine go near the limit switches.
- An Emergency Stop Button is pressed
- The plunger of the Tool Probe is pressed

- Communication with the UC300ETH is lost
- Certain critical faults are detected.

In the RESET state, all outputs are disabled, no motion commands are accepted, and the machine is effectively locked out from operation.

To exit the RESET state, resolve the underlying cause (release the E-Stop, restore communication, or fix the fault) and click the RESET button in UCCNC. The software will re-establish communication with the controller and re-enable I/O. After exiting RESET, you must always re-home the machine before performing any operations, because the machine position reference is no longer reliable.

Note: The tool probe overtravel function also acts as a secondary E-Stop. If the probe plunger is pressed beyond the safe limit (or pressed manually by the operator near the front-right corner of the machine), the overtravel alarm triggers and the machine enters RESET mode immediately.

Servo Health Warnings – ALARM and NOT READY

The M10001 Servo Health Monitoring macro continuously monitors the ALARM and READY signals from all four servo drivers (on the Z axis, only ALARM is monitored, as the pin for the READY was used for controlling the brake on that motor). When a fault is detected and persists for three consecutive readings (approximately 750 ms of debounce), the system transitions to the alarm state.

Symptoms

A warning message appears in the UCCNC status bar identifying which servo axis has the fault and whether it is an ALARM or NOT READY condition. The system blocks the cycle start and prevents spindle starts. You are not allowed to start a normal cutting cycle, but you can try to move the machine for recovery, maintenance, or repairs via the MDI or via JOG command, but the macro will issue a warning when you attempt it.

Common Causes

An ALARM condition typically indicates a following error (the motor has overheated, the motor could not keep up with commands, possibly due to a mechanical obstruction or excessive speed), an overcurrent condition (the motor drew more current than the driver allows), a communication error between the driver and motor encoder, or a hardware fault in the servo driver or motor. A NOT READY condition usually indicates the servo driver has not completed its initialization, the driver is in a fault state that requires a power cycle, or there is a wiring issue with the READY signal.

Resolution

Check the specific servo driver front panel for a fault code. Refer to the DELTA servo driver documentation for the meaning of the fault code. Common remedies include clearing the fault via the servo driver panel or via the ASDA SOFT software, checking for mechanical obstructions on the affected axis, verifying that motor and encoder cables are securely connected, and powercycling the servo driver if the fault persists. Once the fault condition clears, the M10001 monitoring macro automatically returns to normal monitoring and removes all operational blocks.

Spindle Health Warnings – OVERHEAT, ELECTRONICS, FAN

The M10002 Spindle Health Monitoring macro continuously monitors three health sensors on the HSD electrospindle: the thermal alarm sensor (a bimetallic NC switch that opens on overtemperature), the fan status sensor, and the electronics status sensor.

Thermal Alarm (OVERHEAT)

A thermal alarm triggers instantly (no debounce) when the spindle body temperature exceeds the safe operating threshold. The system immediately stops any running cycle, retracts the Z axis to a safe height, stops the spindle, and secures all pneumatic components. The system enters a persistent alarm state that blocks cycle start and spindle operation until the spindle cools down and the thermal switch resets.

Allow the spindle to cool with adequate ventilation. Do not attempt to restart the spindle until the alarm clears automatically. If the thermal alarm triggers frequently, check that the spindle cooling fan is operating correctly, the air intake is not blocked by dust, and the cutting parameters (RPM, feed rate, depth of cut) are within the spindle specifications.

Fan Status Warning

If the spindle cooling fan fails or is not detected (3-read debounce, approximately 750 ms), the system issues a warning and takes the same protective actions as for a thermal alarm. A fan failure will eventually lead to spindle overheating, so the system treats it as a serious condition. Check the fan for mechanical failure, blocked intake, or wiring issues.

Electronics Status Warning

If the spindle electronics monitoring sensor reports a fault (3-read debounce), the system issues a warning and takes protective actions. This may indicate an issue with the spindle encoder, internal electronics, or wiring. Contact the spindle manufacturer or a qualified service technician if this warning persists.

Proximity Sensor Health Warnings

The M10003 Sensor Health Monitoring macro monitors all axis limit switches and detects unexpected activations while the machine is at a known position. The macro is not active before the machine has been homed. Once the machine is homed, it continuously and positively reports its machine coordinates. An unexpected limit switch activation while the machine is out of the limit switch detection envelope may indicate a wiring fault, sensor failure, electromagnetic interference, or a physical object triggering the sensor.

The macro uses soft limit boundaries from the UCCNC configuration for context-aware detection and applies a 3-consecutive-read debounce before triggering an alarm. When triggered, the system blocks the cycle start and performs a Z retraction if needed. The alarm clears automatically when the sensor condition normalizes.

If proximity sensor warnings occur, inspect the affected sensor for physical damage, verify the sensor wiring is secure and properly shielded, check for metal debris near the sensor that might cause false triggers, and verify the sensor LED indicator matches the expected state.

Air Pressure Loss

The M6000 Air Pressure Monitor continuously monitors the pneumatic system pressure via a spring-loaded plunger mechanism with a PNP NC proximity sensor. When air pressure is adequate, the plunger is pushed out, and the sensor detects it. When pressure drops below the minimum threshold, the spring retracts the plunger, and the macro immediately halts all machine motion and alerts the operator.

If an air pressure alarm occurs, check that the compressor is running and has not tripped a breaker, verify the main shut-off valve is open, verify the air supply line is connected and not kinked or damaged, check the pressure regulator setting (minimum 6 bar recommended), drain any accumulated water from the air filter/separator, and check for air leaks in the pneumatic lines or fittings. The system sets an AirPressureOK global key that other macros use to verify pressure before pneumatic operations.

13. SAFETY PROCEDURES AND PRECAUTIONS

Personal Safety

The StratoCNC is a powerful industrial machine with a high-speed rotating spindle, fast-moving axes, and pneumatic actuators. Operators must observe the following personal safety precautions at all times.

Eye protection: Always wear safety glasses or a face shield when the machine is operating. Chips, broken tool fragments, and dust particles can be ejected at high velocity.

Hearing protection: The spindle and dust extraction system generates significant noise levels during operation. Wear hearing protection when operating the machine for extended periods.

Loose clothing and jewelry: Never wear loose clothing, neckties, dangling jewelry, or unsecured long hair near the machine. These can be caught by the rotating spindle or moving components.

Gloves: Do not wear gloves when working near the rotating spindle. Gloves can be caught and pull your hand into the spindle. Gloves may be worn for material handling when the spindle is stopped.

Dust exposure: Wood dust, composite dust, and certain material dusts are hazardous to health. Always operate the dust extraction system during cutting. Wear a dust mask or respirator when the dust extraction is insufficient or during cleanup activities.

Machine Safety Rules

Never leave the machine unattended during operation. Always remain within reach of an Emergency Stop button while the machine is running a cutting cycle.

Verify workpiece clamping. Always ensure the workpiece is securely held by the vacuum table before starting a cut. An unsecured workpiece can shift or be thrown by the cutting forces.

Inspect tools before use. Check each cutting tool for damage, wear, or runout before loading it into the spindle. A damaged tool can break during operation, creating a projectile hazard.

Respect the working envelope. Keep hands and body clear of the machine working area during operation. The gantry, spindle assembly, and dust collector move at high speeds and with significant force.

Do not override safety systems. The macro system includes numerous safety checks and interlocks. Do not attempt to bypass these checks. They exist to protect you and the machine.

Know your Emergency Stop locations. Familiarize yourself with the location of all Emergency Stop buttons and the probe plunger (which also functions as an E-Stop) before operating the machine.

Electrical Safety

The control cabinet contains high-voltage components (220V AC for the servo drives, VFD, and spindle). Never open the control cabinet while the machine is powered on. Always disconnect the main power before performing any electrical maintenance. Only qualified personnel should work on electrical components inside the cabinet.

The cabinet wiring uses shielded cables and ferrites extensively to minimize electromagnetic interference. If you need to add or modify wiring, maintain the same shielding practices to preserve signal integrity.

Pneumatic Safety

The StratoCNC uses compressed air at working pressures of 4 to 7 bar for operating the tool rack, dust collector, tool release collet, pop-up pins, ball bearings, and the probe air cleaning system. Compressed air can be dangerous.

Never direct compressed air at yourself or others. High-pressure air can cause serious injury.

Depressurize before maintenance. Before working on any pneumatic component, turn off the air supply and bleed the remaining pressure from the lines.

Check air quality. The air supply must be clean, dry, and oil-free (or lightly oiled per component requirements). The pneumatic system includes a filter, regulator, and lubricator (FRL) unit. Drain accumulated water from the filter regularly.

Monitor air pressure. The M6000 macro monitors air pressure continuously. If the compressor fails or a line is disconnected, the system will halt operations immediately. Always investigate and resolve air pressure alarms before resuming work.

Spindle Safety

The HSD 7.5 kW electrospindle operates at speeds up to 24,000 RPM. At these speeds, the stored rotational energy is significant, and tool breakage or ejection can be extremely dangerous.

Never reach into the machine while the spindle is rotating. Wait until the spindle has come to a complete stop (verified by the S3 sensor) before any manual intervention.

Ensure tool retention. The ISO30 tool holder must be properly seated in the collet and the collet must be locked. The S1 (tool presence) and S2 (collet status) sensors verify this, but always confirm visually as well.

Use correct RPM settings. Always use the RPM recommended by the tool manufacturer for the material being cut. Excessive RPM can cause tool failure. Insufficient RPM can overload the spindle.

Observe warm-up procedures. Running a cold spindle under heavy load accelerates bearing wear. Always complete the warm-up routine (M20305) at the start of each working day.

Fire Safety

Certain materials (particularly MDF, certain hardwoods, and composites) can generate significant heat during machining. In rare cases, a dull tool combined with excessive feed rates or insufficient dust extraction can produce enough heat to ignite material or accumulated dust.

Keep a fire extinguisher accessible. Have an appropriate fire extinguisher (CO2 or dry powder, suitable for electrical fires) within easy reach of the machine.

Maintain dust extraction. Effective dust extraction reduces the accumulation of combustible dust around the cutting area. Always operate the dust extraction pump during cutting.

Inspect tools regularly. Dull tools generate more heat and friction. Replace or sharpen tools before they become excessively worn.

Clean the machine regularly. Remove accumulated dust from the machine structure, gantry, and control cabinet area at the end of each working day.

14. MAINTENANCE

Regular maintenance is essential to keep the StratoCNC operating at peak performance and to extend the lifespan of all components. This chapter describes the recommended maintenance schedule and procedures.

Daily Maintenance

Clean the machine. At the end of each working day, remove accumulated chips and dust from the table surface, gantry beam, spindle assembly, linear guide rails, and rack-and-pinion drives. Use the dust extraction system and a brush. Do not use compressed air to blow dust into the machine structure or into the control cabinet.

Inspect the spindle collet. Check the ISO30 collet taper and the tool holder tapers for chips, dust, or damage. A contaminated taper will cause runout and reduce tool life. Wipe the taper surfaces clean with a lint-free cloth.

Drain the air filter. Open the drain valve on the pneumatic FRL (filter-regulator-lubricator) unit to release any accumulated water. Moisture in the pneumatic lines causes corrosion and can damage solenoid valves.

Check tool condition. Inspect all tools currently in the tool rack for wear or damage. Replace dull or chipped tools promptly.

Visual inspection. Briefly inspect the machine for any loose bolts, damaged cables, leaking pneumatic fittings, or other obvious issues.

Weekly Maintenance

Lubricate the linear guide rails. Apply a thin film of appropriate machine oil or grease to all linear guide rails (TBI TR25) on the X, Y, and Z axes. Wipe off excess lubricant to prevent dust accumulation. The guide blocks have grease fittings for injecting lubricant.

Lubricate the helical rack. Apply a thin coat of lithium grease or machine oil to the helical rack teeth on the X and Y axes. Manually jog the axes back and forth to distribute the lubricant evenly.

Lubricate the Z-axis ball screw. Apply appropriate ball screw lubricant to the TBI 20x5 ball screw. Jog the Z axis through its full range to distribute the lubricant. Do not over-lubricate, as excess grease attracts dust and can impair performance.

Check the timing belt (Z-axis). Inspect the Optibelt Omega-HP 5M15 timing belt for wear, cracking, or fraying. Check the belt tension. A loose belt can cause Z-axis positioning errors and backlash.

Clean proximity sensors. Wipe all proximity sensors (limit switches, position sensors, pressure sensors) with a clean cloth to remove dust and debris that could cause false triggers.

Monthly Maintenance

Check all electrical connections. With the machine powered off, open the control cabinet and visually inspect all cable connections for looseness, corrosion, or damage. Pay particular attention to the IDC26 ribbon cables connecting the UCBB boards to the UC300ETH, the servo motor power and encoder cables, and the VFD connections.

Check pneumatic system. Inspect all pneumatic lines, fittings, and solenoid valves for leaks. Apply soapy water to fittings and check for bubbles while the system is pressurized. Replace any damaged tubing or fittings.

Inspect the tool rack clamps. Check all 10 ISO30 tool holder clamps for proper spring tension and alignment. Clean the clamp surfaces to ensure tools are held securely.

Test the Emergency Stop circuit. Press each Emergency Stop button and verify that the machine enters RESET state immediately. Verify that the probe overtravel function also triggers a stop. After testing, re-home the machine.

Check the tool probe. Verify the tool probe plunger moves freely and returns to its rest position under spring pressure. Clean the probe surface. Test probe accuracy by measuring a known-length tool and comparing the result with the expected value.

Inspect cabinet cooling. Check that the two cooling fans in the control cabinet are functioning and that the air filter is not clogged with dust. Replace or clean the filter as needed.

Quarterly Maintenance

Verify gantry squaring. Use the Half-Diagonal Beam Compass Method (described in the Adjustment Capabilities chapter) to verify gantry perpendicularity. If the gantry has drifted out of square by more than the acceptable tolerance, perform the mechanical squaring adjustment using the pusher bolts.

Check the spindle tramming. Use a dial indicator or tramming tool to verify spindle perpendicularity in both the X and Y planes. Adjust using the tramming provisions (axis bump for sideways, pusher bolts for forward/backward) if needed.

Inspect the gear reducers. Check the planetary gear reducers (Liming FA-100, 1:7 ratio) on the X and Y axes for unusual noise, vibration, or play. These reducers are generally sealed and do not require regular lubrication, but excessive play or noise may indicate bearing wear.

Inspect the vacuum table. Check the perforated rubber surface of the vacuum table for damage or wear. Verify that all 8 vacuum zones are producing adequate suction. Clean the vacuum channels if suction has decreased.

Annual Maintenance

Full machine leveling check. Using a precision level, verify that the machine structure is level in both the X and Y directions. Adjust the 8 heavy-duty leveling casters (GD-80-S) as needed.

Gantry assembly leveling. Check the gantry assembly level and adjust the 8 linear guide block mounting points using the M8 pusher bolts if needed.

Servo drive parameters. Download and review the servo drive parameters for all four axes. Compare with the reference values in the Configuration Manual. Reset any parameters that have drifted from the reference values.

VFD parameters. Review the DELTA MS300 VFD parameters. Compare with the reference values in the Spindle and VFD Integration Manual.

Replace worn components. Based on visual inspection and operational experience, replace any components showing significant wear: timing belt, linear guide block seals, pneumatic tubing, spindle collet springs, tool rack clamp springs, and so on.

Backup the UCCNC profile. Create a full backup of the UCCNC profile folder, including all configuration files (.pro), macro files (.txt), screen files, and tool tables. Store the backup in a safe location.

Maintenance Log

It is recommended to keep a maintenance log documenting all maintenance activities performed on the machine, including the date, the work performed, any parts replaced, and the name of the person who performed the maintenance. This log is invaluable for tracking component life, identifying recurring issues, and maintaining warranty records.

15. TROUBLESHOOTING

This chapter guides the diagnosis and resolution of common issues that may arise during the operation of the StratoCNC.

The machine does not home properly

If the homing sequence (M20301) fails or produces unexpected results, check the following: verify that no objects are obstructing the axis travel paths; confirm that all limit switches are functioning (check the LED indicators on the UCCNC screen for proper state changes as sensors are triggered); ensure the tool rack is retracted, the dust collector is lifted, and other pneumatic components are in safe positions before homing; check that the servo drives are enabled and not in alarm state. If one Y-axis side homes but the other does not, check the corresponding front limit switch and its wiring.

Tool Change Fails

If an automated tool change (M6 / M20006) fails, the macro system will attempt a SafetyRecovery sequence. Common causes include: the tool rack is not fully extended (check the rack proximity sensor); the rotating base is not locked into the machine (check the base position sensor); the spindle did not fully reach the tool slot position (check for mechanical obstructions); the collet did not open or close properly (check the S2 sensor and the pneumatic solenoid); the tool was not detected after pickup (check the S1 sensor). If a tool is stuck in the spindle or in a rack slot, use the manual tool release button or the solenoid override button to open the collet and remove the tool by hand.

Spindle Does Not Start

If the spindle fails to start, the macro system provides specific diagnostic messages. Common causes include: no tool is detected in the spindle (S1 sensor check fails) -> load a tool before starting the spindle; the target RPM field is empty or set to zero -> enter a valid RPM value; a servo alarm or spindle health alarm is active -> resolve the underlying fault first; the machine is not homed -> home the machine first. If the spindle command is issued but the motor does not rotate, check the VFD for fault codes, verify the spindle control wiring (CW/CCW relay outputs on UCBB-2), and check the VFD analog speed command.

Poor Cut Quality

If the machine is producing poor cut quality (rough surfaces, inaccurate dimensions, or chatter marks), investigate the following: tool condition (dull or damaged tools are the most common cause of poor cuts); verify the correct RPM and feed rate for the material and tool; check for excessive play or backlash in the axis drives (worn pinions, loose gear reducers, worn ball screw nut); verify spindle tramming (a spindle that is not perpendicular to the work surface causes asymmetric cuts); verify gantry squaring (an out-of-square gantry causes dimension errors); check the workpiece clamping (a workpiece that shifts during cutting causes inaccurate cuts).

Communication Errors

If UCCNC reports communication errors with the UC300ETH, check the Ethernet cable connection between the PC and the controller (the green Link LED inside the RJ45 jack should be lit). Ensure the UC300ETH has had at least 3 minutes to initialize after power-on. Verify the PC network settings are correct for the UC300ETH. Check the 5V power supply to the UC300ETH (the green power LED should be lit). If using a network switch or router, verify it is functioning. If communication errors are intermittent, check for electromagnetic interference sources near the Ethernet cable.

Vacuum Table Not Holding

If the vacuum table is not holding the workpiece securely, check that all vacuum motors are running (listen for each zone). Verify that the perforated rubber surface is not damaged or clogged. Ensure the workpiece is lying flat with no debris underneath. For small workpieces, more zones will be uncovered (open to atmosphere), reducing the total suction force – this is normal, but you may need to use additional hold-down methods for very small pieces. Check for air leaks around the table edges and vacuum channels.